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Progress in PDMS modification strategies for high-performance triboelectric nanogenerators: design, applications and future perspective

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ABSTRACT

Eco- and bio-friendly polydimethylsiloxane (PDMS) material-based triboelectric nanogenerators (TENGs) have attracted significant attention for applications in energy harvesting, sensing, and self-powered systems. The performance of PDMS-based TENGs is highly dependent on rational material selection and structural engineering within the PDMS matrix. While conventional components such as dielectrics, metals, two-dimensional (2D) materials, and carbon-based fillers have been widely used, there is a growing demand for advanced materials to enhance efficiency and broaden the scope of applications. In this context, a variety of fillers have been explored in PDMS-based TENGs owing to their favorable properties, including biocompatibility, high sensitivity, non-toxicity, and superior electron mobility. These features make them particularly suitable for next-generation energy harvesting technologies. Unlike previous reviews, this work provides an in-depth and systematic overview of PDMS-based TENGs incorporating diverse fillers, with a focus on their impact on triboelectric performance and practical applications. It highlights the influence of these fillers on output efficiency and expands on their use in areas such as energy harvesting, self-powered sensing, human–machine interaction, and wearable devices. The review concludes by outlining current challenges, emphasizing the need for further research, and offering perspectives to guide future developments in this rapidly advancing field.

1. Introduction

1.1. Background of energy harvesting technologies

Due to rapid development, the global energy demand and heavy dependence on fossil fuels have resulted in a severe energy and environmental crisis. To this end, an alternative technologies like energy harnessing from the eco- and bio-friendly and renewable sources are highly in demand to achieve sustainable growth [1,2]. It has the ability to power different electronic devices and systems by gathering energy from naturally available phenomena or sources and converting it into usable electrical power [3–5]. These innovations exhibit great promise for applications where conventional power sources, especially Li-based batteries, are limited by their drawbacks, such as limited lifespan, environmental harm, and frequent replacement requirements [6]. Energy harvesting is all about generating electricity for low-power electronics [7,8] such as wearable sensors [9], IoT devices [10,11], biomedical implants [12], and microelectronics [10,13,14] by tapping into natural sources like light, electromagnetic waves, temperature differences, and mechanical vibrations [15]. In particular, the widespread presence of mechanical energy in our environment has made it a key focus of interest. Some of the most effective technologies for capturing mechanical energy from sources like human motion [16–22], wind [23–26], and ocean waves [27–30] include piezoelectric [31–34], triboelectric [35,36], and electromagnetic generators [37–41]. In recent years, triboelectric nanogenerators (TENGs) have emerged as a promising approach that converts mechanical energy into electrical power through contact electrification and electrostatic induction. TENG's

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simple design and its ability to integrate seamlessly with lightweight, flexible materials make it ideal for wearable technology and Internet of Things (IoT) applications.

1.2. Overview of TENGs

TENGs represent an innovative way to capture mechanical energy from the environment and turn it into electrical power through the triboelectric effect and electrostatic induction. The phenomenon of triboelectrification, or contact electrification, was first observed by Plato around 400BCE Then, around 300 CE, the Chinese philosopher Pu Guo observed that amber would become electrically charged through friction [42]. Since Prof. Zhong Lin Wang first introduced TENGs in 2012 [43], they have become known as an attractive choice for developing self-powered and environmentally friendly electronics. They are especially well-suited for use in wearable technology, sensors, and the rapidly expanding IoT systems [44,45].

TENGs work based on a simple yet fascinating concept called contact electrification [46]. When two different materials come into contact and then pull apart, they become electrically charged. This happens because electrons move from one material to the other, depending on how easily they gain or lose electrons (as determined by their positions in the triboelectric series). Once these materials separate, a difference in electrostatic potential forms between them, which can cause electrons to flow through an external circuit, generating electricity. When the layers are in a short-circuit condition, where the load resistance is much lower than the TENG's input impedance, the potential difference causes electrons to move between the electrodes, generating a short-circuit current (I_{sc}) that can be used for power. In contrast, under open-circuit conditions, the open-circuit voltage (V_{oc}) is measured, indicating how much triboelectric energy has been produced [47–51]. The amount of triboelectricity generated depends on the surface charge density after contact, and the performance of the TENG is usually evaluated by the peak voltage or current generated when the layers are separated/in contacted with a specific contact force, frequency, and load resistance. As the layers pull apart completely, both the V_{oc} and I_{sc} decrease because the gap between them increases.

A continuous flow of current is generated whenever there's mechanical motion, allowing TENGs to efficiently harvest energy [52–60]. There are four distinct ways that TENGs produce electricity, and they are all dependent on mechanical motions. In the contact-separation mode [61–63], two surfaces repeatedly touch and pull away, creating electricity. The sliding mode [64–69] generates energy through side-to-side surface friction. The single-electrode mode [70–72] is advantageous for harvesting energy from freely moving objects, and the freestanding mode allows for contactless energy generation, making it suitable for applications like wind and water energy harvesting. Fig. 1 shows a schematic diagram illustrating the four different operating principles of TENGs.

TENGs are great for capturing energy from various mechanical sources, but what really sets them apart is their lightweight, flexible design. In addition, they're affordable and easy to scale, making them perfect for portable and wearable devices [73–84]. At larger scales, TENGs can be integrated into floor tiles/smart roads [85], wave-energy buoys [86], or wind-harvesting [87] systems to capture environmental energy, although such applications often require energy storage or hybrid systems to supply continuous power. Furthermore, TENGs can be developed with biodegradable and biocompatible materials [88–96], making them a promising option for sustainable energy generation. Even though TENGs have a lot of potential, there are still issues with enhancing their durability and performance, particularly in changing environmental conditions [97–101]. Researchers are working hard to enhance their mechanical stability [102–105], energy conversion efficiency, and flexibility [106–110] for applications in everyday life.

2. Strategies in triboelectric material selection

The overall performance and efficiency of TENGs are highly dependent on the selection of triboelectric materials. TENGs operate on the principle of contact electrification between two triboelectric materials with different electron affinities: the material with stronger electron affinity acts as the electron acceptor, while the other functions as the electron donor [46,111]. Fig. 2 illustrates representative donor and acceptor materials commonly employed in TENGs. The position of materials in the triboelectric series provides a useful guideline to predict their polarity and potential for charge transfer [112]. In addition to the

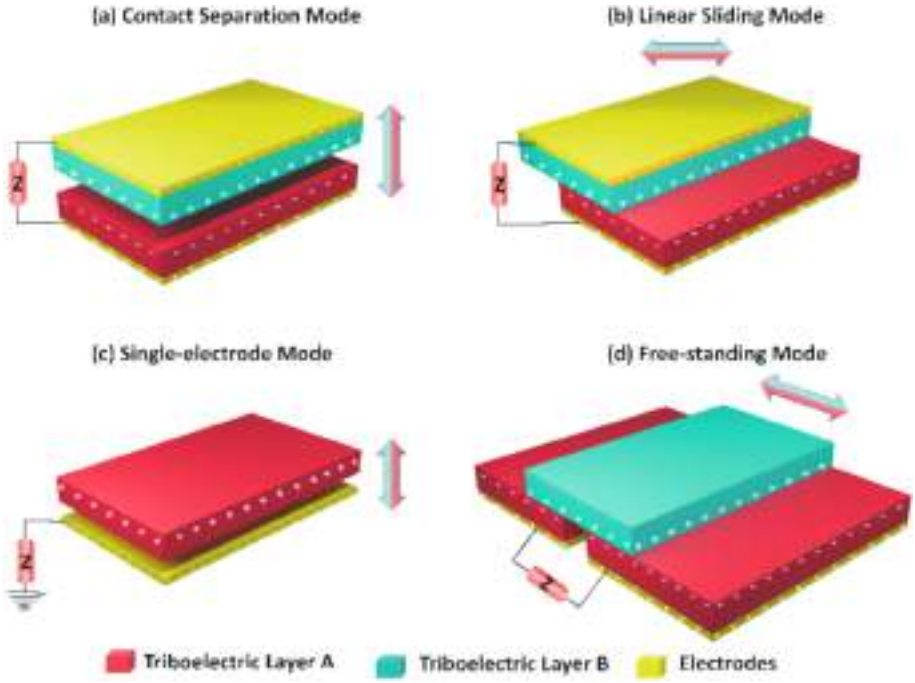


Fig. 1. Schematic diagram of the four fundamental working modes of TENG.

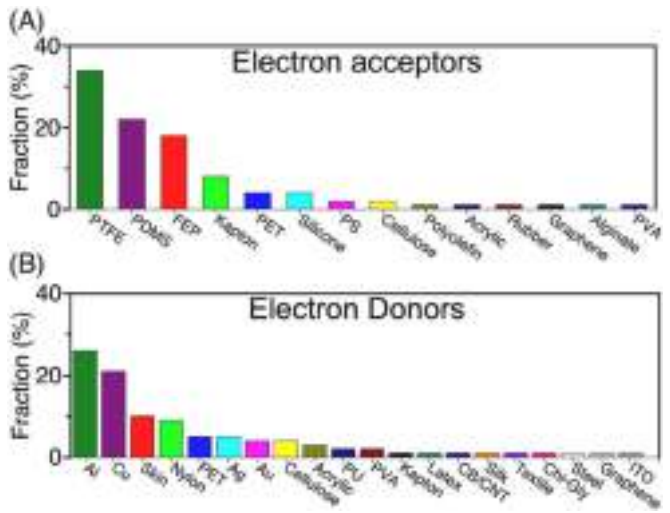


Fig. 2. Materials of choice in the triboelectric series. (A) Electron-acceptor materials (highly electronegative, tending to gain electrons). (B) Electron-donor materials (highly electropositive, tending to lose electrons). . Reproduced with permission from Ref. [116]

electronegativity of materials, their flexibility, durability, and conductivity are equally critical factors. Flexibility ensures reliable performance in wearable or deformable devices, where TENGs are subjected to bending and stretching [113]. Durability is essential for withstanding repeated contact–separation cycles without degradation, enabling stable long-term operation [114]. Conductivity, though not inherent to the triboelectric layers themselves, is crucial for the efficient transfer and collection of charges in external circuits. Thus, selecting triboelectric materials requires a balanced consideration of both electrical and mechanical properties, as well as long-term reliability [113,115]. Historically, however, research in this field prioritized maximizing charge density and output power, often at the expense of flexibility or environmental considerations. In this context, fluorinated polymers emerged as the dominant tribo-negative materials due to their superior electron affinity and charge retention.

2.1. Limitations of fluorinated polymers

Fluorinated polymers such as polytetrafluoroethylene (PTFE) [117,118], fluorinated ethylene propylene (FEP) [119], perfluoroalkoxy alkane (PFA) [120], and polyvinylidene fluoride (PVDF) [100,101] have long served as benchmark tribo-negative materials. Positioned at the far negative end of the triboelectric series, these polymers exhibit strong electron affinity, high dielectric strength, and mechanical durability. PTFE, for instance, consistently yields high-output voltages, while PVDF combines negative polarity with ferroelectric polarization, further enhancing triboelectric performance [117,119,121,122]. Yet, these advantages come with significant drawbacks. Fluorinated polymers are highly resistant to chemical degradation, making them non-biodegradable in the environment [123,124], prone to bio-accumulation [125], and potentially toxic [126]. Their durability, which benefits device stability [127], simultaneously creates waste management challenges [128]. Moreover, their synthesis often relies on hazardous precursors such as hydrofluoric acid and perfluorooctanoic acid, introducing risks to human health and the environment [129,130]. With regulatory pressures mounting against persistent fluorinated compounds [131,132], their long-term suitability for sustainable TENG development has been called into question. These limitations highlight the urgent need for environmentally benign, mechanically flexible, and versatile alternatives to fluorinated tribo-negative materials.

2.2. Why focus on PDMS?

While fluorinated polymers remain dominant in high-performance applications, PDMS has attracted increasing attention as an alternative due to its versatility, low environmental impact, and excellent mechanical flexibility and compliance. PDMS is a non-fluorinated elastomer composed of a silicon–oxygen backbone with methyl side groups, as shown in Fig. 3A. This molecular structure provides a unique balance of stability and flexibility: the strong Si–O bonds confer high thermal and chemical stability, while the flexible bond angles maintain mechanical elasticity [133,134]. The surface methyl groups render PDMS hydrophobic, contributing to its water-repellent behavior. Moreover, the ionic character of Si–O and Si–C bonds allows PDMS to undergo bond scission under certain conditions, generating reactive radicals that can be advantageous in energy conversion applications [135,136]. In addition, PDMS exhibits optical transparency, low dielectric constant, and intrinsic elasticity, which make it particularly suitable for flexible and wearable TENGs. Its material properties can be readily tuned by varying the base-to-curing agent ratio or by incorporating other additives, enabling precise control over stiffness, dielectric response, and overall mechanical performance in Fig. 3B [137].

To better contextualize the significance of PDMS as an alternative material, it is important to compare its properties with other tribo-negative materials, especially fluoropolymers. As summarized in Table 1, fluorinated polymers such as PTFE and PVDF generally exhibit superior TENG performance due to their strong electron-withdrawing nature and high charge retention capability. In contrast, PDMS offers distinct advantages in terms of flexibility, ease of processing, and lower environmental impact as discussed earlier. However, its relatively lower intrinsic charge density and susceptibility to environmental factors such as humidity, temperature variations, surface contamination, and mechanical wear during repeated operation may limit its performance in high-output applications. This difference primarily arises from the lower electron affinity and weaker electron-withdrawing capability of PDMS, which restrict its charge saturation density and accelerate charge dissipation under ambient conditions. To better visualize this comparison, a schematic diagram is presented in Fig. 4. The comparison is qualitative and based on widely reported trends in the literature for triboelectric nanogenerator materials. The bar intensity represents relative performance (low to high) and is intended for illustrative comparison rather than quantitative evaluation.

The evolution of PDMS-based TENGs is summarized in Fig. 5, highlighting the transition from foundational studies to application-driven systems. Early work (2012–2013) focused on establishing the working mechanism and proof-of-concept energy harvesting using pristine PDMS. This was followed by engineering advancements (2013–2015), including surface microstructuring and improved device architectures, enabling self-powered sensing. Subsequent optimization (2015–2018) through material modification and hierarchical structuring significantly enhanced performance and enabled wearable applications. More recent developments (2018–present) emphasize system-level integration, including electronic skin, wireless sensing, and intelligent self-powered systems, clearly distinguishing early foundational contributions from later application-oriented advances.

2.3. Strategies to enhance PDMS performance

In the triboelectric series, PDMS is positioned as a negative material, acting as an effective electron acceptor during contact electrification. Despite its advantages, pristine PDMS also suffers from limitations such as low surface energy and mechanical softness, which can restrict charge density and long-term durability [133]. To overcome these challenges, various modification strategies have been developed. For example, surface morphology engineering approaches such as micro/nano-patterning (e.g., pyramids or pillars) play a role in increasing the effective surface contact area and enhancing charge generation

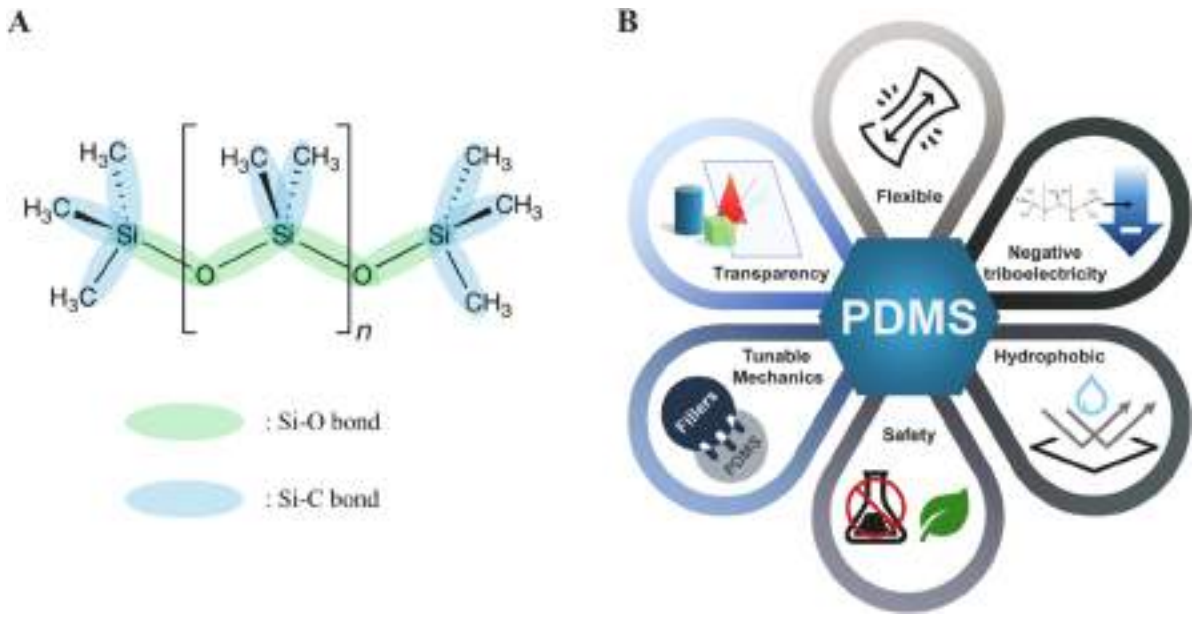


Fig. 3. (A) Chemical structure of PDMS. (B) Representative properties of PDMS.

Table 1
Comparative assessment of PDMS and fluorinated polymers (PTFE/PVDF) for TENG applications.

Property	PDMS	PTFE/PVDF
Charge density/ output [49,72]	Moderate(enhanced via engineering)	High
Electron affinity [72]	Moderate	Very high
Flexibility [138]	Excellent	Limited flexibility (especially PTFE)
Processability	Easy (casting, molding)	More complex
Environmental impact	Relatively benign	Environmentally persistent; recycling challenges [126,139]
Cost	Low	Higher
Mechanical durability	Good	Excellent
Stability (humidity)	Moderate (susceptible to humidity-induced charge decay)	Better

[140,141]. Surface functionalization improves electron affinity and surface energy. Composite blending with high-dielectric fillers (e.g., barium titanate ($BaTiO_3$)), conductive additives (e.g., carbon nanotubes (CNTs)) enhances dielectric strength, triboelectric output, and mechanical robustness [142,143]. Furthermore, hybrid and multilayer designs such as stacking PDMS layers with different properties or pairing PDMS with tribo-positive materials can significantly optimize charge transfer efficiency [144–147]. Overall, these strategies aim to improve PDMS-based TENGs performance through two primary routes: (1) modifying the physical surface structure, and (2) blending with other materials. As reflected in the rapid growth of publications on PDMS-based TENGs (Fig. 6), this material has become a focal point of recent research. In the scientific community, major strategies are commonly employed in the PDMS-based TENGs, as shown in Fig. 7. These strategies include PDMS surface modification, bulk composites, structure-based hybrid composites, layer-by-layer deposition, and encapsulation.

From an application perspective, fluoropolymers remain preferable for systems requiring maximum electrical output and long-term charge stability. On the other hand, PDMS is particularly well suited for flexible, stretchable, and wearable devices, where mechanical compliance and ease of fabrication are critical. Consequently, the choice between these materials should be guided by specific application requirements rather than a single performance metric. Based on this evolution, the following sections of this review will provide a detailed discussion of the modification strategies and their impact on PDMS-based TENG device performance. To the best of our knowledge, this is the first review that focuses exclusively on PDMS-based TENGs.

3. Modification strategies

3.1. Surface engineering strategies

Surface engineering/modification strategies of PDMS is a powerful strategy to enhance its performance in TENGs, making it more effective for energy harvesting. In general, the performance of a TENG is proportional to the charge density on the contact surface, so increasing charge generation has been a key approach to improving output electrical power. The square of the charge density is often used as a material figure of merit to quantify a TENG's performance [148–152]. The first strategy for enhancing TENG performance involves modifying the surface of the PDMS film. Surface modification mainly improves performance through three mechanisms: (1) increasing surface roughness, (2) enhancing localized electric fields, and (3) reducing contact stiffness. First, as shown in Fig. 8a, introducing patterns or roughness on the PDMS surface increases the real contact area (A_r) between the triboelectric layers [153–155]. If the local triboelectric surface charge density is $\sigma(C/m^2)$, the total charge can be approximated as $Q \approx \sigma A_r$ [149,150,154]. Therefore, increasing A_r directly boosts charge generation and subsequently enhances the short-circuit charge transfer and output performance. Second, as illustrated in Fig. 8b, micro- or nanoscale edges, tips, and asperities can induce localized electric-field enhancement [156,157]. These intensified fields facilitate more efficient charge separation during contact-electrification, thereby increasing the surface charge density [158]. Third, the inherently soft and compliant nature of PDMS results in a lower effective elastic modulus, allowing the material to conform more intimately to the counter-surface under applied load [159–161]. This improved conformity further increases the real contact area, making the interface more uniform and enhancing triboelectric charge transfer, as shown in Fig. 8c. One of the simplest and most widely adopted ways to increase TENG output is by expanding the effective contact area through surface microstructuring. Because the real contact area between two surfaces of

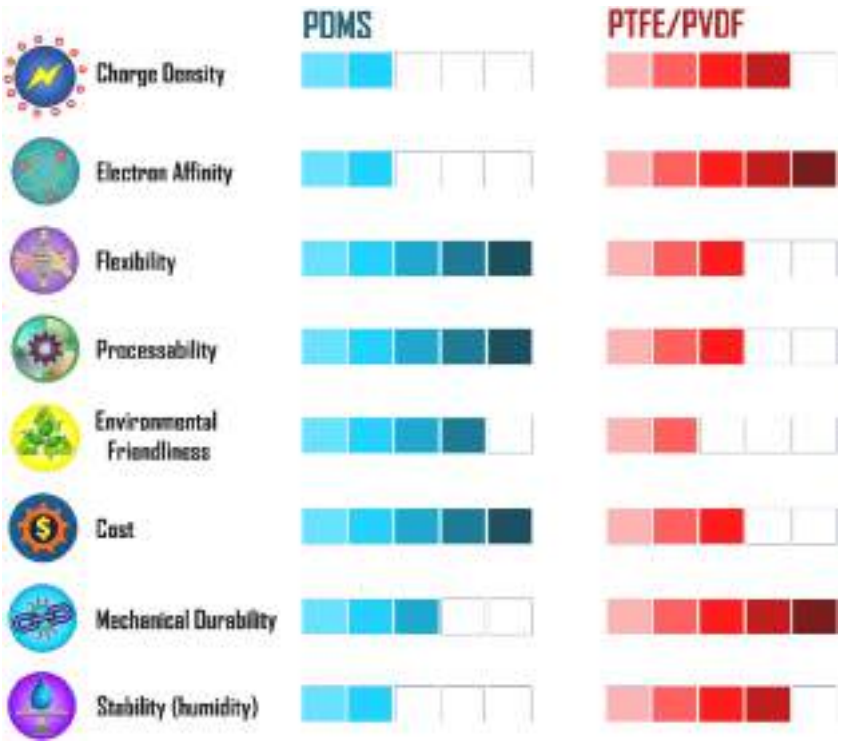


Fig. 4. Trade-off comparison between PDMS and fluorinated polymers for TENG applications *[Bar intensity represents qualitative performance (low → high)]*.

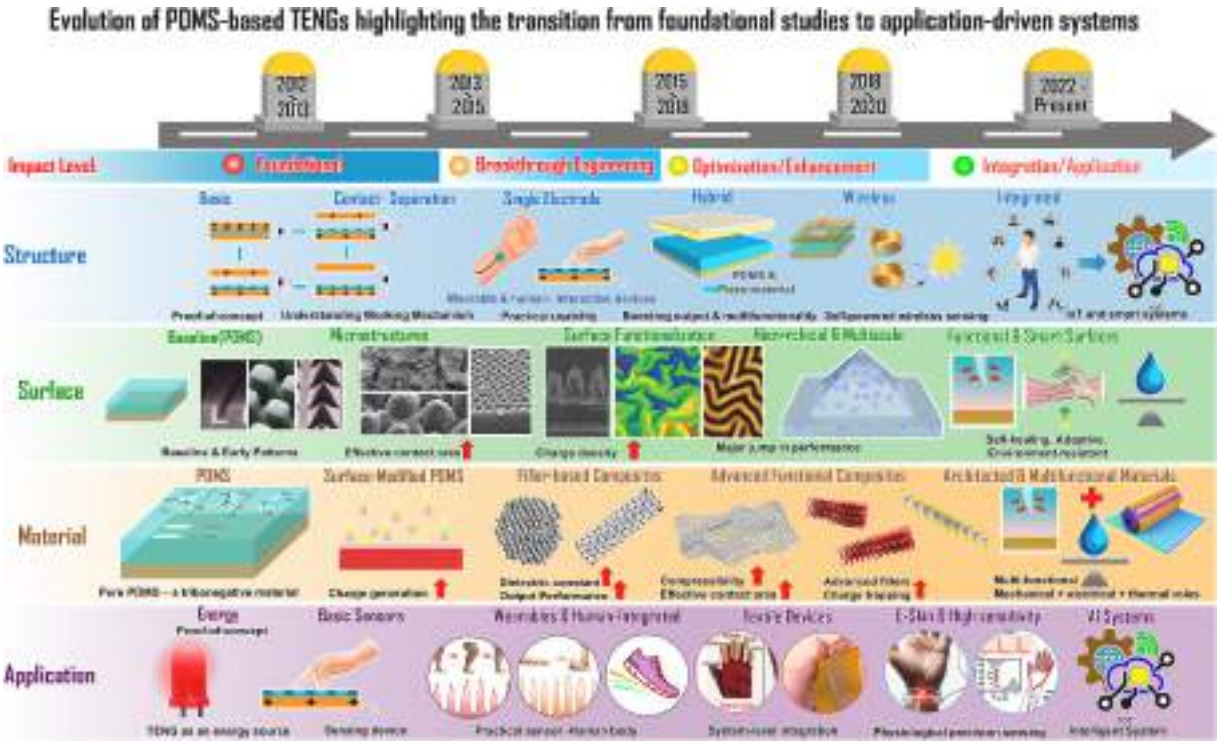


Fig. 5. Evolution of PDMS from foundational level to application level.

triboelectric layers under pressure is typically much smaller than their apparent area, researchers aim to employ micro- and nano-texturing to overcome this limitation. A variety of fabrication techniques, such as soft lithography, photolithography, laser micro-patterning, colloidal self-assembly, and biomimetic replication are commonly used. Additionally, chemical and plasma treatments can be applied to tune surface morphology and improve contact characteristics.

3.1.1. Nano/micro patterns

Yang et al. [140] investigated the fundamental behavior of the effective contact area in micro- and nanotextured surfaces for TENGs through both theoretical modelling and experimental validation

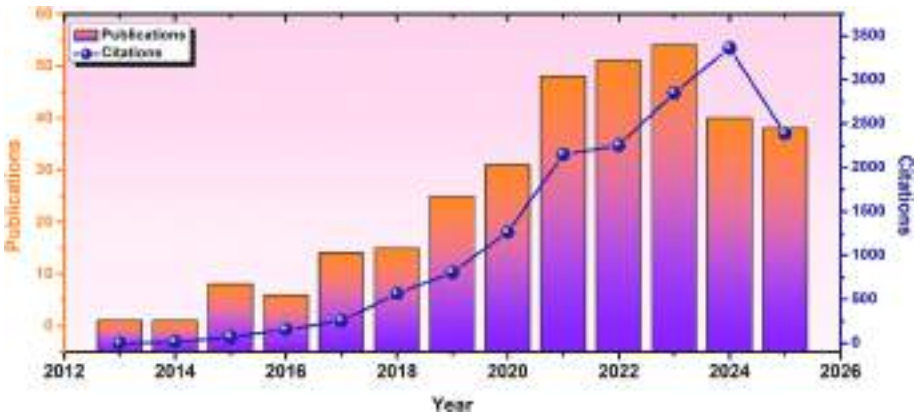


Fig. 6. Year-wise publications & citations [.
Source: WoS search - "PDMS composites Triboelectric Nanogenerators"]

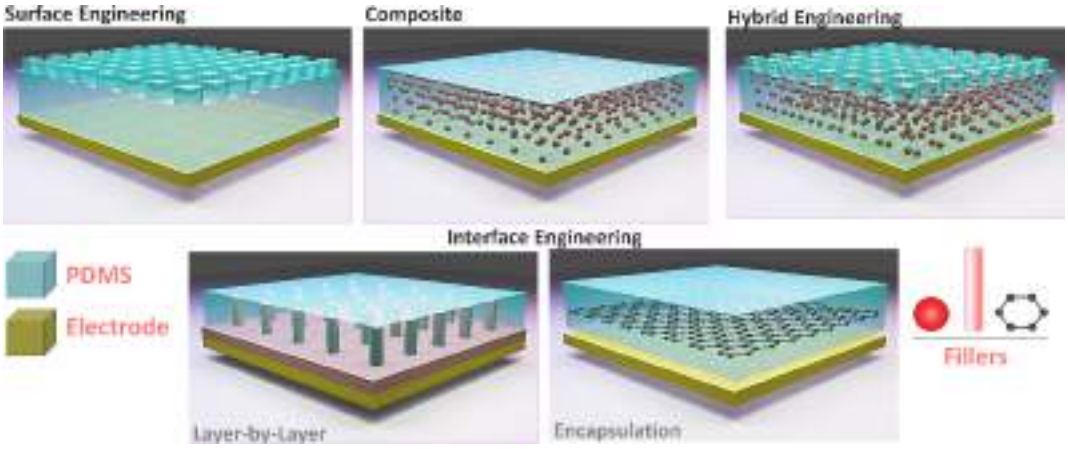


Fig. 7. Schematic illustration of different strategies following for PDMS-based TENGs for enhancing output performance.

(Fig. 8d). They developed an adhesive contact model to investigate how applied force and texture size influence the effective V_{oc}, using numerical methods such as the Newton method, Bi-CGSTAB, and FFT. Experimentally, pyramid-patterned textures were fabricated to validate the model. Their results show that the V_{oc} increases with applied load up to 70 N due to the expanding contact area on textured and flat regions, and then saturates beyond 70 N once full contact is achieved (Fig. 8e and f).

One of the most widely used techniques for fabricating well-defined micro- and nanostructures involves pouring PDMS onto a patterned silicon mold (mother pattern, created via lithography) and then peeling it off (Fig. 8g) [166]. The process begins with photolithography to pattern silicon wafers, followed by anisotropic etching to form recessed structures. PDMS elastomer and cross-linker are mixed thoroughly, poured onto the molds, degassed under vacuum to remove air bubbles, and spin-coated to achieve a uniform layer. After curing, the PDMS film is peeled off, replicating the mold features [138]. Feng-Ru Fan et al. [141] fabricated line, cube, and pyramid-shaped PDMS patterns using the soft lithography method, as shown in Fig. 8h–j. Among these, the pyramid-patterned devices demonstrated enhanced output, likely due to the three mechanisms described earlier. Similar pyramid PDMS structures have also been explored for TENG devices by other researchers, including Chandrashekar et al. [163] Zong-hong Lin et al.,[162] and Harris Varghese et al. [164] as illustrated in Fig. 8(k–m), respectively. Additionally, in 2021, Hoang Si Hong et al. [165] fabricated a corrugated PDMS pattern using soft lithography, shown in Fig. 8n. These studies collectively highlight the effectiveness of nano/micro patterning in enhancing TENG performance by increasing the triboelectric contact area and optimizing surface interactions.

3.1.2. *Alternative method for PDMS microstructure fabrication*

Beyond the silicon mold and peeling technique, researchers have explored several alternative approaches to fabricate PDMS micro- and nanostructures. Dongjin Jang et al. [167] utilized polystyrene (PS) colloids to create hierarchical PDMS surfaces, producing two distinct patterns: intaglio and embossed PDMS (Fig. 9a and b). Similarly, Ju Hyun Lee et al. [168] fabricated a micropillar PDMS structure (Fig. 9c) using a photoresist-based mold.

Seol et al. [169] drew inspiration from natural surfaces such as lotus leaves, rose petals, and cicada wings to fabricate PDMS surface patterns. They created a nano-in-micro morphology through a two-step replication process. First, a hard PDMS (h-PDMS) layer (~10 μm) was spin-coated onto the original natural template and cured. A soft PDMS (s-PDMS) layer was then applied on top to form the first mold. This mold captured the negative features of the original surface. To prepare for the second replication, the first mold was treated with oxygen plasma to generate hydroxyl groups on the surface, followed by the formation of a self-assembled monolayer (SAM). This step made the surface superhydrophobic, minimizing adhesion between h-PDMS layers during the second molding. The second molding process mirrored the first, but without the addition of the plasma and SAM treatments. After completing the double-layer h-PDMS/s-PDMS replication, a positive replica of the original natural surface was obtained, shown in Fig. 9(d–f). Finally, electrodes and spacers were added to integrate the patterned surface into a functional TENG device. In addition, Yue Sun et al. (2022) [170] demonstrated a method to fabricate waterproof TENG devices using 3D-structured PDMS patterns. They first created a negative mold by engraving micropatterns onto an acrylic plate with a laser cutter. The

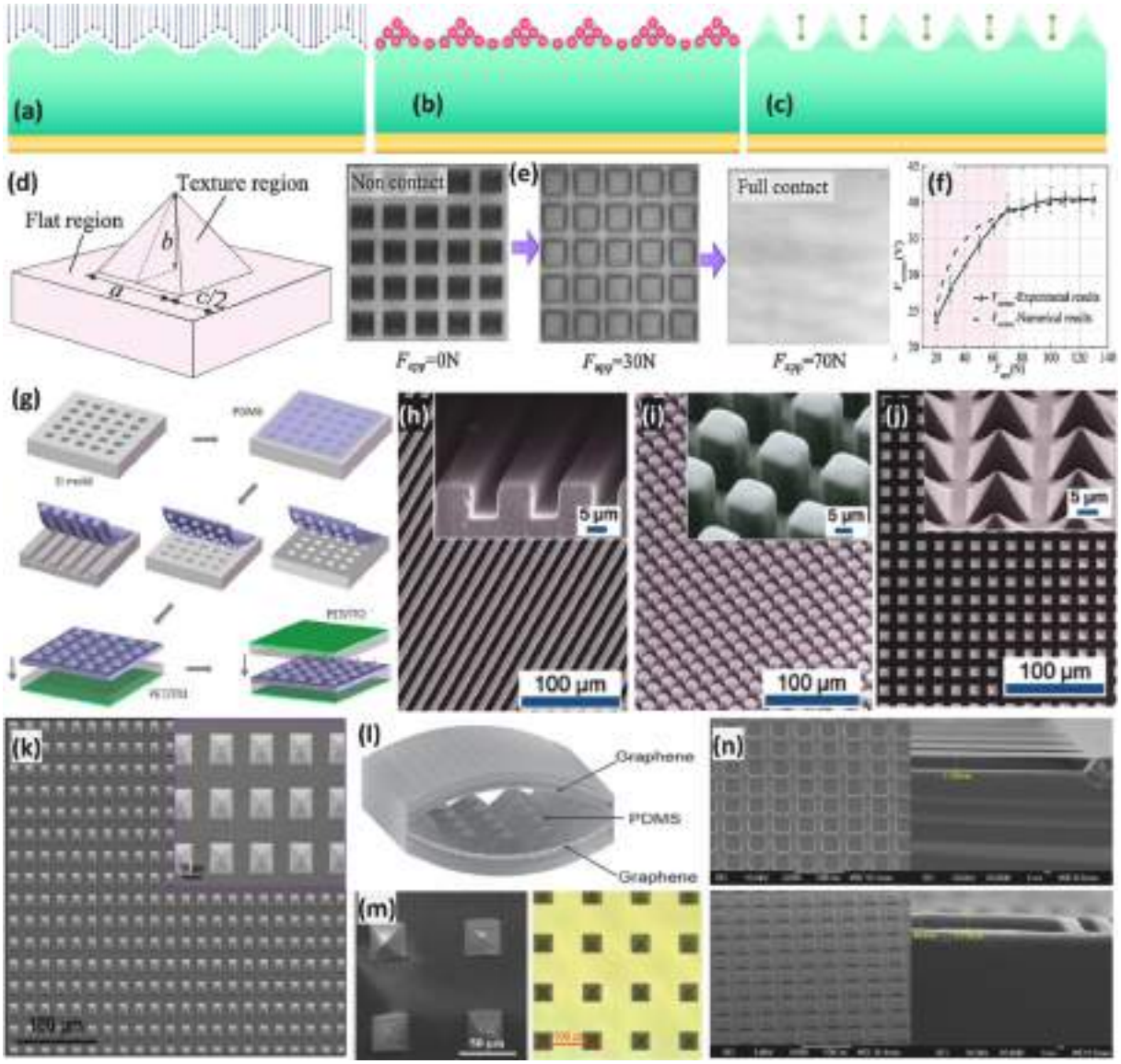


Fig. 8. PDMS surface patterning using the soft-lithography approach. (a–c) Schematic illustrations showing three key mechanisms expected from surface modification: (a) increased real contact area, (b) enhancement of localized charges, and (c) reduced contact stiffness. (d) Schematic comparison between flat and textured PDMS regions, (e) contact images of textured PDMS surfaces under different applied forces, and (f) variation in I_{sc} with applied force. Reproduced with permission from Ref. [140]. (g) Schematic representation of the fabrication process and (h–j) SEM images of patterned PDMS thin films with line, cube, and pyramid structures, respectively (insets show higher magnifications). Reproduced with permission from Ref. [141]. (k) SEM images of the PDMS pyramid array (inset: magnified view). Reproduced with permission from Ref. [162]. (l) Schematic of the pyramid-patterned PDMS layer used in a graphene-based triboelectric nanogenerator, Reproduced with permission from Ref. [163] and (m) pyramidal surface-modified PDMS film. Reproduced with permission from Ref. [164]. (n) SEM images of the silicon mold after wet etching and the corresponding corrugated PDMS surface after peel-off. Reproduced with permission from Ref. [165].

PDMS solution was then poured into this custom mold, and after curing, the structured PDMS film was carefully peeled off, producing a patterned layer ready for integration into TENG devices, as illustrated in Fig. 9g. Young Jung et al. [171] developed a novel approach to fabricate spherical micro- and nanoscale hierarchical PDMS patterns using the thermal expansion of microspheres. In this method, the microspheres consist of a thermoplastic acrylonitrile (AN) copolymer shell surrounding a liquid hydrocarbon core. When heated to a specific temperature, the AN shell softens while the hydrocarbon core vaporizes. The resulting gas expands through the thin polymer shell, causing the microspheres to grow three to six times their original size. After cooling, the expanded microspheres retain their shape due to the plastic deformation of the shell. These expanded microspheres embedded in PDMS create micro- and nanoscale spherical patterns, as illustrated in Fig. 9h–i.

3.1.2.1. Plasma treatments. In addition to micro- and nanoscale structuring, PDMS surface properties can be further enhanced through treatments such as plasma surface modification, offering an effective strategy to improve TENG output, as illustrated in Fig. 10a. Choonghyun Lee et al. [172] functionalized PDMS surfaces using a two-step RIE plasma treatment. In the first step, Ar plasma was used to break weak molecular chains on the PDMS surface. In the second step, a $CF_4 + O_2$ plasma treatment functionalized these broken bonds, which are highly reactive, introducing chemical groups with higher electron affinity [173,174]. This functionalization improves the surface's roughness (nano-textured) and chargeability to attract additional charges during TENG operation, enhancing device performance, as shown in Fig. 10b. Fig. 10(c–d) present SEM images of the plasma-modified PDMS surface and the corresponding current density measurements, respectively. Similarly, Guang-Gui Cheng et al. [175] fabricated a micropillar-array

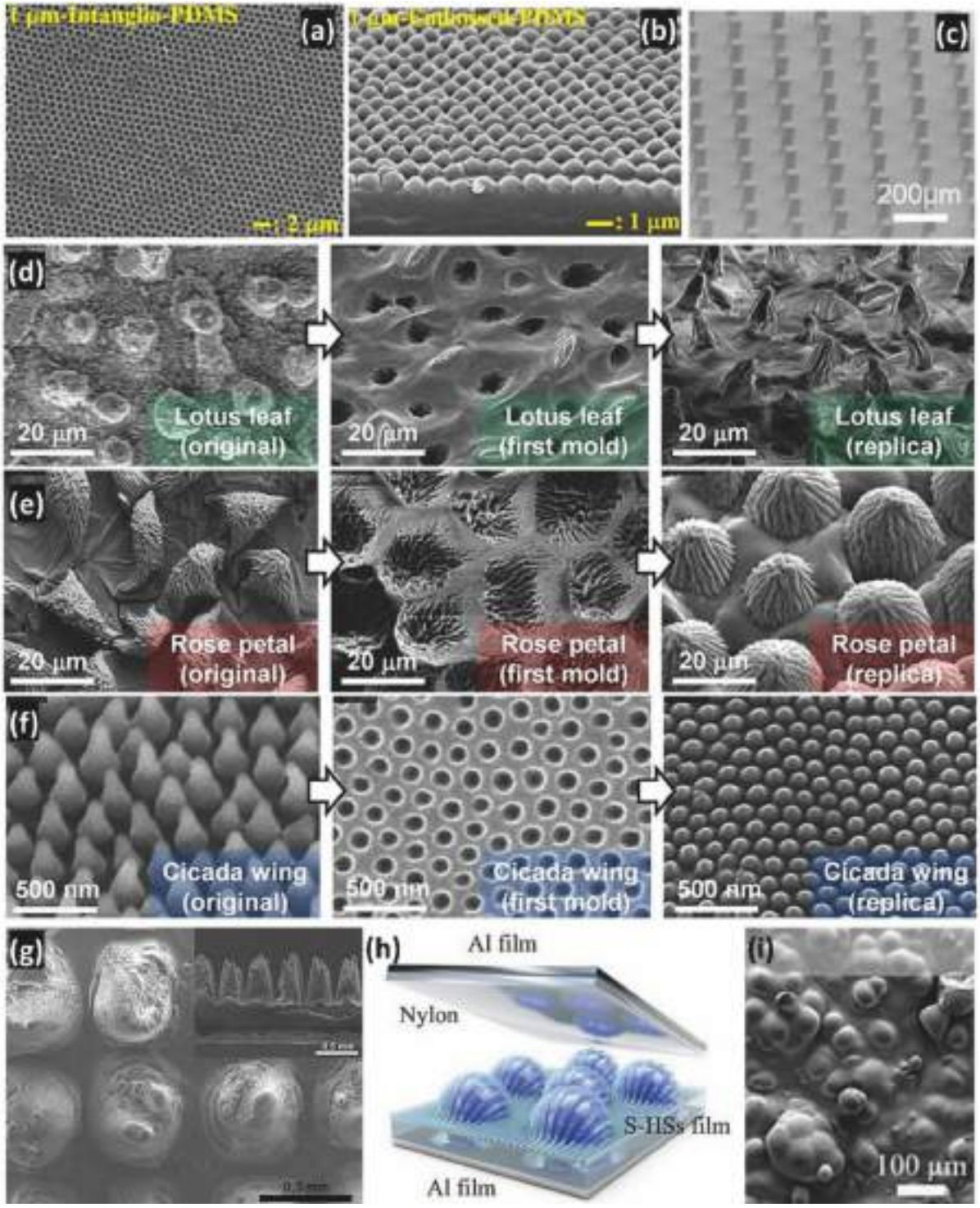


Fig. 9. Various surface patterning approaches for PDMS films. (a, b) SEM images showing intaglio and embossed PDMS structures replicated from a PS monolayer-coated Si wafer. Reproduced with permission from Ref. [167]. (c) Tilted SEM image highlighting the well-defined PDMS pillar morphology. Reproduced with permission from Ref. [168]. (d–f) Biomimetic surface replications of lotus leaf, rose petal, and cicada wing textures, demonstrating nature-inspired micro/nano-structures. Reproduced with permission from Ref. [169]. (g) SEM image of a 3D PDMS architecture fabricated using an acrylic laser-cutting technique. Reproduced with permission from Ref. [170]. (h) Schematic illustration of a microsphere-assisted spherical surface design for TENG application, and (i) corresponding SEM image of the fabricated surface. Reproduced with permission from Ref. [171].

PDMS layer treated with Ar plasma for use in a TENG device, as shown in Fig. 10e and f. They found that the plasma power had a significant influence on both the etching efficiency of PDMS and the resulting output performance of the device. Another effective surface modification approach involves creating nanoscale wrinkle structures in combination with plasma treatment. In this method, PDMS films are first pre-stretched, then exposed to fluorocarbon (C_4F_8) plasma, and finally released to form a wrinkled surface structure, as illustrated in Fig. 10g and i. Xiaoliang Cheng et al. [176] utilized this technique to fabricate nanoscale wrinkle patterns on PDMS surfaces through fluorocarbon

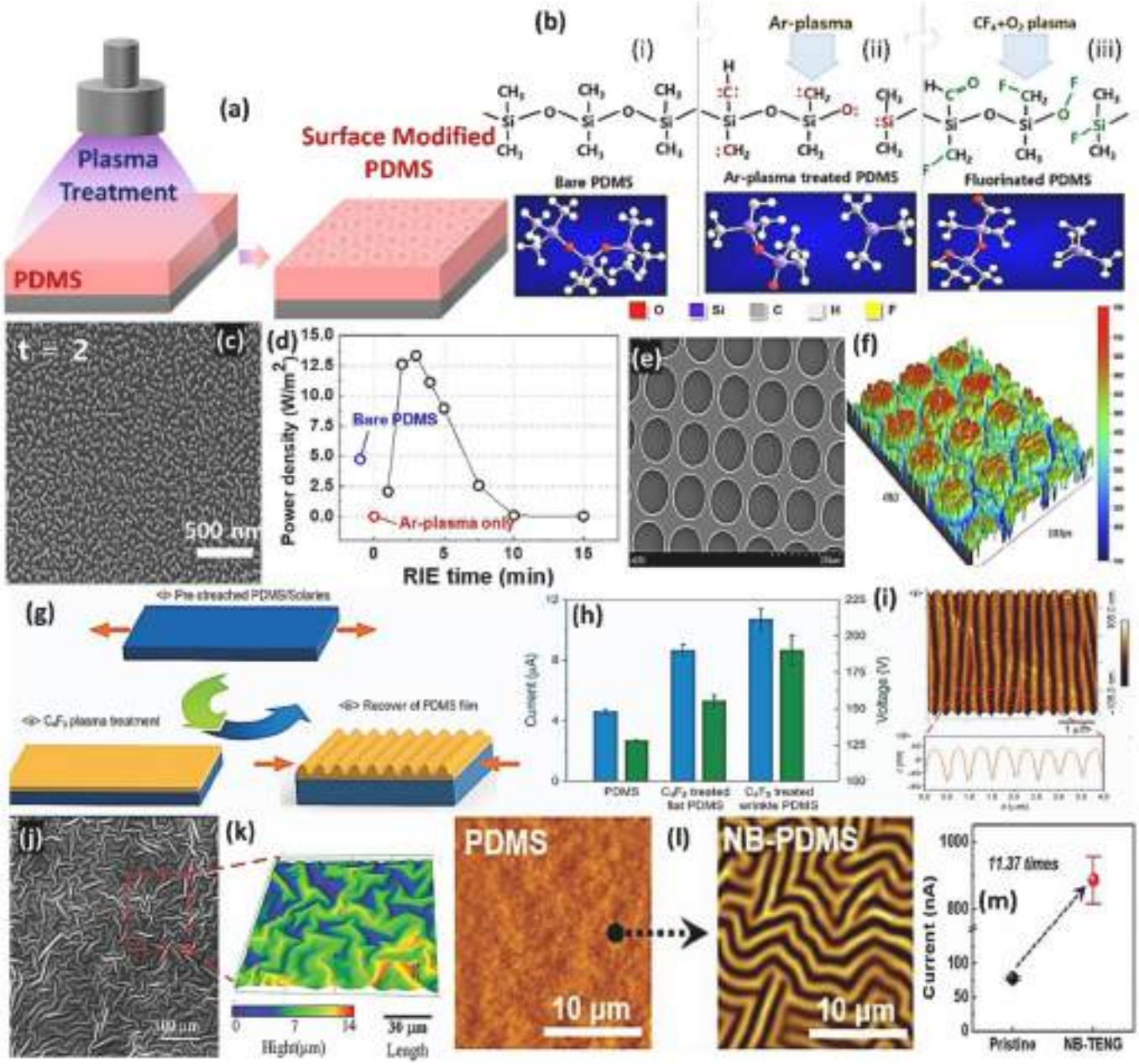


Fig. 10. PDMS surface modification via plasma treatment techniques. (a) Schematic illustration of the plasma treatment process for surface modification. (b) Proposed mechanism depicting chemical changes on the PDMS surface after plasma exposure. (c) SEM image of the plasma-treated PDMS surface. (d) Power density versus RIE treatment time plot showing the correlation between plasma exposure and surface modification. Reproduced with permission from Ref. [172]. (e) SEM image of the Si mold with micro-pillar features, and (f) corresponding AFM image of the plasma-treated PDMS micro-pillar surface. Reproduced with permission from Ref. [175]. (g) Schematic representation of the steps involved in forming wrinkled PDMS surfaces. (h) Graph showing the enhanced current and voltage output of wrinkled and plasma-treated PDMS-based TENGs. (i) AFM image displaying the wrinkled PDMS surface topography. Reproduced with permission from Ref. [174]. (j, k) SEM and AFM images of the C_4F_8 plasma-treated wrinkled PDMS surface. Reproduced with permission from Ref. [177]. (l) AFM images comparing flat and wrinkled PDMS surfaces obtained through neutral beam treatment, and (m) corresponding output graph showing an elevenfold enhancement in electrical performance. Reproduced with permission from Ref. [178].

plasma treatment. The treated PDMS films exhibited significantly enhanced electrical output, as shown in Fig. 10h. During the C_4F_8 plasma treatment, a thin fluorocarbon polymer layer is deposited onto the stretched PDMS surface. This stiff coating induces mechanical stress due to the difference in elastic properties between the deposited layer and the underlying PDMS. When the pre-stretched film is released, the elastic recovery of PDMS leads to the spontaneous formation of wrinkle topographies across the surface. A similar study by Xiaoliang Cheng et al. [177] also demonstrated the fabrication of wrinkled PDMS surfaces, as shown in Fig. 10 (j, k). Building on this concept, Wook Kim et al. [178] introduced a neutral beam (NB) treatment technique to create wrinkle patterns on PDMS surfaces. In their work, the performance of the NB method was compared with that of conventional plasma treatment. While traditional plasma treatments often introduce surface defects and degrade material properties, the NB approach effectively minimizes such damage. This is achieved by neutralizing charged particles as they pass through an aperture plate and collide with its sidewalls before reaching the PDMS surface. The researchers employed both oxygen (O_2) and nitrogen (N_2) gases to study the influence of ambient gas on the triboelectric performance. The resulting wrinkle structures,

shown in Fig. 10l, exhibited significantly improved electrical outputs. After NB treatment, the charge density, voltage, and current (Fig. 10m) of the PDMS-based TENG increased by approximately 5-, 6-, and 11-fold, respectively.

To provide a clearer comparison of surface engineering approaches beyond individual studies, the key strategies are summarized in Table 2 in terms of structure, performance range, advantages, and limitations.

Among the various approaches, microstructured surfaces (e.g., pyramids and pillars) offer the most practical balance between performance enhancement and fabrication simplicity, making them the preferred choice for pressure sensing and wearable applications. Hierarchical and bio-inspired structures deliver the highest electrical output through enhanced charge trapping and multi-scale roughness, but their fabrication complexity limits scalability. In contrast, surface functionalization methods provide a scalable route to performance improvement without altering bulk properties, though long-term stability remains a concern. Overall, microstructuring is the most application-ready strategy, while hierarchical designs maximize performance at the cost of complexity.

3.2. Composite materials strategies

PDMS-based composites incorporating functional additives have emerged as an effective and practical approach to improve dielectric properties, offering advantages such as durability, compositional flexibility, and ease of implementation. This strategy typically involves the use of (i) high-dielectric-constant (high-k) fillers, (ii) conductive fillers, and (iii) hybrid filler systems within polymer matrices (Fig. 11a). Such compositional tuning enhances dielectric constant, charge storage capability, mechanical robustness, and output stability during prolonged operation [179,180].

Dielectric-based TENGs have attracted considerable attention due to their tunable electrical properties, mechanical flexibility, and compatibility with conventional fabrication techniques [182–187]. Importantly, rational dielectric design provides a cost-effective means to improve device performance without complex structural modifications, making these systems promising for scalable energy harvesting and self-powered sensing applications [188–191].

3.2.1. Fundamental mechanisms of performance enhancement

The enhancement in triboelectric performance of composite materials can be understood through several fundamental mechanisms involving dielectric enhancement, interfacial polarization, charge trapping, and conductive network effects.

(i) *High-k dielectric enhancement:*

The incorporation of high-dielectric-constant (high-k) fillers such as barium titanate ($BaTiO_3$) increases the effective dielectric constant of the composite. A higher dielectric constant enables the material to store a greater amount of electrostatic charge during the contact–separation process, thereby increasing the surface charge density generated through triboelectrification [192–194]. From a device perspective, this enhanced charge storage leads to a higher induced potential difference between electrodes, resulting in improved V_{oc} and output power. Additionally, permittivity mismatch between the filler and polymer matrix creates localized electric field enhancement, which facilitates efficient charge separation and suppresses recombination, thereby improving both charge generation and retention.

(ii) *Interfacial polarization and charge trapping (MWS effect):*

Another key mechanism arises from interfacial polarization, also known as Maxwell–Wagner–Sillars (MWS) polarization, which occurs in heterogeneous systems composed of materials with different dielectric constants and electrical conductivities. Under an applied electric field, charge carriers accumulate at these interfaces due to restricted transport across phase boundaries, forming space charge regions (figure 11b). These accumulated charges act as trapped charges, significantly enhancing charge retention and reducing recombination losses during repeated contact–separation cycles [191,195–199]. As a result, MWS polarization contributes not only to increased instantaneous output but also to improved operational stability and durability. This effect is particularly pronounced in nanocomposites and multilayer structures with strong contrasts in dielectric permittivity (ε_r) and electrical conductivity (σ). Increasing the interfacial area through nanoscale dispersion or structural design further amplifies this mechanism.

(iii) *Conductive filler-induced micro-capacitive effect:*

Conductive fillers, such as CNTs, graphene, and metal nanoparticles (NPs), introduce an additional mechanism when dispersed below the percolation threshold. In this regime, conductive fillers remain electrically isolated and form discrete conductive domains within the insulating polymer matrix. These domains act as localized micro-capacitors, where charges accumulate at the filler–matrix interfaces, increasing the effective capacitance of the composite (figure 11c). This micro-capacitive network enhances electrostatic induction and facilitates efficient charge storage and redistribution, leading to improved dielectric response and output performance [200,201].

(iv) Percolation threshold and trade-off:

As the filler concentration increases toward the percolation threshold, the distance between conductive fillers decreases, eventually leading to the formation of continuous conductive pathways. Beyond this threshold, the composite transitions from an insulating to a conductive regime, allowing charges to flow freely through the material. This results in charge leakage and reduced charge retention, leading to a decline in output performance. For example, in multiwalled carbon nanotube (MWCNT)–PDMS composites, a percolation threshold of ~4

Table 2
Summary of surface engineering strategies in PDMS based TENGs.

Surface strategy	Typical structures	Fabrication methods	Output range (Voc/Isc)	Key advantage	Key limitation	Best application	Ref.
Micro-structured	Pyramid, pillar, dome	Lithography, soft molding	~18–440 V / ~0.7–10 μA	Increased contact area, high sensitivity	Transparency loss, tip wear	Pressure sensors, tactile sensing	[140,164,166]
Hierarchical (Micro + Nano)	Nano-on-micro (lotus, dual-scale)	Nanoimprint, replication	~140–220 V / ~2–10 μA	Highest charge density (multi-scale roughness)	Complex fabrication	E-skin, harsh environments	[167,169,171]
Bio-inspired Surfaces	Lotus leaf, rose petal	Natural templating	~140–200 V / ~10–20 μA	Low-cost, superhydrophobic	Reproducibility issues	Environmental energy harvesting	[169]
Advanced Architectures	Film-covered pillars, suspended films	Multi-step molding	~200–220 V / ~10 μA	Ultra-high sensitivity at low pressure	Fabrication complexity	Pulse, vibration sensing	[168]
Functionalized Surfaces	Plasma-treated, chemically modified, nano-wrinkled	O_2/CF_4 plasma, ICP, neutral beam, fluorocarbon treatment	~130–230 V / ~1–11 μA	Enhanced charge density vis surface tuning	Aging effects; requires specialized equipment	General performance enhancement	[172,174–178]

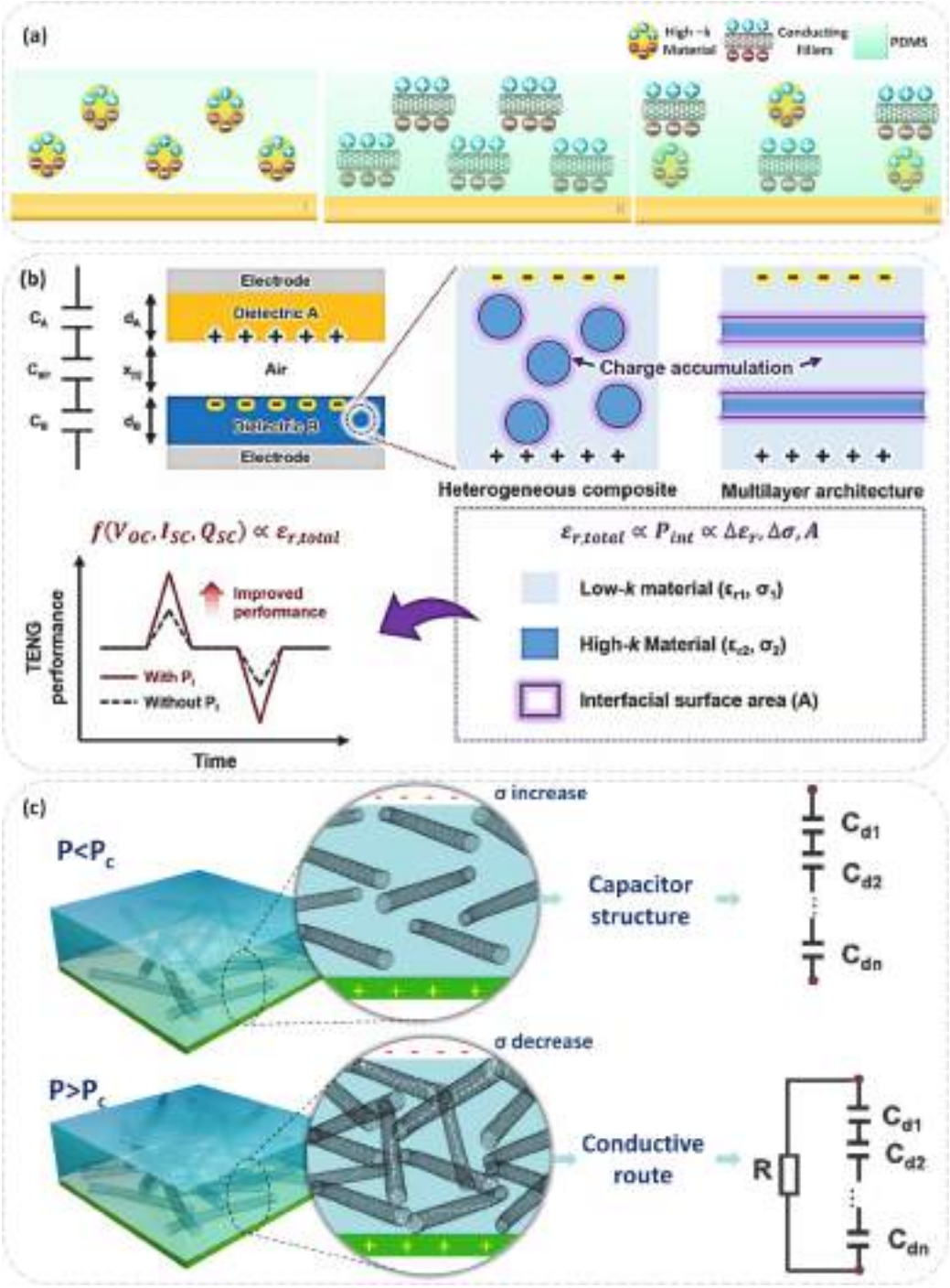


Fig. 11. (a) Schematic diagrams of three PDMS-based composite design strategies: (i) addition of high-k materials, (ii) inclusion of conductive fillers, and (iii) hybrid composites combining both high-k materials and conductive fillers. (b) Schematic illustration of the interfacial polarization mechanism at heterogeneous interfaces between materials with differing dielectric permittivity and electrical conductivity, known as the Maxwell–Wagner–Sillars (MWS) effect. This mechanism enhances surface charge density and contributes to improved stability in triboelectric devices. Reproduced with permission from Ref. [181]. (c) Schematic of conductive fillers in a PDMS matrix illustrating the percolation threshold, capacitive behavior below percolation, conductive pathways above percolation, and the corresponding output and equivalent circuits.

wt% has been reported, beyond which the I_{sc} voltage decreases [202]. Therefore, maintaining filler concentration below the percolation threshold is critical to maximizing micro-capacitive effects while minimizing charge leakage.

(iv) *Coupled enhancement mechanisms in hybrid systems:*

Hybrid composites that combine high-k dielectric fillers and conductive fillers can exploit synergistic effects by integrating multiple enhancement mechanisms within a single composite. High-k fillers improve dielectric polarization and charge storage, while conductive fillers contribute to micro-capacitive behavior and enhanced charge redistribution. The coexistence of multiple interfaces with strong contrasts in ε_r and σ significantly amplifies interfacial (MWS) polarization. Additionally, conductive fillers can act as charge transfer bridges between high-k domains, facilitating efficient charge transport while

maintaining localized charge trapping. This multi-scale interaction leads to improved charge generation, enhanced separation efficiency, and superior charge retention. As a result, hybrid composites often exhibit higher output power, improved stability, and better long-term durability compared to single-filler systems.

Overall, these mechanisms demonstrate that the performance of composite triboelectric materials is governed by a delicate balance among dielectric enhancement, interfacial polarization, charge trapping, and electrical conductivity. Accordingly, precise control over filler type, concentration, dispersion, and interfacial architecture is essential for designing high-performance triboelectric nanogenerators. These fundamental principles provide the basis for understanding the

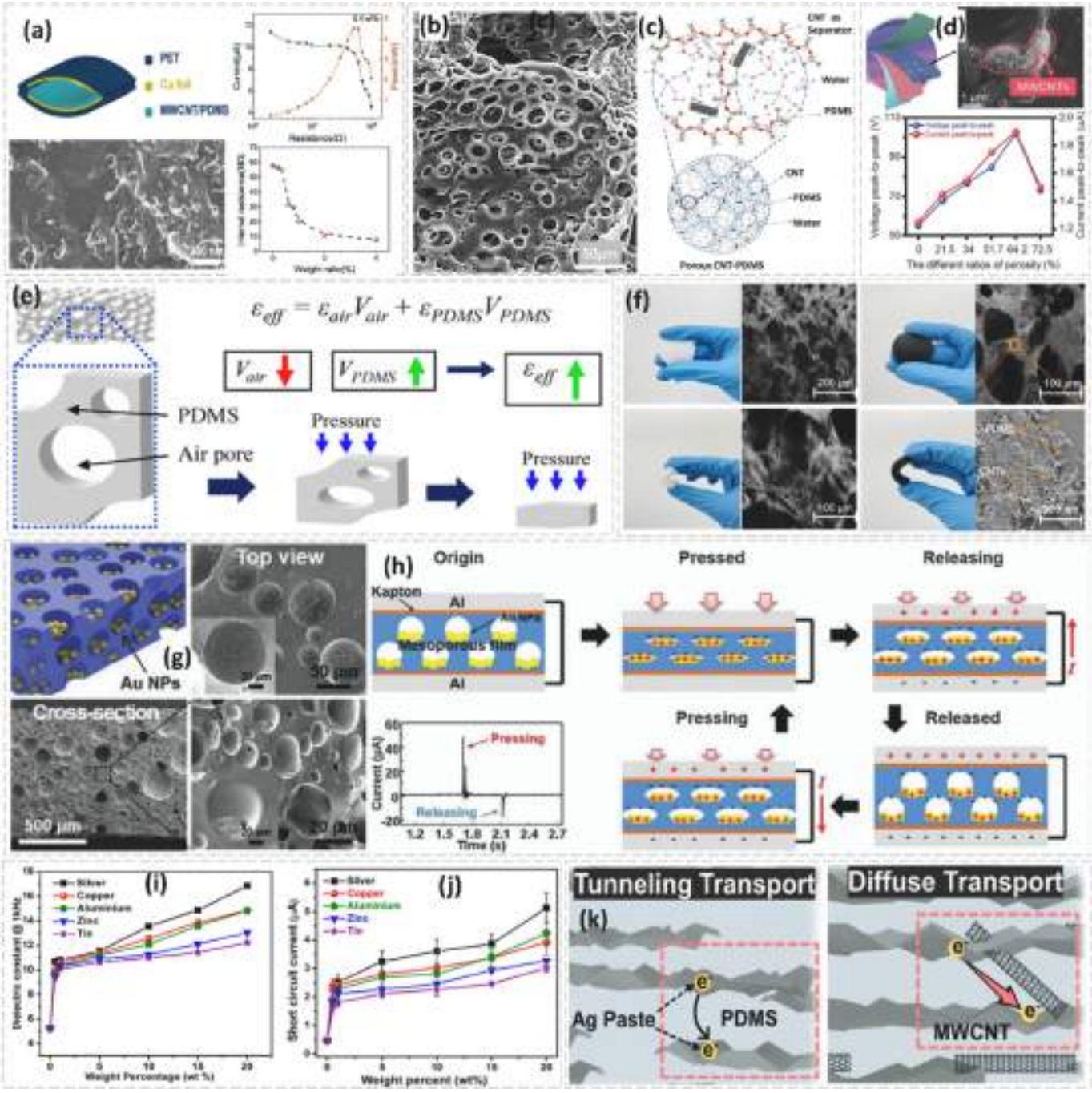


Fig. 12. (a) Schematic diagram of the MWCNT/PDMS TENG device and its respective SEM image and output performance. Reproduced with permission from Ref. [201]. (b) SEM image of the cross-section of the porous CNT-PDMS composite. (c) Formation of pores containing water molecules inside the CNT-PDMS network, where CNTs act as separators to create smaller pores. Reproduced with permission from Ref. [203]. Schematic of the MWCNT–PDMS TENG device, SEM image of the composite, and variation of electrical output with porosity. Reproduced with permission from Ref. [204]. (e) Structural deformation behavior of porous PDMS under compressive stress. Reproduced with permission from Ref. [142]. (f) Optical photograph of flexible porous structures based on pure PDMS and CNT-PDMS composite films, along with their corresponding SEM images. Reproduced with permission from Ref. [205]. (g) Top and cross-sectional SEM images of mesoporous PDMS, with and without Au NPs, respectively. (h) Schematic of the charge generation mechanism in the mesoporous Au NPs–PDMS TENG. Reproduced with permission from Ref. [207]. (i) Dielectric constant of metal/PDMS composite films at 1 kHz as a function of metal filler weight percentage. (j) Electrical conductivity of metal/PDMS composites at 1 kHz versus metal filler content. Reproduced with permission from Ref. [208]. (k) Schematic of electron transport behavior upon addition of MWCNTs. Reproduced with permission from Ref. [209].

performance trends observed across different material systems discussed in the following sections.

3.2.2. Conducting fillers-based composites

Composites combining conductive fillers such as CNTs, graphene, graphene oxide (GO), and metal nanoparticles (NPs) with PDMS have been extensively explored for enhancing TENG performance. As discussed earlier, below the percolation threshold, conductive fillers enhance the dielectric and capacitive properties of the polymer through interfacial polarization and micro-capacitor formation (Fig. 11c), leading to improved charge accumulation and electrostatic induction while preserving mechanical flexibility. Zhanqi Liu et al. [201] investigated the mechanisms behind the non-monotonic output behavior induced by conductive fillers, attributing the enhanced performance to an optimal balance between the capacitive effect and the leakage effect of CNTs on triboelectric charges (Fig. 12a). Beyond electrical enhancement, researchers have also explored the overall performance of such composites as influenced by porosity. Porosity is another key factor influencing device performance. Vafaiee et al. [203] reported the development of a CNT–PDMS foam designed as a self-powered humidity sensor, fabricated through a simple process that introduced controlled porosity. They observed that the inclusion of CNTs played a critical role in preserving small pore structures, as the nanotubes acted as spacers and prevented the collapse of pores by interacting with water molecules. As shown in Fig. 12b, the SEM image reveals the porous structure of the CNT–PDMS foam, while Fig. 12c presents a schematic illustration of its porous network. The functional groups (C=O & O–H) on the CNT surface can form hydrogen bonds with water molecules, effectively trapping moisture within the polymer matrix during mixing between CNT and PDMS. Consequently, a limited but stable amount of water remains confined within the CNT–PDMS network, which plays a significant role in the material's distinctive humidity-sensing properties. In another study, Changwoo Cho et al. [142] developed a self-powered pressure sensor based on a CNT/PDMS composite. The researchers fabricated a porous CNT–PDMS structure using the well-established sugar-templating method, where sugar served as the sacrificial material. Within the composite, CNTs functioned as micro-capacitive elements, enabling measurable capacitance variations when the porous PDMS was employed as the dielectric layer. The porosity of the PDMS increased proportionally with the amount of sugar used, and the pore size distribution remained relatively uniform. Upon applying external pressure, the air-filled pores within the dielectric layer collapsed, leading to an increase in the PDMS volume fraction. This transition effectively replaced the low dielectric constant of air with the higher dielectric constant of CNT-PDMS, thereby increasing the effective dielectric constant and enhancing the capacitance response of the device compared to that of non-porous, bulk PDMS (Fig. 12e). In 2023, Xie et al. [204] systematically studied the influence of MWCNT loading and composite porosity on the performance of the TENG active layer (Fig. 12d). A composite containing 10 wt% MWCNTs with 64% porosity demonstrated superior sensitivity. Similarly, Weidong Yang et al. [205] fabricated a porous CNT–PDMS sponge using the sacrificial template method (Fig. 12f) and investigated its performance as a piezoresistive carbon-based TENG. Nima Mohamadbeigi et al. [206] developed a 3D microporous PDMS–graphene oxide nanocomposite and evaluated its performance as a self-powered humidity sensor.

In addition to carbon-based conductive fillers, metal NPs have also been incorporated into bulk composites as conductive additives, often combined with various modification strategies to enhance dielectric and triboelectric performance. Jinsung Chun et al. [207] developed a mesoporous PDMS film embedded with Au NPs to enhance nanogenerator performance (Fig. 12g). The pores were formed by curing a PDMS–water mixture, after which the film was soaked in a colloidal Au suspension to fill the cavities. During device operation, the interaction between the Au NPs and the PDMS matrix generates alternating electron flow: pressing induces electrons from the top to the bottom electrode, while release drives them back, producing alternating current. The presence of Au NPs amplifies charge induction, and the output scales almost linearly with the active area, demonstrating the effectiveness of this porous nanocomposite design for high-performance TENGs (Fig. 12h). Similarly, Lazar K. Anlin et al. [208] investigated the influence of metal NPs, including silver, copper, aluminum, zinc, and tin, on the performance of PDMS based TENGs. Incorporating metal NPs into the PDMS matrix was aimed at enhancing the dielectric constant of the triboelectric layer (Fig. 12i), which in turn led to a linear increase in I_{sc} with increasing filler content (Fig. 12j). Among the metals tested, silver NPs exhibited the most significant improvement in device performance. In another study, Chenhao Cong et al. [209] introduced a novel approach for fabricating dispenser-printed electrodes using Ag/MWCNT/PDMS composite inks. The electron transport mechanism in the presence of MWCNTs is illustrated schematically in Fig. 12k. The gradient bridge architecture formed by the MWCNT/Ag polymer network endows the electrode with a high specific surface area, enhanced surface roughness, and mechanical elasticity, which collectively influence the contact resistance under mechanical deformation. To strengthen the conductive pathways, creating a dense network of conductive bridges between the metallic and carbon fillers is an effective strategy to maintain continuous electrical connectivity. In this composite, the MWCNTs serve as bridges between Ag flakes, facilitating electron migration and improving the overall conductivity of the electrode.

A comparison in Table 3 reveals the performance of various conductive fillers, revealing that dielectric amplification and micro-capacitive charge networks play a key role in enhancing TENG output. Among them, CNT/MWCNT-based composites consistently deliver the highest electrical performance due to their high aspect ratio and efficient conductive pathways at low percolation thresholds. In contrast, metal nanoparticles provide moderate output but are advantageous for their simple fabrication, cost-effectiveness, and tunable dielectric properties, albeit requiring higher loading and facing oxidation-related limitations. Graphene oxide and its hybrid systems achieve balanced performance at very low concentrations, offering improved flexibility and surface charge density, making them suitable for wearable applications. Overall, CNT-based fillers are most effective for maximizing output performance, while metal nanoparticles favour scalability, and GO-based systems are optimal for low-loading flexible devices.

3.2.3. High-dielectric (high-k) material composites

High-k ceramics possess a significantly higher relative permittivity (ε_r) compared with PDMS. When incorporated into the PDMS matrix via mixing, casting, and curing [209] (Fig. 13a), these particles become strongly polarized under the local electric field generated during contact electrification or mechanical deformation. Each polarized particle develops a dipole moment, and the collective dipole alignment enhances the effective dielectric response of the composite. In 2021, Doldet Tantraviat et al. [143] reported a $BaTiO_3$/porous PDMS composite film for TENG applications that delivered an output voltage of approximately 2500 V and a short-circuit current of 150 μA. The dielectric constant of the composite increased linearly with increasing $BaTiO_3$ loading, while the electrical output declined at higher loadings (over 7 wt%), as shown in Fig. 13b. Although a higher volume fraction of high-k fillers results in an increased effective dielectric constant, excessive filler loading can deteriorate TENG performance due to particle agglomeration, reduced PDMS elasticity, and increased composite stiffness. Therefore, identifying an optimal filler concentration is crucial to balance dielectric enhancement and mechanical compliance, thereby maximizing overall device performance. In addition, both particle size and filler fraction play critical roles in composite design [143,214]. A similar trend, where the incorporation of high k materials enhances the dielectric properties of the film, was observed by Jingjing Jiao et al. [215] for a $ZnO@ZnSnO_3/ZnSnO_3$/PDMS composite film (Fig. 13e).

Poling is another effective strategy to improve performance. Yun et al. [216] conducted a study on the poling effect in $BaTiO_3$/PDMS

Table 3
Comparative summary of conductive fillers in PDMS-Based TENGs.

Filler type	Structure	Optimal concentration	Output performance	Key advantage	Limitation	Ref.
MWCNTs	Porous PDMS	~0.5 wt%	V_{oc} ~ 131 V, I_{sc} ~ 0.62 μA	High dielectric (~21.6), dual-mode sensing	Percolation risk	[142]
MWCNTs	Dense PDMS	~0.4 wt%	V_{oc} ~ 720 V, I_{sc} ~ 18.28 μA	Very high output	Reduced flexibility	[201]
MWCNTs	Bulk composite	up to 6 wt%	—	Tunable Dielectric properties (FEA supported)	Agglomeration at high loading	[202]
CNTs	Foam (porous)	~4 mg/ml	V_{oc} ~ 27 V, I_{sc} ~ 1.6–2.9 μA	Humidity sensing, fast response	Lower output	[203]
MWCNTs	Spongy	—	V_{oc} ~ 18 V, I= ~0.45 μA	Wide pressure sensing	Low power output	[204]
CNTs	Hierarchical porous	—	Sensitivity ~ 2.59 V/kPa	Dual-mode sensing	Moderate output	[205]
MWCNTs	Micropillar structured	~1 wt%	V_{oc} ~ 155 V, I_{sc} ~ 10.5 μA	Structure + dielectric synergy	Mechanical instability (collapse risk)	[210]
CNTs	High-loading dielectric	up to 10 wt%	>10 × sensitivity improvement	Ultrawide sensing range	Dielectric loss	[211]
Metal NPs (Ag, Al, Cu, Zn, Sn)	PDMS composite	~1 wt% *(output)* up to 20 wt% *(dielectric)*	V_{oc}: ~28–33.6 V Isc: up to ~ 5.1 μA	Simple fabrication, tunable dielectric	Lower output than CNTs, oxidation issues	[208]
GO	Porous PDMS + plasma	~0.5 wt%	V_{oc} ~ 43–140 V, I_{sc} ~ 0.45–2.57 μA	High surface charge density	Requires surface treatment	[206,212]
GO + PANI	Solid PDMS	~0.5 wt%	V_{oc} ~ 24 V, I_{sc} ~ 0.67 μA	Low loading, good flexibility	Sensitive to composition ratio	[213]
GO + PPy	Solid PDMS	—	Moderate	Improved interfacial polarization	Lower performance	[213]

composites for TENG applications. They applied a simple electric field switching method to control the orientation and alignment of the composite. The results revealed that poling facilitated the alignment of $BaTiO_3$ particles along one direction, forming nanopillar like structures that significantly enhanced the electrical output from 1.45 V (random dispersion) to 14.5 V (oriented and aligned particles). Fig. 13c shows the schematic illustration of this orientation process and the corresponding SEM image of the aligned NPs. Similarly, Sugato Hajra et al. [217] investigated the poling effect in $Bi_4Ti_3O_{12}$ (BiTO)/PDMS composite films. They observed that the P E hysteresis loop of the poled composite (≈0.015) was notably higher than that of pure PDMS (Pr ≈ 0.002), confirming that poling effectively enhanced internal polarization by introducing a high dielectric (colossal) material into the PDMS matrix (Fig. 13d). In another study, Guoguan Suo et al. [218] examined the dual effects (piezoelectric and triboelectric) of $BaTiO_3$/PDMS composite films. They found that the V_{oc} and I_{sc} increased under positive (forward) poling, while both decreased under reverse poling, compared to the unpoled sample (Fig. 13f). Furthermore, they demonstrated that the hybrid (piezo-tribo) output exhibited a synergistic enhancement in overall electrical performance. In hybrid piezoelectric–triboelectric systems, the enhancement in electrical output originates from the coupling between contact electrification and piezoelectric polarization. During mechanical deformation, the triboelectric effect generates surface charges due to contact and separation between two materials. Simultaneously, the embedded piezoelectric fillers (e.g., $BaTiO_3$, ZnO) undergo mechanical stress–induced polarization, producing an internal piezoelectric potential. This piezoelectric potential interacts with the triboelectric field, acting as an additional driving force that promotes more efficient charge separation and suppresses charge recombination. As a result, more electrons are transferred and retained on the triboelectric surfaces. Therefore, the combined action of triboelectric charge generation and piezoelectric-induced internal electric field leads to a synergistic enhancement, where the overall electrical output exceeds the sum of the individual contributions.

The morphology of high-k fillers also plays a key role. Paydari et al. [219] also explored the influence of morphology induced variations on TENG performance. They synthesized ZnO materials with various morphologies such as rod, ellipsoidal, and pyramid shaped structures and embedded them into PDMS matrices. Among these, the pyramid shaped ZnO PDMS composite exhibited the highest open circuit voltage (Voc), while the ellipsoidal ZnO PDMS composite showed the highest short circuit current (Isc), as presented in Fig. 13g. In another investigation, Xianming He et al. [220] utilized ZnO spheres composed of aggregated NPs as templates to fabricate a nest like porous PDMS film for TENG applications. Fig. 13h presents the cross-sectional schematic diagram of the fabricated device and the corresponding porous PDMS structure. Beyond $BaTiO_3$ and ZnO, other high-k ceramics such as $NiFe_2O_4$ and $FeNbO_4$ have been incorporated into PDMS composites for TENG applications [221,222], synthesized via solid-state reactions [224] (Fig. 13i and j).

Poling in high-k composites aligns internal dipoles, producing a net polarization that increases surface-bound charges at the PDMS/ceramic interface, enhances the effective dielectric constant (ε_{eff}), and strengthens the interfacial electric field (Fig. 13k). The result is higher charge storage and improved electron transfer during triboelectrification, leading to increased V_{oc} and I_{sc}. The polarity of the applied electric field determines the orientation of dipoles, allowing tuning of the resulting polarization and TENG performance [223].

Insights from Table 4 indicate that high-k dielectric fillers significantly improve TENG performance by increasing dielectric constant, interfacial polarization, and charge storage capability. Among these, $BaTiO_3$-based systems are the most versatile and widely studied, achieving exceptionally high output through strategies such as porosity engineering, dipole alignment, and electrical poling. However, their performance is highly sensitive to filler loading, with excessive concentrations leading to reduced flexibility and diminishing returns. Complex oxides (e.g., $Bi_4Ti_3O_{12}$, $FeNbO_4$) provide stable and relatively high output at moderate loadings due to their strong intrinsic polarization. Hybrid dielectric systems further enhance performance by combining multiple polarization mechanisms. Overall, $BaTiO_3$-based fillers are the most effective for achieving high voltage output, while complex oxides offer stable performance at moderate loading, and structural engineering plays a critical role alongside material selection.

3.2.4. *Hybrid composite strategies (conducting fillers + high-k materials)*

As discussed earlier, researchers have also investigated hybrid composites that combine both high dielectric (high k) materials and conductive fillers within the triboelectric polymer matrix. Kangpyo Lee et al. [225] studied a $BaTiO_3$/MWCNT/PDMS bulk hybrid composite, where Fig. 14a illustrates the schematic diagram of the preparation process, and Fig. 14b shows the SEM image of the composite film. The incorporation of both $BaTiO_3$ (BTO) and MWCNTs significantly enhanced the dielectric properties of the composite compared to the individual MWCNT PDMS and BTO PDMS films (Fig. 14c), and the

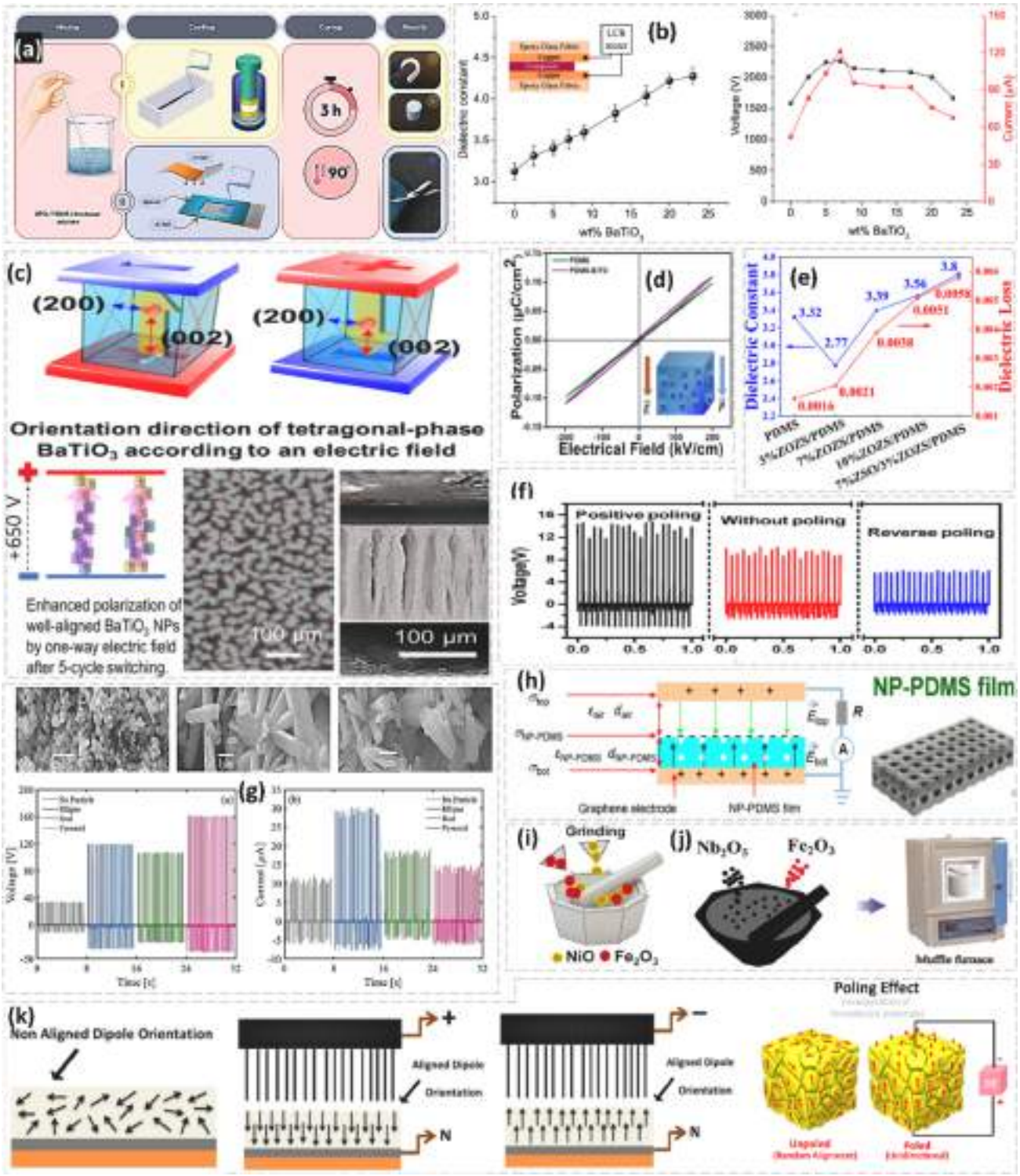


Fig. 13. (a) Schematic illustration of the fabrication process for high-k material–PDMS composite films. Reproduced with permission from Ref. [214]. (b) Dependence of the dielectric constant and electrical output of the composite films on $BaTiO_3$ loading content. Reproduced with permission from Ref. [143]. (c) Schematic and corresponding SEM image demonstrating the alignment of $BaTiO_3$ NPs under the electrode-switching direction during poling. Reproduced with permission from Ref. [216]. (d) Polarization–electric field (P–E) hysteresis loops of pristine PDMS and BiTO-filled PDMS composites measured at room temperature. Reproduced with permission from Ref. [217]. (e) Variation trends of dielectric constant and dielectric loss with different high-k filler concentrations at 10^3 Hz. Reproduced with permission from Ref. [215]. (f) Influence of poling on the I_{sc} of $BaTiO_3$/PDMS composite-based TENGs. Reproduced with permission from Ref. [218]. (g) Effect of morphological modulation on the electrical output performance of TENG devices. Reproduced with permission from Ref. [219]. (h) Schematic of a ZnO-assisted nanoporous PDMS structure. Reproduced with permission from Ref. [220]. (i, j) Schematics of various high-k fillers incorporated into PDMS composites synthesized via solid-state reaction routes. Reproduced with permission from Refs. [221,222]. (k) Conceptual diagram showing the post-poling treatment and its influence on composite polarization behavior. Reproduced with permission from Ref. [223].

Table 4
High-k filler-based PDMS composites for TENG applications.

Filler type	Structure/feature	Optimal concentration	Output performance	Key advantage	Limitation	Ref.
$BaTiO_3$ (BT)	Porous PDMS	~7 wt%	V_{oc} ~ 2500 V, I_{sc} ~ 150 μA, Power ~ 1.2 W·m^{-2}	Extremely high output (porosity + dielectric effect)	Moderate dielectric (~4.8) despite high output	[143]
$BaTiO_3$ (BTO)	Particle size effect (80–700 nm)	23 vol%	V_{oc} ~ 15 V, J_{sc} ~ 0.15 μA·cm^{-2}	High dielectric (~18); size-dependent tuning	Very high loading required; low output	[214]
$BaTiO_3$ (BTO)	Aligned nanoparticles	~15 wt%	V_{oc} ~ 22.1 V, I_{sc} ~ 1.26 μA	2.6 × improvement via dipole alignment	Complex fabrication (electric field alignment)	[216]
$Bi_4Ti_3O_{12}$ (BITO)	Hybrid piezo-triboelectric	~15 wt%	V_{oc} ~ 230 V, I_{sc} ~ 11.2 μA, Power ~ 2.1 W·m^{-2}	Very high dielectric (~315); strong polarization	Material complexity	[217]
ZnO@$ZnSnO_3$ (ZSO)	Core–shell hybrid nanocubes	~10 wt%	V_{oc} ~ 445 V, I_{sc} ~ 13.2 μA	Synergistic semiconductor + ferroelectric effect	Moderate dielectric (~7.3)	[215]
$BaTiO_3$ (BTO)	Hybrid piezo/triboelectric	~20 wt%	V_{oc} ~ 110 V, I_{sc} ~ 9 μA, Power ~ 6.13 W·m^{-2}	Combined piezo + tribo enhances output	Higher filler loading	[218]
ZnO	Morphology-controlled (pyramid/rod/ellipse)	~10 wt%	V_{oc} ~ 15.2 V, I_{sc} ~ 0.95 μA	Shape-dependent performance tuning	Relatively low output	[219]
$NiFe_2O_4$ (NFO)	Porous composite	~10 wt%	V_{oc} ~ 115 V, I_{sc} ~ 11.5 μA, Power ~ 2.66 W·m^{-2}	Magnetic + dielectric collective effect	Moderate output	[221]
BT-based ceramic	Temperature-stable filler	~15 wt%	V_{oc} ~ 215 V, I_{sc} ~ 11.5 μA, Power ~ 2.4 W·m^{-2}	Excellent thermal stability (up to 170°C)	Slightly lower dielectric (~5)	[222]
$FeNbO_4$ (FNO)	High-k oxide composite	~10 wt%	V_{oc} ~ 200 V, I_{sc} ~ 11.2 μA, Power ~ 2.4 W·m^{-2}	Eco-friendly; high polarization	Moderate dielectric, dispersion	[223]
$BaTiO_3$ (BTO)	Post-poled composite	~10 wt%	V_{oc} ~ 135 V, I_{sc} ~ 11.5 μA, Power ~ 2.6 W·m^{-2}	3.5 × enhancement via dipole alignment	Requires high-voltage poling	[224]

hybrid exhibited excellent stability over 2000 cycles (Fig. 14d). In addition, plasma treatment was applied to modify the surface properties, further enhancing the electrical performance of the device. Similarly, Saichon Sriphan et al. [226] fabricated a BTO/PDMS composite film incorporating $Ti_{0.8}O_2$ nanosheets and Ag nanopowder as co-fillers (Fig. 14e), and the resulting electrical performance showed that the hybrid configuration significantly enhanced outpit (Fig. 14f). Shan Feng et al. [227] incorporated BTO NPs of different sizes (70 nm and 500 nm) into MWCNT/PDMS composite films, with the BT-70 – MWCNT/PDMS film achieving the highest peak current and surface charge density (Fig. 14g). Further developments include PDMS composites incorporating barium titanate (BTO) and graphene quantum dots (GQDs), as demonstrated by Hatta et al. [228]. The optimized films exhibited wrinkled surfaces (Fig. 14h&i) indicative of enhanced interfacial interactions. GQDs promoted uniform nanoscale dispersion and an interconnected dielectric network among BTO, GQDs, and PDMS, improving dielectric performance and electrical output (Fig. 14j). Natthawadi Buatip et al. [229] (2024) fabricated a bulk hybrid composite film consisting of $(Ba_{0.85}Ca_{0.15})(Ti_{0.09}Zr_{0.10})O_3$ (BCZT) ceramic particles, MWCNTs, and PDMS, where the 50 wt% BCZT exhibited the highest density and cross-linking density, yielding a piezo-triboelectric nanogenerator (P-TENG) that maintained stable output over ~ 6900 cycles. Kim et al. [230] incorporated ZnO nanoflowers with MWCNTs into PDMS (Fig. 14k). The resulting hybrid composite exhibited superior electrical performance, outperforming the individual piezoelectric and triboelectric components in terms of overall output efficiency. Kavarthapu et al. [231] employed barium sodium niobium oxide ($Ba_2NaNb_5O_{15}$) (BNNO) microparticles with MWCNTs in a PDMS polymer matrix. The resulting bulk hybrid composite film was integrated into a prototype self-powered smart inhaler, designed for health monitoring and alert applications. Kangpyo Lee et al. [232] demonstrated laser-driven growth of $ZnSnO_3$ (ZTO) on surface-modified CNTs for a bulk hybrid composite (Fig. 14l), where enhanced dielectric constant, press-induced polarization, and increased frictional area contributed to superior triboelectric output. Table 5 summarizes the hybrid composite fillers used in PDMS-based TENGs.

Table 5 demonstrates that combining conductive and high-k materials enable simultaneous enhancement of charge generation and transport through synergistic effects. In these systems, high-k components increase dielectric polarization and charge density, while conductive fillers form micro-capacitive networks that facilitate efficient charge transfer. As a result, hybrid composites generally outperform single-filler systems in both voltage and current output. However, their performance is highly dependent on precise control of filler ratios, as excessive conductive loading can induce leakage currents, while complex multi-filler compositions increase fabrication difficulty. Overall, hybrid systems represent the most effective strategy for maximizing overall TENG performance, provided that filler composition and dispersion are carefully optimized.

3.3. *Hybrid engineering strategies (surface engineering + composite)*

A third structural approach in PDMS-based TENGs involves integrating surface modification with composite formation, often referred to as structure-based hybrid composites. This strategy effectively enhances both electrical output and operational durability, leveraging the synergistic interplay between surface morphology and embedded functional fillers.

In a study by Junsu Seong et al. [144], a $BaTiO_3$/PDMS micropyramidal structure was fabricated, as shown in Fig. 15a. When the two triboelectric layers are pressed together, the micro pyramidal PDMS deforms, generating oppositely charged surfaces, positive on the Al layer and negative on the PDMS layer. Simultaneously, the embedded $BaTiO_3$ NPs experience compressive stress, inducing additional piezoelectric charges due to their pseudocubic crystal structure. Upon poling, the Ti^{4+} cations shift off center, enhancing overall polarization. The synergistic interaction between triboelectric and piezoelectric effects thereby increases the total surface charge density, resulting in an enhanced electrical output, as illustrated in Fig. 15b. Tao Zhang et al. [145] (2024) developed a wearable TENG-based sensor for monitoring human pulse and blood pressure. In this work, a CNT/PDMS microcolumn structure composite was introduced (Fig. 15c and d) to optimize the device architecture. The composite demonstrated superior sensing performance compared to flat structures, exhibiting enhanced sensitivity and signal response, as illustrated in Fig. 15e. Chaeeun Lee et al. [233] developed a highly flexible triboelectric nanogenerator (TENG) employing patterned MWCNTs/PDMS/AgNWs composite electrodes, as depicted in Fig. 15f and g. Similarly, a patterned PDMS–TiO_2 composite [234] was utilized to design a TENG-based humidity sensor, where the surface morphology of the hybrid composite is shown in Fig. 15h. The charge transfer behavior of the device is illustrated in Fig. 15i, where the areas under the current peaks during the pressing and releasing cycles correspond to the amount of charge exchanged between the two electrodes. By integrating the current peaks over time, the total surface charge was determined. As

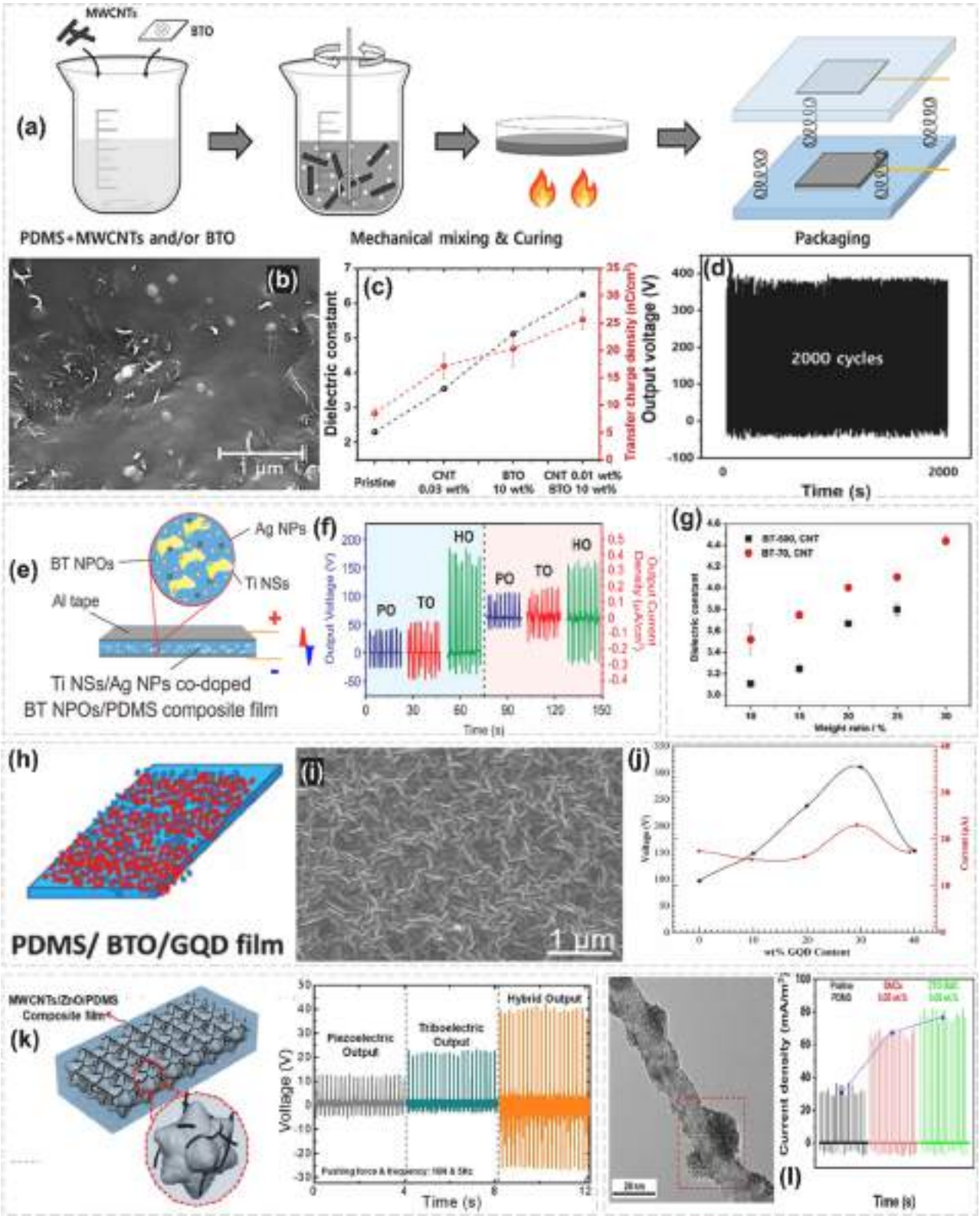


Fig. 14. (a) Schematic illustration of the fabrication process for $BaTiO_3$/MWCNT/PDMS bulk hybrid composites. (b) SEM image of the resulting hybrid composite. (c) Dielectric constant and transferred charge density of the composites prepared with different additives. (d) Long-term stability performance of the optimized composite. Reproduced with permission from Ref. [225]. (e) Structural diagram of the Ti NSs/Ag NPs–BT NPOs/PDMS multilayer film and (f) corresponding extracted piezoelectric, triboelectric, and hybrid electrical outputs. Reproduced with permission from Ref. [226]. (g) Comparison of dielectric properties between BT-500/PDMS and BT-70/PDMS films with varying BT loadings. Reproduced with permission from Ref. [227]. (h, i) Schematic and SEM image of the PDMS/BTO/GQD hybrid film. (j) Output of the device showing maximum performance at 30 wt% filler. Reproduced with permission from Ref. [228]. (k) Schematic depiction of the ZnO nanoflower/MWCNT/PDMS composite and its enhanced electrical output. Reproduced with permission from Ref. [230]. (l) SEM image showing the laser-induced formation of the $ZnSnO_3$/CNT heterostructure and the corresponding improvement in current density for the bulk hybrid composite. Reproduced with permission from Ref. [232].

Table 5
Hybrid composite fillers (Conducting + High-k) in PDMS-based TENGs.

Filler system	Structure/feature	Optimal concentration	Output performance	Key advantage	Limitation	Ref.
$BaTiO_3$ + MWCNTs	Binary hybrid + RF plasma-treated surface	10 wt% BTO + 0.01 wt% MWCNTs	V_{oc} ~ 394.75 V (421.06 V with plasma), I_{sc} ~ 28.24 μA (32.33 μA with plasma), Power ~ 4.76 $W{\cdot}m^{-2}$	Dielectric- conductive synergy; enhanced surface charge	Limited durability	[225]
$BaTiO_3$ + $Ti_{0\cdot8}O_2$ + Ag	Triple-filler micro-capacitor system	20 wt% BTO + 0.7 wt % TO + 0.1 wt% Ag	V_{oc} ~ 180 V, J_{sc} ~ 0.42 $\mu A{\cdot}cm^{-2}$, Power ~ 0.23 $W{\cdot}m^{-2}$	Multi-interface polarization; enhanced dielectric via micro-capacitors	Complex composition; lower power density	[226]
$BaTiO_3$ + MWCNTs	Size-dependent BT + conductive network	10 wt% BT + 0.1 wt% MWCNTs	V_{oc} ~ 115 V, I_{sc} ~ 4.5 μA	CNTs act as bridge-linkers improving polarization	Moderate output	[227]
$BaTiO_3$ + GQDs	Conductive quantum dot hybrid	10 wt% BTO + 30 wt % GQDs	V_{oc} ~ 11.4 V, I_{sc} ~ 0.31 μA, Power ~ 0.25 $W{\cdot}m^{-2}$	Nano-interface dielectric enhancement	Very high GQD loading; low output	[228]
BCZT + MWCNTs	Lead-free piezoelectric + conductive network	50 wt% BCZT + 3 wt % MWCNTs	V_{oc} ~ 39.7 V, I_{sc} ~ 1.9 μA, Power ~ 157.7 μW	Strong piezoelectric contribution; eco-friendly	Very high filler loading	[229]
ZnO (nanoflower) + MWCNTs	Morphology-enhanced hybrid	~4.8 wt% ZnO + trace MWCNTs	V_{oc} ~ 75–400 V, I_{sc} ~ 3.2–30 μA	Nanoflower structure increases surface area + piezo effect	Output depends on external force	[230]
BNNO + MWCNTs	Ferroelectric + conductive hybrid	10 wt% BNNO + 0.1 wt% MWCNTs	V_{oc} ~ 185 V, I_{sc} ~ 10.5 μA, Power ~ 1.81 $W{\cdot}m^{-2}$	Improved polarization + charge transport	Material complexity	[231]
$ZnSnO_3$/CNT	Heterostructure (PLAL synthesized)	10 wt% composite	V_{oc} ~ 215 V, I_{sc} ~ 11.5 μA, Power ~ 2.4 $W{\cdot}m^{-2}$	Ohmic-like interface reduces resistance; strong charge trapping	Complex fabrication (laser-based)	[232]

evident from the results, the patterned PDMS–TiO_2 nanocomposite TENG exhibited the highest surface charge density, confirming its superior charge generation capability. In a related study, Xiao Meng et al. [235] fabricated $SrTiO_3$/PDMS composite films with nanopillar and nanocone surface morphologies using the anodic aluminum oxide (AAO) template method, as illustrated in Fig. 15j & k. The TENG incorporating nanocone-structured composites exhibited markedly superior electrical performance compared to the nanopillar-based design. Moreover, the enhancement in the aspect ratio of the nanostructures further promoted the triboelectric output. Increasing the $SrTiO_3$ content within the PDMS matrix also contributed significantly to improving the device's electrical characteristics.

Table 6 summarizes composite–structure hybrid strategies, highlighting that integration of material modification with surface engineering provides a combined route to enhance both electrical properties and contact mechanics. In these systems, fillers improve dielectric and polarization properties, while micro/nano-structured surfaces increase the effective contact area and stress concentration, leading to enhanced charge transfer efficiency. Compared to single-strategy approaches, these hybrid designs enable simultaneous optimization of sensitivity, output performance, and multifunctionality. However, the requirement for additional fabrication steps such as templating or lithography increases complexity and limits scalability. Overall, composite–structure hybridization is highly effective for high-sensitivity sensing applications, particularly where both mechanical and electrical optimization are required.

3.4. Interface engineering

Another effective strategy explored by researchers is interface engineering within the PDMS matrix. In general, interface engineering can be classified into two approaches: (i) layer-by-layer structures, which are realized by depositing functional fillers onto a substrate followed by casting a PDMS layer, and (ii) functional encapsulation, in which the fillers are confined between polymer layers. In encapsulated configurations, the surrounding PDMS layers actively regulate interfacial charge polarity and electric field distribution. In the work of Gang Jian et al. [236], TiO_2 particles were deposited on stainless steel textiles, followed by PDMS casting to form a hybrid PDMS–TiO_2–SS textile TENG (Fig. 16a and b). Fig. 16c shows confocal images of the device surface, revealing an uneven morphology that enhances permittivity (Fig. 16d) and promotes interfacial charge generation. Similarly, Wen He et al. [237] synthesized ZnO nanoflakes (NFs) on an Au-coated PET film, subsequently coated with PDMS (Fig. 16e and f). This hybrid nanogenerator exhibited both a high piezoelectric current output, attributed to the enhanced surface piezoelectricity of ZnO NFs, and a strong triboelectric voltage arising from pronounced triboelectrification at the Au–PDMS interface. The device achieved a peak-to-peak voltage of ~470 V, a current density of ~60 $\mu A{\cdot}cm^{-2}$, and an average power density of ~28.2 $mW{\cdot}cm^{-2}$ (Fig. 16g). In 2021, Raguel Barras et al. [146] fabricated a hybrid TENG by growing ZnO nanorods (NRs) on carbon fiber (CF) yarns, followed by PDMS coating and curing using a novel method that involved passing an electrical current through the CF yarns. Fig. 16i depicts the schematic of the CF/ZnO NRs/PDMS composite, while Fig. 16j shows the SEM image of the ZnO NRs grown on CF surfaces. The device operates via a coupled piezoelectric–triboelectric mechanism: upon applying pressure, the ZnO NRs deform, generating piezoelectric charges and inducing current across the CF electrode, while triboelectric charges form at the PDMS interface. Releasing the pressure reverses both the piezoelectric and triboelectric currents, producing continuous alternating outputs during deformation–recovery cycles (Fig. 16h). A comparable approach was followed for CNT/Carbon Nanofiber-deposited thermoplastic polyurethane (TPU) fibers prepared via centrifugal spinning, which were then coated with PDMS (Fig. 16k) [238]. Following the same strategy, Yonteng Qian et al. [239] fabricated 3D bilayer graphene/carbon cloth–PDMS composites in an encapsulated configuration (Fig. 16l), further demonstrating the versatility of the encapsulating method in enhancing mechanical robustness, charge generation, and overall device performance.

Embedding isolated filler layers within the PDMS matrix represents an important encapsulation strategy explored by several researchers. In a related study, Chaiyanut Jirayupat et al. [240] developed a piezo-induced triboelectric nanogenerator (PI-TENG) by growing ZnO nanowires (NWs) on an Au/PDMS substrate, followed by PDMS encapsulation (Fig. 17 a & b). The positive piezopotential generated by the ZnO NWs induces an external electric field from the PDMS surface toward the metal electrode, enhancing electron transfer and reinforcing the internal built-in potential (Fig. 17c). The synergistic coupling between ZnO's piezoelectric polarization and the triboelectric behavior of PDMS significantly improves charge separation and electrical output performance. Wonjea Shin et al. [147] fabricated a single-crystalline SnS-based piezo-assisted TENG (S-TENG) by depositing a SnS epilayer on a CNT membrane via vacuum filtration, transferring it onto a PDMS film, and encapsulating it with PDMS (Fig. 17d). The distribution of triboelectric charges on the PDMS surface, corresponding to the CNT and SnS sides, was analyzed using contact potential difference (CPD)

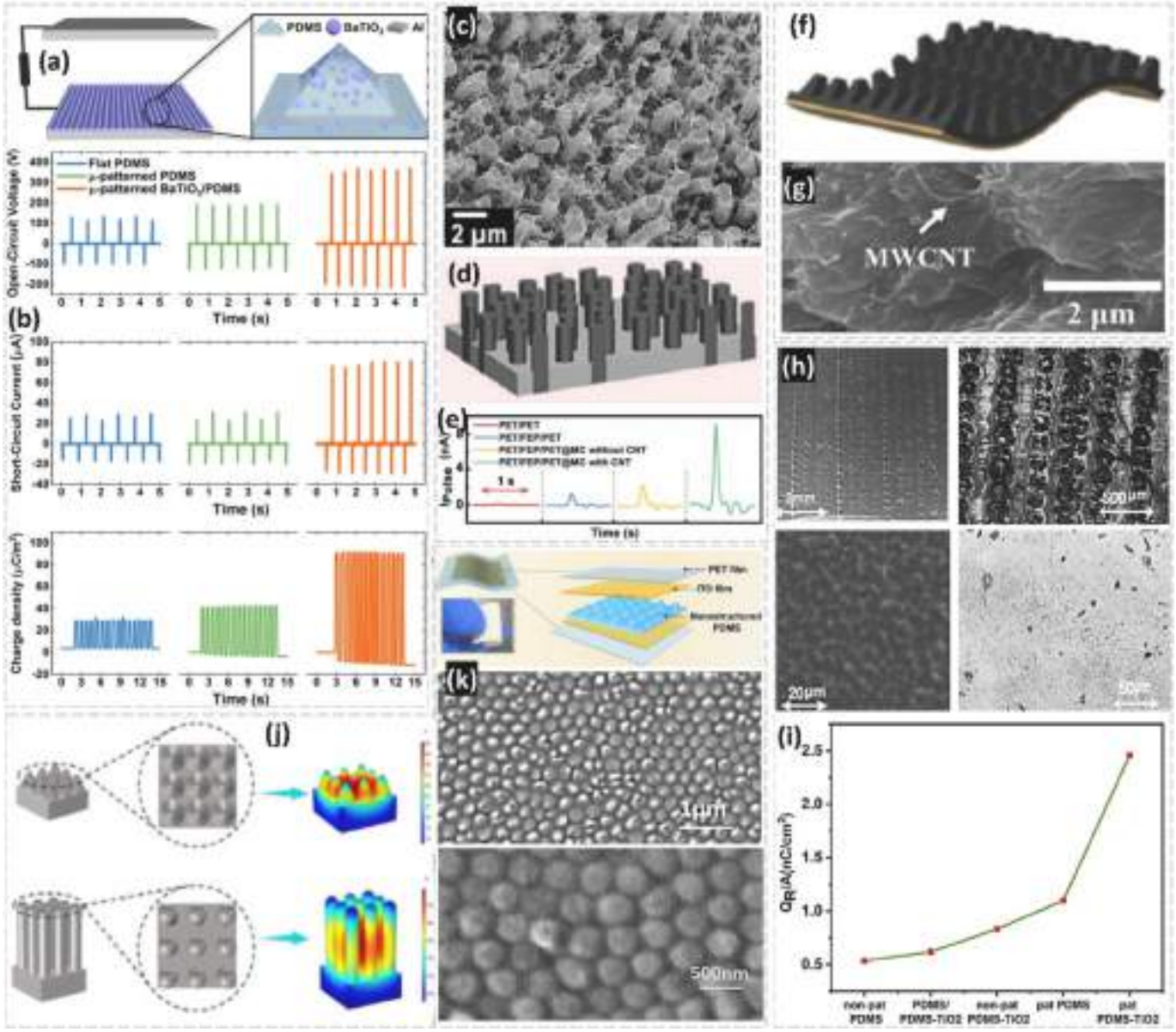


Fig. 15. (a) Schematic illustration of the $BaTiO_3$ nanoparticle-embedded pyramidal TENG structure. (b) Comparison of the V_{OC}, I_{SC}, and charge density for flat PDMS, micro-pyramidal PDMS, and micro-pyramidal $BaTiO_3$/PDMS TENGs. Reproduced with permission from Ref. [144]. (c, d) SEM image and corresponding schematic of the CNT/PDMS micropillar substrate. (e) Pulse-wave signal comparison demonstrating the enhanced sensing capability of the CNT/PDMS micropillar structure. Reproduced with permission from Ref. [145]. (f, g) Schematic and SEM image of the patterned MWCNTs/PDMS/AgNWs composite. Reproduced with permission from Ref. [233]. (h) SEM image of pure and patterned PDMS–TiO_2 nanocomposites. (i) Enhanced charge-transfer behavior of the patterned PDMS–TiO_2 structure. Reproduced with permission from Ref. [234]. (j) Simulation models and potential distributions of various nanostructured TENG configurations. (k) Structural diagram and SEM image of patterned $SrTiO_3$ (STO)/PDMS composite films. Reproduced with permission from Ref. [235].

Table 6
Hybrid engineering strategies (composite + surface engineering) in PDMS-based TENG.

Filler system	Surface structure/ feature	Optimal concentration	Output performance	Key advantage	Limitation	Ref.
$BaTiO_3$ (BT)	Micropyramidal structure	~10 wt%	V_{oc} ~ 110 V, I_{sc} ~ 10 μA, Sensitivity ~ 3.71 V·kPa^{-1}	Tribo-piezo synergy; enhanced contact area	Requires microfabrication	[144]
CNTs	Microcolumn structure	~5 wt%	V_{oc} ~ 11.5 V, I_{sc} ~ 0.4 μA, Sensitivity ~ 0.31 V·kPa^{-1}	High sensitivity for low-pressure sensing (biomedical)	Low electrical output	[145]
MWCNTs + AgNWs	Electrospun nanofiber friction layer	5 wt% MWCNTs + 1 wt% AgNWs	V_{oc} ~ 112 V, I_{sc} ~ 2.3 μA, Power ~ 0.13 W·m^{-2}	Highly flexible electrode; improved conductivity and dielectric	Complex multi-layer fabrication	[233]
TiO_2	Microstructured surface	~10 wt%	V_{oc} ~ 210 V, I_{sc} ~ 11 μA	Enhanced dielectric + surface roughness; multifunctional (humidity sensing)	Moderate fabrication complexity	[234]
$SrTiO_3$ (STO)	Nanocone / nanopillar structure	~9 wt%	V_{oc} ~ 358.8 V, I_{sc} ~ 11.1 μA	High aspect ratio nanostructures improve charge density	Template-based fabrication required	[235]

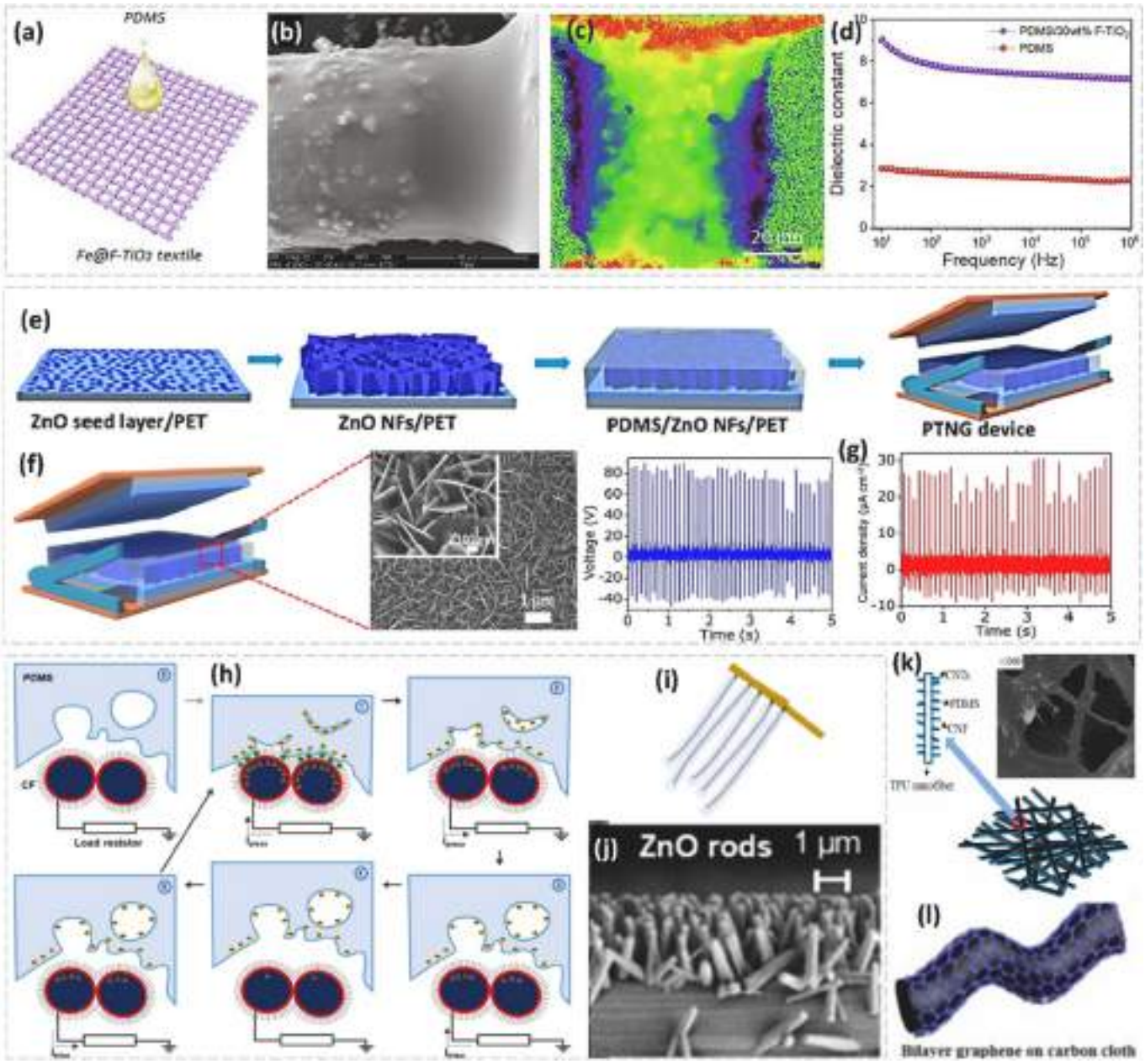


Fig. 16. (a) Schematic of the electrophoretic deposition of Fe@F-TiO_2/PDMS on SS textile, and (b) corresponding SEM image showing the layered composite. (c) Confocal image displaying the uneven deposited surface. (d) Frequency-dependent dielectric constant of pure PDMS and PDMS/30 wt% F-TiO_2. Reproduced with permission from Ref. [236]. (e, f) Schematic and SEM image of ZnO nanoflakes/PDMS, with (g) peak-to-peak VOC and JSC. Reproduced with permission from Ref. [237]. (h) Mechanism of triboelectric energy generation through friction-induced charge transfer. (i,j) Schematic and SEM image of ZnO nanorod/PDMS/CF yarn. Reproduced with permission from Ref. [146]. (k) Schematic and SEM image of CNT/CNF/PDMS/TPU nanofiber film. Reproduced with permission from Ref. [238]. (l) Schematic of the 3D-bilayer graphene/CC-PDMS encapsulated composite. Reproduced with permission from Ref. [239].

measurements via KPFM. The SnS side exhibited a surface potential of ~536 mV, 198 mV higher than that of the CNT side (Fig. 17e & f). Interface modifications using insulating or semiconducting materials act as interlayers, trapping or blocking charges to enhance performance. The SnS layer functions as a blocking layer, preventing charge neutralization and maintaining high surface charge density. Incorporating CNT or CNT–SnS increases the dielectric constant and electronegativity of the friction layer, enhancing charge induction, transfer, and triboelectric potential (Fig. 17g), thereby promoting greater electron transfer to the SnS side upon friction. The combined effects of piezoelectric polarization and electron-blocking substantially improve device performance, highlighting the critical role of material selection and interface engineering in high-performance TENGs. Similarly, Masahiro Matsunaga et al. [241] developed a flexible PDMS/CNT/PDMS film using a spray-coating method (Fig. 17h) and fabricated a large-area S-TENG integrated with chip LEDs (Fig. 17i & j). Fig. 17k and l illustrate a glove embedded with S-TENGs and LEDs connected via stretchable CNT thin-film interconnections, showing the glove in operation with LEDs powered by a hand clap. This study highlights the potential of encapsulated PDMS/CNT structures for flexible, wearable TENG devices capable of directly powering electronic components through mechanical stimuli.

As shown in Table 7, interface engineering improves TENG performance by enhancing interfacial charge transfer, electrical conductivity, and contact efficiency. Strategies such as layer-by-layer assembly, conductive interlayers, and nanostructured interfaces enable more effective charge separation and transport compared to bulk composites.

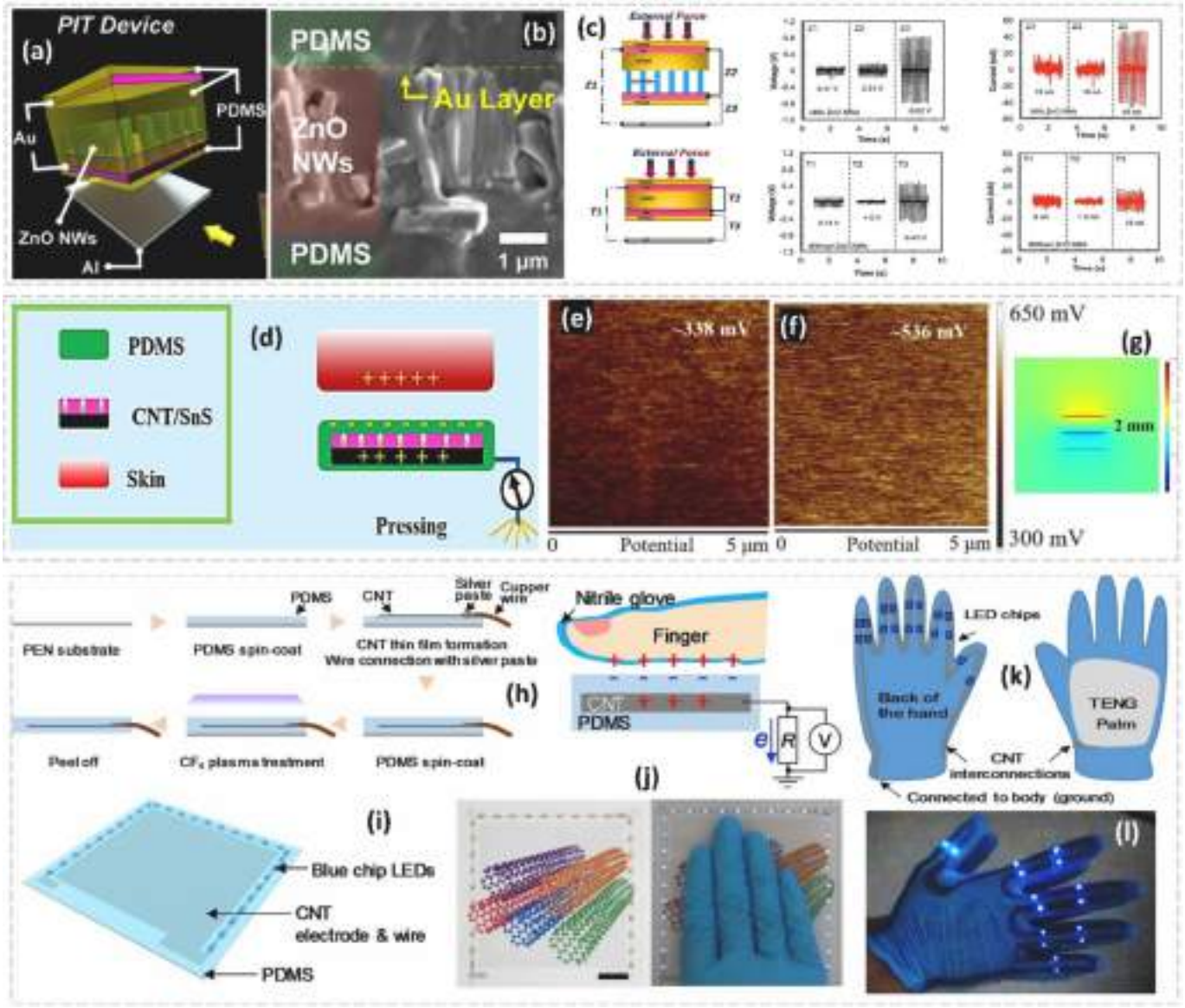


Fig. 17. (a) Schematic illustration of ZnO nanowires (NWs) encapsulated within PDMS. (b) SEM image of the encapsulated structure. (c) Electrical output peaks of the encapsulated nanogenerator. Reproduced with permission from Ref. [240]. (d) Schematic diagram of the operating principle of the CNT/SnS-encapsulated TENG. (e, f) Surface potential distributions of the PDMS on the CNT side and SnS side, respectively. (g) Theoretically calculated electrical potential distributions. Reproduced with permission from Ref. [147]. (h) Schematic of the fabrication procedure for CNT-embedded PDMS. (i, j) Schematic illustration and photograph of the CNT electrode and wire-bonded blue chip LEDs. (k, l) Schematic illustration and photograph of the TENG-integrated LED gloves. Reproduced with permission from Ref. [241].

The integration of conductive networks (e.g., CNTs, graphene) with functional nanostructures (e.g., ZnO nanowires) further introduces synergistic triboelectric–piezoelectric effects, resulting in improved output and flexibility. However, these approaches often involve complex fabrication processes and require precise control of interface uniformity. Overall, interface engineering is particularly advantageous for flexible and textile-based TENGs, where efficient charge transport and mechanical adaptability are critical.

3.5. Advanced Materials

Advanced materials, including 2D materials, ionic liquids, and liquid metals, have emerged as effective fillers in PDMS-based TENGs, enabling performance enhancement through dielectric polarization, interfacial charge modulation, and multi-physical coupling effects. These materials introduce various mechanisms beyond conventional filler strategies, offering route toward high-performance and multifunctional energy harvesting systems.

Among 2D materials, hexagonal boron nitride (h-BN) has attracted significant attention due to its high mechanical strength and excellent electrical properties. In 2025, Panicker et al. [242] developed a wearable h-BN/PDMS TENG that utilizes the synergistic coupling of triboelectric, piezoelectric, and flexoelectric effects, resulting in enhanced energy output. The uniformly dispersed h-BN–PDMS composite (Fig. 18a) exhibited an enhanced piezoelectric response under mechanical stress (Fig. 18b), leading to improved energy generation for wearable and self-powered devices. The enhanced output was attributed to strong electrostatic interactions between PDMS and electronegative nitrogen atoms in h-BN [243], intrinsic piezoelectricity and flexoelectric polarization of monolayer h-BN activated under deformation [244–252], increased surface charge density due to facilitated electron transfer, and internal contact electrification arising from nanoscale

Table 7
Interface engineering strategy in PDMS-based TENGs.

System type	Interface Strategy/ Structure	Optimal condition	Output performance	Key advantage	Limitation	Ref.
ZnO nanorods on CF yarn + porous PDMS	In-situ functionalized 1D fiber interface	Multi-yarn integration	V_{oc} ~ 72 V, I_{sc} ~ 10 μA, Power ~ 74.1 μW·cm^{-2}	Conformal coating on fibers; enhanced surface area and charge transfer	Moderate output; yarn integration complexity	[146]
SnS + CNT hybrid	Binary interface-engineered composite	0.7 wt% SnS + 0.01 wt% CNTs	V_{oc} ~ 275 V, I_{sc} ~ 11.5 μA, Power ~ 2.56 W·m^{-2}	High piezo + conductive synergy; reduced internal resistance	Requires precise filler control	[147]
Flower-like TiO_2 on textile	3D interface deposition via EPD	Tunable deposition	V_{oc} ~ 110–120 V, I_{sc} ~ 25–30 μA, Power ~ 0.85–1.05 W·m^{-2}	Uniform coating on flexible textiles; morphology-enhanced performance	Fabrication complexity (EPD process)	[236]
ZnO nanoflakes	High surface-area filler interface	~10 wt%	V_{oc} ~ 300 V, I_{sc} ~ 12 μA	Enhanced piezo + tribo synergy via nanoflake morphology	Moderate complexity in synthesis	[237]
3D bilayer graphene + carbon cloth	Layered conductive network interface	—	V_{oc} ~ 70 V, J ~ 9.3 μA·cm^{-2}, Power ~ 0.65 mW·cm^{-2}	High conductivity + large surface area; flexible electrode	CVD-based fabrication complexity	[239]
ZnO nanowires (PIT system)	Piezo-induced interface enhancement	~2 μm NWs	Power ~ 50 μW·cm^{-2} (≈100 × improvement)	External piezopotential boosts internal electric field	Requires nanowire growth + device integration	[240]
CNT thin film (sandwiched PDMS)	Layer-by-layer encapsulated conductive film	—	V_{oc} ~ 160 V, Power ~ 8 W·m^{-2}	High transparency, stretchability, and conductivity	Requires multilayer processing + plasma treatment	[241]

interactions between h-BN fillers and the PDMS matrix. Similarly, Umapathi et al. [253] reported a high-performance h-BN-based composite with a V_{oc} of 198.6 V and a power density of 7.86 W/m^2, primarily due to an increased effective dielectric constant and improved charge storage. These results highlight the potential of h-BN for multi-mode energy harvesting, although the overall output remains moderate compared to hybrid systems.

Transition metal dichalcogenides, particularly MoS_2, provide an alternative route for enhancing dielectric and triboelectric properties. Mahapatra et al. [254] reported a MoS_2–PDMS textile electrode, where uniform flake distribution and strong interfacial adhesion (Fig. 18c) contributed to improved dielectric behavior. KPFM analysis (Fig. 18d) showed that 6 wt% MoS_2 yielded the highest surface potential, indicating enhanced charge accumulation and improved output performance. Zhao et al. [260] further demonstrated that MoS_2 incorporation significantly improves durability, achieving a 146.6% increase in operational cycles and stable performance over 10,000 cycles, highlighting strong interfacial compatibility. Earlier studies explored multifunctional designs, including Ag nanowire/MoS_2 hybrid structures for stretchable sensing [261] and MoS_2-based systems driven by water flow [262], although the latter lack comprehensive durability evaluation. Overall, MoS_2-based composites provide a balance between electrical performance, mechanical robustness, and multifunctionality.

MXenes (e.g., $Ti_3C_2T_x$, where T_x = O, OH, or F) have recently gained attention as advanced conductive fillers for PDMS-based TENGs because of their outstanding electrical conductivity, high electronegativity, and superior mechanical flexibility [263]. Their incorporation enhances the capacitance and charge storage capability of the composite while lowering its internal resistance, leading to a significant improvement in output performance. He et al. [255] demonstrated a MXene/PDMS hybrid film with uniform coating (Fig. 18e, f), resulting in increased surface roughness and charge-trapping capability. Similarly, Jiang et al. [256] fabricated a stretchable MXene-based electrode (Fig. 18g) with excellent conductivity for wearable applications. The MXene–PDMS system can be described as a metal–insulator–metal-like structure, where efficient charge accumulation at internal interfaces enhances dielectric polarization (Fig. 18h). Recent studies highlight multifunctional applications, including self-powered sensing devices for heart rate and UV detection (Fig. 18i) [257], and bio-inspired systems integrating MXene-based TENGs with synaptic transistors for motion sensing, achieving high sensitivity (0.197 kPa^{-1}) [264]. Overall, MXene incorporation provides an effective strategy for enhancing charge transport, interfacial polarization, and multifunctionality in PDMS-based TENGs, although challenges related to stability and oxidation remain.

Ionic liquids (ILs) have emerged as effective functional fillers in PDMS-based TENGs, enabling improved charge dynamics, interfacial behavior, and device stability. Zhou et al. [258] reported IL/PDMS composites exhibiting morphological and compositional changes after operation (Fig. 18j), with increased fluorine and sulfur content indicating IL migration and surface polarization. The dipolar orientation of IL molecules under an electric field (Fig. 18k) enhances surface charge accumulation and output performance (Fig. 18l). In another study, Zhao et al. [265] demonstrated that IL incorporation can effectively tune the triboelectric polarity of PDMS/CNT composites, mitigating the adverse effects of high conductivity and enhancing charge generation. Similarly, Qing et al. [266] developed an IL@PDMS/rGO-based flexible single-electrode sensor array, where ILs improved charge-trapping capability and mechanical stability, enabling applications in material recognition and trajectory tracking. IL incorporation provides a versatile strategy for optimizing charge generation, polarity control, and device robustness in polymer-based TENG systems.

Zhu et al. [259] investigated liquid metal (LM)–PDMS composites, demonstrating that gravity-driven phase separation leads to the formation of interconnected conductive networks within the polymer matrix [267–269]. By comparing solid, porous, and hybrid porous/array structures (Fig. 18m), they showed that integrating a hemispherical array with porous architecture significantly enhances sensor sensitivity, particularly at low pressures. Cross-sectional analysis (Fig. 18n) revealed a hierarchical structure with distinct LM-rich conductive regions and insulating zones, as illustrated in Fig. 18o. Additionally, incorporating CNTs further improved electrical performance. These findings highlight the potential of LM-based hierarchical composites for developing flexible, high-sensitivity TENGs and pressure sensors.

Table 8 summarizes the performance of advanced material systems, highlighting the diverse mechanisms governing their behavior. MXene-based systems generally exhibit among the highest electrical outputs due to superior conductivity and charge trapping [255–257,264,270–272], whereas MoS_2-based materials offer a balance between performance and multifunctionality. h-BN systems enable multi-mode energy harvesting with moderate output, while IL-based composites provide effective control over charge dynamics and polarity [265,273]. Liquid metal systems contribute to enhanced conductivity and sensitivity but require careful design to avoid charge screening effects [274]. Overall, these advanced materials provide complementary strategies for optimizing PDMS-based TENGs, with selection dependent on the targeted balance between output performance, stability, and functional integration.

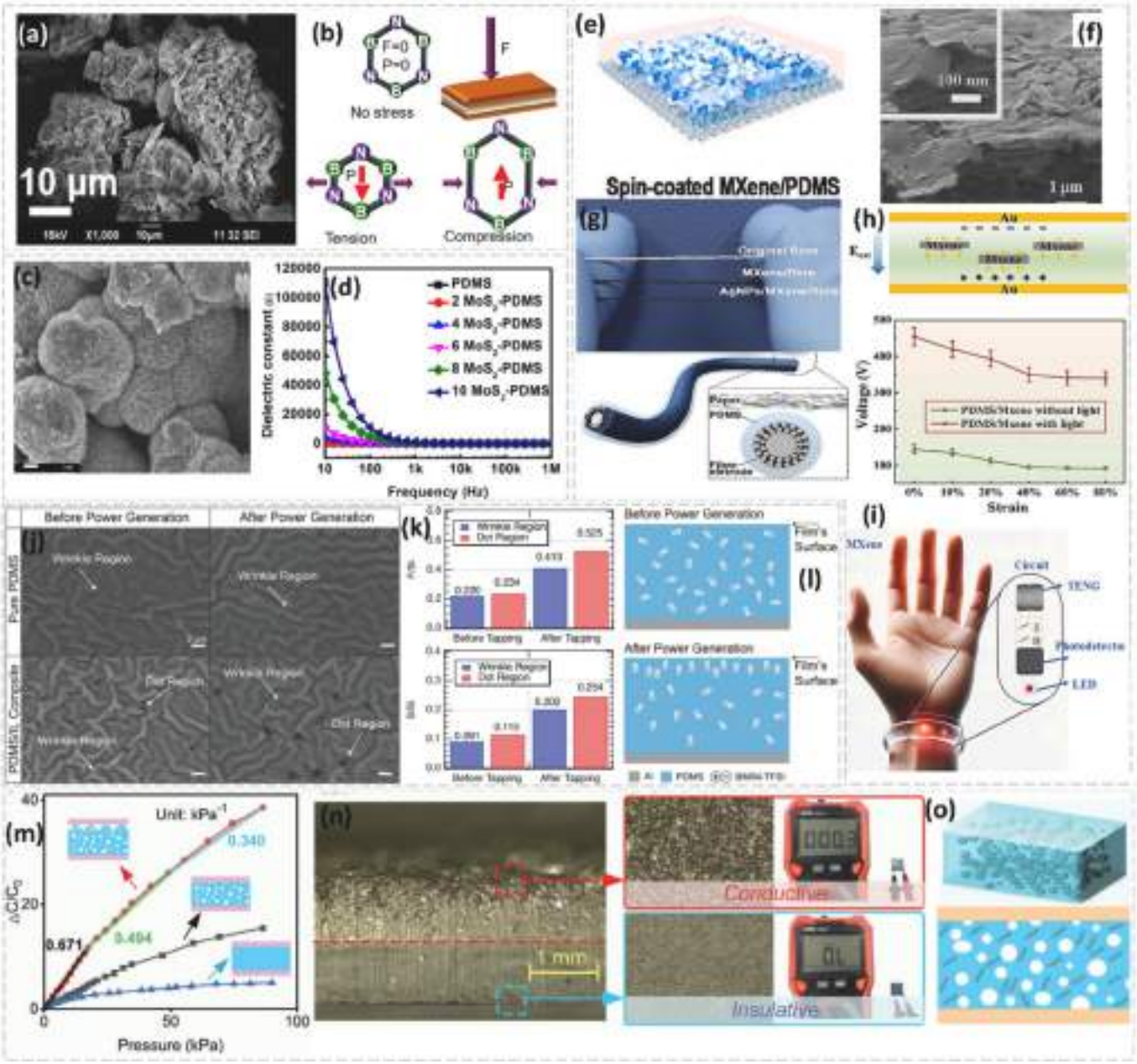


Fig. 18. (a) SEM image of the hBN–PDMS composite film. (b) Schematic diagram of the piezoelectric effect in hBN. Reproduced with permission from Ref. [242]. (c) SEM image of MoS_2 grown on textile. (d) Frequency-dependent dielectric constant variation of MoS_2–PDMS composites with different MoS_2 concentrations. Reproduced with permission from Ref. [254]. (e, f) Schematic diagram and SEM image of the spin-coated MXene/PDMS film. Reproduced with permission from Ref. [255]. (g) Photograph of the Ag NPs/MXene/PDMS/fiber composite. (h) Schematic illustration showing MXene microcapacitor behavior and the voltage output of MXene/PDMS with and without light under different strain conditions. Reproduced with permission from Ref. [256]. (i) Schematic diagram of the MXene/PDMS self-powered TENG photodetector. Reproduced with permission from Ref. [257]. (j) SEM image of the ionic liquid/PDMS surface before and after power generation. (k) Concentration of Si and Fe before and after power generation. (l) Schematic showing the alignment of ionic liquids on the surface after power generation. Reproduced with permission from Ref. [258]. (m) Pressure versus charge plot of the liquid metal structure under different configurations. (n) Image showing the dual-layer formation of the film with conductive and insulative layers due to gravity-induced segregation. (o) Schematic of the composite film (liquid metal-PDMS) with and without CNTs. Reproduced with permission from Ref. [259].

4. Applications of PDMS-Based TENGs

PDMS-based TENGs have demonstrated broad applicability in energy harvesting and self-powered sensing due to their intrinsic flexibility, tunable surface properties, and compatibility with various modification strategies. However, the suitability of a given PDMS modification strategy strongly depends on the specific requirements of the target application. Therefore, it is essential to correlate application requirements with appropriate material design approaches and underlying mechanisms.

Wearable and human-motion sensing systems require high flexibility, mechanical durability, and sensitivity to low-frequency mechanical stimuli. In this regard, surface-engineered PDMS and composite-based systems are particularly effective. For instance, Kaur et al. [202] developed a smart glove based on PDMS TENGs for motion sensing. The inherent elasticity of PDMS ensures conformal contact with human skin, while surface microstructuring enhances effective contact area, thereby increasing charge generation under small deformations.

Table 8
Advanced materials in PDMS-based TENGs.

Filler type	Optimal concentration	Output performance	Key advantage	Limitation	Ref.
hBN nanoflakes	~7.5 wt%	V_{oc} ~ 81.9 V, I_{sc} ~ 8.4 μA	Multi-mode energy harvesting	Moderate output	242
hBN / defective hBN	~ 0.75 wt%	V_{oc} ~ 198.6 V, I_{sc} ~ 13.5 μA	High power density (~7.86 W/m^2)	Processing complexity	253
MoS_2	~6 wt%	V_{oc} ~ 320 V, J_{sc} ~ 15.4 μA/cm^2	High output, textile integration	Complex mechanism	254
MoS_2 nanosheets	~3 wt%	V_{oc} ~ 206 V, I_{sc} ~ 10.5 μA	Improved wear resistance	Moderate dielectric	255
AgNWs + MoS_2 (1D–2D hybrid)	—	Power density ~ 1.16 W/m^2	High flexibility (70% strain), multifunctional sensing (GF ~ 215.4)	Moderate electrical output, fabrication complexity	256
MoS_2 nanoflowers	~5 wt%	~23 V, J ~ 13 mA/m^2	Energy + catalysis	Low voltage	257
MXene (Ti_3C_2Tx)	Variable	V_{oc} ~ 225 V, J_{sc} ~ 30 μA/cm^2	High conductivity, wearable	Stability concerns	258
MXene + AgNPs	—	V_{oc} ~ 7.7 V, I_{sc} ~ 0.2 μA	High stretchability (300%)	Low output	259
MXene	~5 wt%	V_{oc} ~ 130 V, I_{sc} ~ 1.4 μA	Good balance of performance	Moderate output	260
MXene (wrinkled)	~5 wt%	V_{oc} ~ 112 V, I_{sc} ~ 1.4 μA	High sensitivity	Limited durability cycles	264
MXene powder	Low (mg scale)	V_{oc} ~ 900 V, I_{sc} ~ 22 μA	Extremely high voltage	Uniformity issues	270
MXene	~20 wt%	V_{oc} ~ 453 V, I_{sc} ~ 131 μA	Multi-energy harvesting	Stability	271
MXene (wrinkled)	~0.4 wt%	Sensitivity ~ 0.18 V/Pa	High sensitivity (E-skin)	Low power output	272
IL-coated SWCNTs	0.1 wt% SWCNTs + IL (0.2)	Voc ~ 90 V, Isc ~ 2.5 μA	Improved dispersion and dielectric enhancement + polarity shift	Moderate current output	265
Ti_2CN + IL	Optimized IL + MXene	Isc ~ 400—500 μA	Self-power bio sensing device (bacterial)	Stability concerns	273
Liquid metal (EGaIn) network	~40 vol% LM	High conductivity (2.6×10^4 S/m)	Excellent electrical pathways and mechanical adaptability	Excess conductivity may suppress charge accumulation	274
IL + rGO hybrid composite	Optimized ILs + rGO	Voc ~ 110 V, Isc ~ 1.2 μA; Sensitivity ~ 1.14 V/kPa	High sensitivity, enhanced charge trapping and conductivity	Moderate electrical output	266

Similarly, Shan et al. [264] demonstrated a bioinspired kinesthetic system, where flexible PDMS-based TENGs enable real-time motion sensing and human–machine interaction. These behaviours are likely associated with enhanced contact electrification and mechanical compliance.

For physiological monitoring applications, such as pulse and blood pressure detection, high sensitivity and signal stability are critical. Zhang et al. [145] reported a nanopore-templated CNT/PDMS micro-column TENG capable of detecting subtle arterial signals. In this case, the incorporation of conductive CNT fillers are likely to enhance sensitivity through micro-capacitive effects below the percolation threshold, while the microstructured architecture amplifies signal output by increasing contact efficiency. This combination contributes to the enhanced detection of weak physiological signals in a self-powered manner.

Environmental sensing applications, such as humidity detection, require high surface area and strong interaction with ambient conditions. Nima et al. [206] developed a porous PDMS–graphene oxide composite TENG, where the porous structure significantly increases the effective surface area for moisture adsorption. The presence of graphene oxide enhances dielectric response and charge trapping, improving sensitivity and response speed. Here, both structural design and composite engineering contribute synergistically to performance enhancement. In multifunctional sensing systems, hybrid material strategies are highly beneficial. Ali et al. [257] demonstrated a wearable PDMS–MXene TENG integrated with a MoS_2 photodetector for simultaneous heart rate and UV monitoring. In such systems, hybrid composites enable efficient energy harvesting while supporting additional sensing functionalities. The coexistence of multiple materials enhances interfacial polarization and charge transport, enabling stable operation across different sensing modalities.

In the context of smart agriculture, Jiang et al. [256] developed TENG yarns integrated into agrotextiles for monitoring environmental parameters such as soil moisture and temperature. These applications require mechanical robustness, scalability, and environmental stability. Fiber-based and composite PDMS systems are particularly suitable due to their durability and ability to harvest energy from ambient mechanical sources such as wind or plant movement. Similarly, Kavarthapu et al. [231] reported a BNNO-MWCNTs/PDMS composite TENG integrated into an inhaler for monitoring usage patterns. In this case, conductive filler networks enhance signal generation, while maintaining flexibility for integration into medical devices. However, such systems also highlight challenges related to long-term stability and reliability in practical environments. These application-specific requirements and corresponding material strategies are summarized in Table 9.

4.1. Synthesis and design perspective

From the above analysis, it is evident that different PDMS modification strategies are suited for specific application domains. Surface engineering approaches are practically suitable for wearable and motion sensing applications due to their ability to enhance contact area and sensitivity to small mechanical inputs. In contrast, composite-based strategies, particularly those incorporating high-k or conductive fillers, are generally suitable for sensing applications requiring enhanced dielectric response and signal stability. Hybrid systems further extend the functionality of PDMS-based TENGs by combining multiple enhancement mechanisms, enabling multifunctional sensing and improved overall performance.

Despite these advances, several challenges remain. Issues such as long-term durability, environmental stability, and scalable manufacturing continue to limit the practical deployment of PDMS-based TENGs. Addressing these challenges will be critical for translating laboratory-scale demonstrations into real-world applications. In addition to application-specific systems, a significant portion of reported TENG studies focuses on demonstrating energy harvesting capability through simple electrical outputs, such as lighting light-emitting diodes (LEDs) or charging capacitors. These demonstrations are primarily used to validate device performance, power density, and energy conversion efficiency under controlled conditions. While such studies provide important benchmarks for comparing material and structural designs, they often do not reflect application-specific requirements or long-term operational stability. While such demonstrations provide useful performance benchmarks, they often do not reflect real-world operating conditions. Bridging this gap requires addressing key challenges such as durability, scalability, and system-level integration.

5. Performance landscape of modification strategies in PDMS-based TENGs

Fig. 19 provides a comprehensive performance overview of PDMS-based TENGs employing diverse material modification strategies, correlating power density with electrical output parameters, filler concentration, operating frequency, and surface charge density.

Table 9
Application-oriented design strategies for PDMS-based TENGs.

Application	Key Requirements	Strategy Used	Underlying Mechanism(s)	Advantages	Remaining Challenges
Wearable motion sensing	Flexibility, sensitivity	Surface microstructure	Increased contact area – surface charge ↑	High sensitivity	Mechanical durability
Physiological monitoring	High sensitivity, stability	CNT/PDMS composite	Micro-capacitor effect	Detect weak signals	Noise, calibration
Humidity sensing	Surface interaction	Porous PDMS–GO	Adsorption + dielectric enhancement	Fast response	Long-term stability
Multifunctional sensing	Multi-signal detection	Hybrid composites	MWS + synergy	Multi-functionality	Complexity
Smart agriculture	Durability, scalability	Fiber-based PDMS	Mechanical robustness	Large-scale use	Environmental degradation
Medical devices	Flexibility, reliability	Conductive composites	Charge enhancement	Compact integration	Stability

Collectively, these data establish a general performance hierarchy among material systems, including single-filler composites, engineering-based approaches, and hybrid strategies, thereby enabling rational selection of design methodologies for energy harvesting applications. It is important to note that while these trends illustrate relative performance, direct comparison of absolute electrical output values across studies should be approached with caution, as they are highly sensitive to experimental conditions, device configuration, and measurement protocols. Key considerations and challenges in comparing TENG performance are discussed in detail in Section 5.1.

As shown in Fig. 19a and b, hybrid composite strategies consistently define the upper performance boundary, achieving high power densities across a wide range of filler concentrations and normalized current densities. Unlike systems based solely on conductive or high-k fillers, hybrid composites maintain elevated output even at relatively high filler loadings. This behavior can be attributed to the simultaneous operation of complementary mechanisms. High-k fillers enhance dielectric polarization and charge storage, while conductive fillers introduce localized micro-capacitive networks that facilitate charge accumulation and redistribution. Importantly, the presence of conductive fillers helps homogenize local electric fields and improve charge transport, while the high-k domains suppress excessive leakage by maintaining dielectric integrity. This synergistic balance prevents the formation of fully percolated conductive pathways, enabling sustained high output without significant charge loss even at increased filler concentrations.

In contrast, conducting and high-k filler systems define a baseline performance tier, as evidenced in Fig. 19a–c. Conductive fillers can sustain moderate current densities due to enhanced charge transport; however, their power density is often limited by leakage pathways, particularly near or above the percolation threshold. High-k fillers, on the other hand, generate high I_{sc} owing to increased dielectric polarization, but their current output is often inconsistent, resulting in lower overall power density. These observations confirm that single-filler systems are intrinsically constrained without additional structural or interfacial optimization.

Engineering-driven strategies, including surface engineering, interface engineering, hybrid engineering, and advanced material design, consistently outperform single-filler systems by improving charge transfer efficiency and mechanical–electrical coupling (Fig. 19b–d). Among these, interface engineering demonstrates the broadest operational envelope, spanning multiple orders of magnitude in both current density and voltage (Fig. 19d). This versatility highlights the critical role of interfacial control in enabling stable and adaptable TENG performance under varying mechanical input conditions.

The strong diagonal trend observed in the power density versus normalized current density plots (Fig. 19a and b) indicates that current density is the primary determinant of power output. Notably, hybrid composite and hybrid engineering strategies shift this relationship upward, delivering higher power densities at comparable current levels. This suggests that performance enhancement is governed more by material architecture and interfacial design than by filler concentration alone.

The relationship between performance and filler concentration further reinforces an efficiency-driven design strategy. As shown in Fig. 19a, b, and e, high-performance hybrid composites typically operate at moderate to high filler loadings, whereas hybrid engineering and advanced material strategies achieve competitive performance at lower concentrations. At higher filler loadings (>10 wt%), many systems exhibit performance saturation or increased variability, likely due to agglomeration, dielectric loss, or interfacial degradation.

The observed spread in performance data across different strategies provides important insights into reproducibility and reliability. A relatively narrow distribution of data points indicates consistent performance across different studies, suggesting robust material design and fabrication processes. In contrast, a wider data spread reflects sensitivity to parameters such as filler dispersion, interfacial quality, and measurement conditions. For example, conductive filler systems often show higher variability due to challenges in controlling percolation behavior and uniform dispersion, whereas interface-engineered systems tend to exhibit more consistent and reproducible performance. This highlights the importance of precise material control and fabrication optimization in achieving reliable device performance.

Frequency-dependent behavior (Fig. 19f) confirms that most systems operate within the ultra-low to low-frequency regime (0.5–10 Hz), which is highly relevant for biomechanical and environmental energy harvesting. Even under these conditions, hybrid composite strategies maintain high power density, while surface and interface engineering approaches provide stable and consistent performance across varying frequencies. Furthermore, surface charge density analysis (Fig. 19g) supports these observations, confirming the decisive role of interfacial engineering and hybridization in enhancing charge generation and retention.

From a practical design perspective, Fig. 19 serves as a valuable guideline for selecting appropriate material strategies based on application requirements. For applications requiring maximum power density, hybrid composite and hybrid engineering approaches are preferable. For systems prioritizing stability and reproducibility, interface engineering offers a reliable solution. In contrast, single-filler systems may be suitable for simplified or cost-effective designs but are inherently limited in performance. Thus, the figure can be interpreted as a design map that links material strategy to expected performance, enabling informed decision-making for PDMS-based TENG development.

These findings provide a data-driven foundation for rational material and structural design. Fig. 20 further extends this analysis by presenting a decision tree for selecting appropriate modification strategies based on performance requirements and application constraints. Overall, the analysis reveals that achieving high-performance PDMS-based TENGs requires not only material selection but also careful integration of interfacial design, structural engineering, and compositional optimization.

5.1. Challenges in comparing TENG performance

Despite the rapid progress in triboelectric nanogenerators, direct comparison of performance metrics reported across different studies remains challenging due to the lack of standardized testing protocols. Key output parameters such as open-circuit voltage, short-circuit current

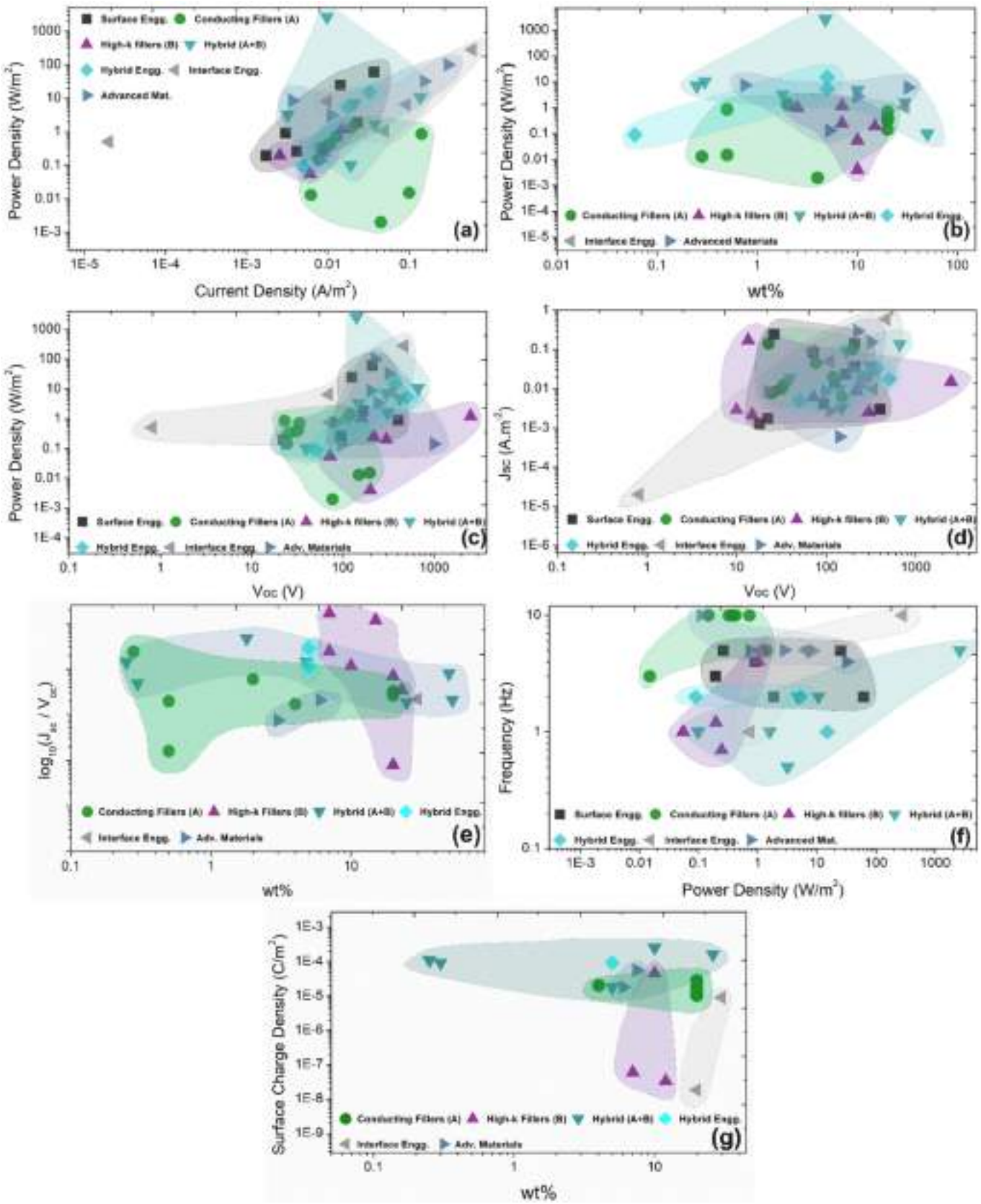


Fig. 19. Performance comparison of material modification strategies for PDMS-based TENGs: (a) Power density vs. normalized current density. (b) Power density vs. filler concentration. (c) Power density vs. open-circuit voltage. (d) Current density vs. voltage operational ranges. (e) Efficiency metric log (J_{sc}/V_{oc}) vs. filler concentration. (f) Frequency-dependent power density. (g) Surface charge density vs. filler concentration.

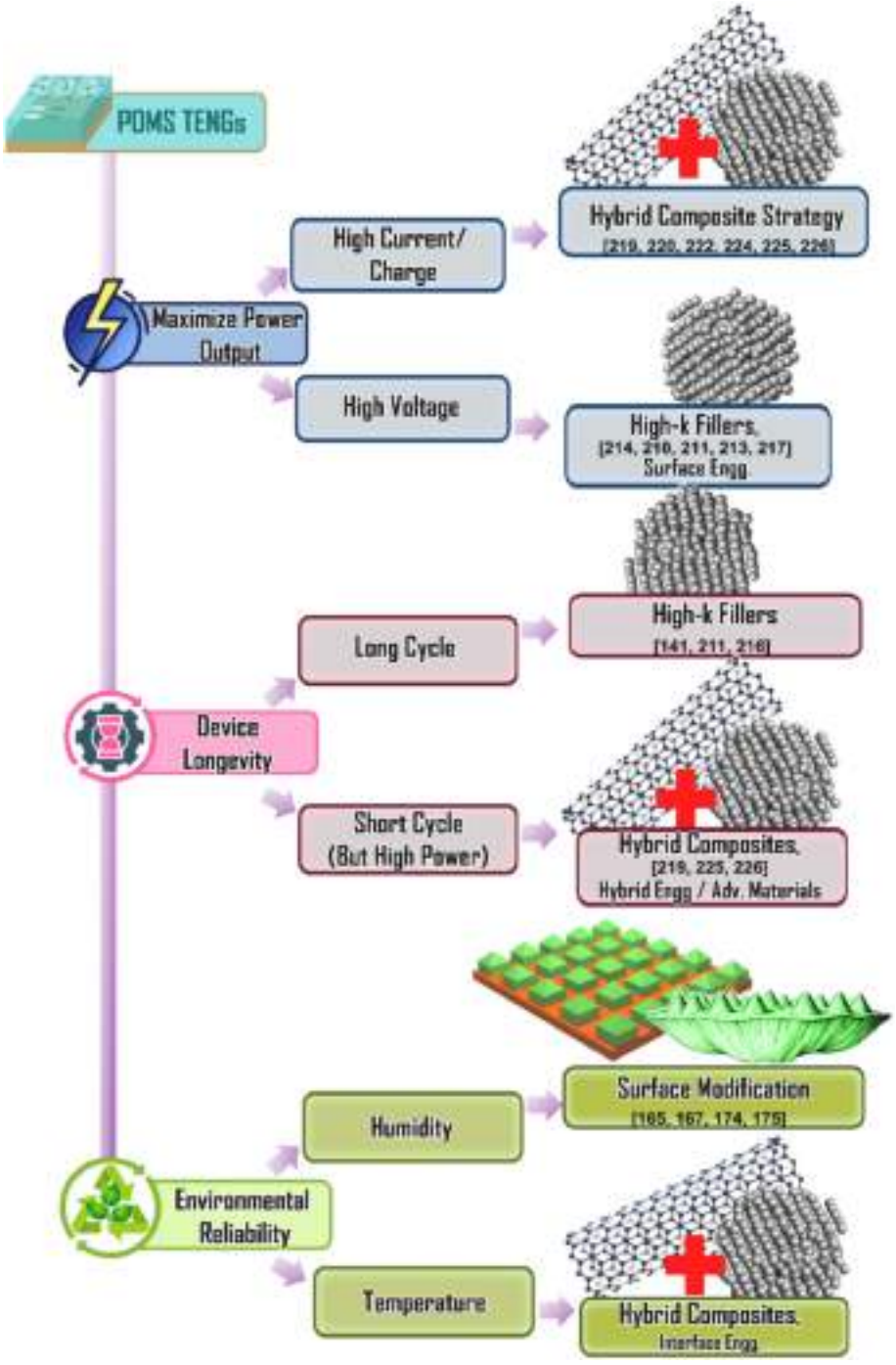


Fig. 20. Decision tree illustrating modification strategies in PDMS-based TENGs.

density, and power density are highly sensitive to experimental conditions, including applied force, contact area, operating frequency, and device configuration.

In particular, variations in mechanical input such as contact pressure and motion mode (vertical contact–separation, sliding, or single-electrode mode) can significantly alter charge generation and transfer behavior. Similarly, environmental factors such as humidity and temperature strongly influence surface charge retention, often leading to reduced performance under high humidity due to charge dissipation. Furthermore, differences in measurement methodology, including external load resistance, data acquisition systems, and normalization methods (e.g., area-normalized vs. device-level output), introduce additional variability. As a result, reported performance values can differ by orders of magnitude even for similar material systems.

These inconsistencies highlight the need for standardized testing protocols and reporting guidelines in the TENG field. Until such standards are widely adopted, caution must be exercised when comparing performance metrics across studies. Instead of relying solely on absolute values, it is more meaningful to evaluate relative improvements within the same experimental framework and to consider the underlying material design strategies and operating conditions.

6. Perspective

Despite significant progress in PDMS-based TENGs, several critical challenges remain that hinder their practical deployment. These include

(i) limited long-term mechanical and electrical durability, (ii) challenges in scalable and reproducible fabrication, and (iii) environmental instability, particularly under humid conditions. Addressing these issues requires a combination of material innovation, structural design, and system-level integration, as discussed below. Fig. 21 presents a radar roadmap comparing the current and short-term target readiness levels across five key domains: materials, surface engineering, interface physics, device architecture, and applications, based on the literature trends discussed above.

In the Materials domain, PDMS-based TENG composites remain largely empirical, with limited deterministic control over charge generation and retention. This limitation primarily arises from insufficient control over interfacial polarization and the absence of stable charge-trapping sites. To address this, recent studies have shown that incorporating ferroelectric fillers, defect-engineered dielectric materials, and deep trap states can enhance interfacial polarization and suppress charge recombination, thereby improving both output performance and long-term charge retention [268,269,275,276]. Furthermore, self-healing and covalently crosslinked PDMS elastomers have been developed to mitigate mechanical degradation by restoring damaged microstructures and maintaining electrical stability under repeated deformation [275,276].

For Surface engineering, current approaches rely heavily on small-area patterning methods such as soft lithography and plasma treatments, which are inherently batch-based and limited in throughput, making them unsuitable for large-area manufacturing. To overcome these limitations, scalable fabrication techniques such as roll-to-roll nanoimprint lithography, laser patterning, and continuous template-assisted processing have been explored. These approaches enable continuous and uniform micro/nanostructure replication over large areas, reduce fabrication cost, and are compatible with flexible substrates, thereby providing a practical pathway toward industrial-scale and reproducible surface engineering of TENG devices [277,278].

Within Interface Physics, the fundamental mechanisms of tribo-electrification remain debated, particularly regarding the relative contributions of electron transfer, ion transfer, and material polarization. This lack of clarity limits predictive design. Recent advances indicate that combining molecular-level surface characterization with first-principles calculations and data-driven models can establish quantitative relationships between surface chemistry, electronic structure, and charge transfer behavior. In particular, machine learning frameworks trained on material descriptors have shown potential for predicting triboelectric polarity and output performance, enabling a transition from empirical material selection to rational interface design [279–282].

At the device level, PDMS-based TENGs are predominantly at the proof-of-concept stage, with limited integration of power management and environmental protection. One key limitation is the mismatch between high voltage output and low current stability, which restricts direct application. This can be addressed by integrating rectifiers, capacitors, and energy storage units to stabilize and store the generated power. In addition, flexible encapsulation layers and multilayer packaging strategies have been demonstrated to effectively reduce environmental degradation, particularly moisture-induced charge dissipation. Scalable fabrication methods such as printing-based electronics and textile-integrated architectures further enable the development of large-area, mechanically robust, and self-sustaining energy systems [283,284].

From an application perspective, PDMS-based TENGs face challenges including humidity sensitivity, mechanical wear, and performance instability under real operating conditions. Humidity-induced charge leakage, for example, occurs due to the adsorption of water molecules that facilitate surface charge dissipation. This issue can be mitigated through hydrophobic surface modification, surface fluorination, and encapsulation techniques that limit moisture interaction. Similarly, mechanical degradation of microstructures can be reduced by using elastic and self-healing composites, while output instability can be improved through composite dielectric engineering that stabilizes charge density. These strategies collectively enhance durability, environmental resistance, and operational reliability in wearable systems, biomedical devices, and industrial sensing applications [100,102,285,286].

Overall, while increasing material and structural complexity generally leads to improved electrical performance, it also introduces challenges related to scalability, reproducibility, and long-term durability. Therefore, future development should focus on balancing performance enhancement with manufacturability and system integration. Even though PDMS has been widely used in TENGs since the early stages of triboelectric nanogenerator research (approximately 2012 to the present), it continues to attract significant scientific attention [51]. This is primarily due to its intrinsic advantages, including eco-friendliness, excellent mechanical flexibility, strong negative tribopolarity, compatibility with micro- and nano-patterning, high chemical and mechanical stability, low material cost, and scalability for large-area fabrication. These attributes make PDMS one of the most versatile and reliable materials for triboelectric applications, as discussed in Section 2.1.

However, PDMS-based TENGs are far from reaching saturation in terms of performance and technological maturity. Several fundamental challenges persist, including charge leakage in humid environments, intrinsic limitations in charge saturation levels, degradation of surface microstructures during long-term cycling, insufficient electrical output for practical electronics, and incomplete system-level integration. To systematically address these challenges, a future development roadmap for PDMS-based TENGs is outlined in Fig. 22. This roadmap highlights three key stages: (1) rational surface engineering to maximize charge generation and retention, (2) functionalized PDMS composites designed to enhance dielectric properties and environmental stability, and (3) system-level integration, where optimized TENG units are combined with energy storage and sensing modules. Together, these strategies provide a clear and practical pathway for advancing PDMS-based TENGs from laboratory-scale demonstrations toward reliable, scalable, and commercially viable energy solutions.

7. Conclusion

PDMS has established itself as a cornerstone material in triboelectric nanogenerator TENG research due to its intrinsic flexibility, strong negative triboelectric polarity, mechanical durability, and excellent processability. Over the past decade, significant performance enhancements have been achieved through surface micro/nanostructuring, bioinspired hierarchical designs, and composite engineering with high-k fillers and functional additives. These strategies have markedly improved charge generation, sensitivity, and environmental adaptability of PDMS-based TENGs.

Despite these advances, challenges such as tribo-charge saturation, humidity-induced charge dissipation, mechanical wear of surface structures, and limited continuous power output continue to impede large-scale and long-term applications. Addressing these limitations requires a shift from empirical optimization toward deterministic materials and interface engineering supported by advanced simulations, data-driven design, and multiscale structural control. Looking forward, the integration of rationally designed PDMS composites, humidity-resilient architectures, and standardized durability protocols will be critical for bridging the gap between laboratory demonstrations and practical deployment. With continued progress in materials design, structural engineering, and system-level integration, PDMS-based TENGs are expected to evolve into reliable, high-performance, and scalable self-powered systems, enabling widespread applications in wearable electronics, sensing, and sustainable energy harvesting.

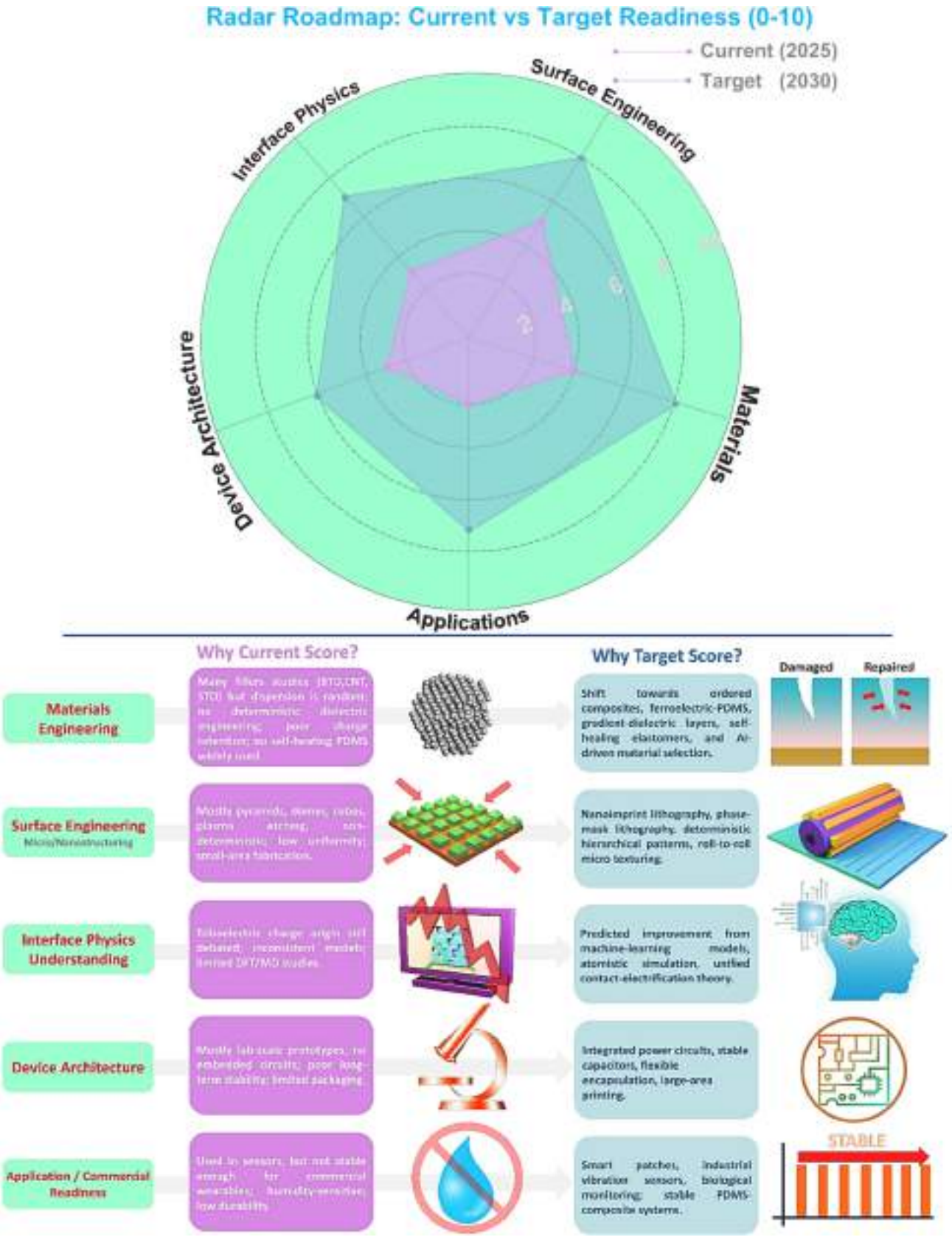


Fig. 21. Short term radar roadmap on PDMS-based TENG: current vs target readiness.

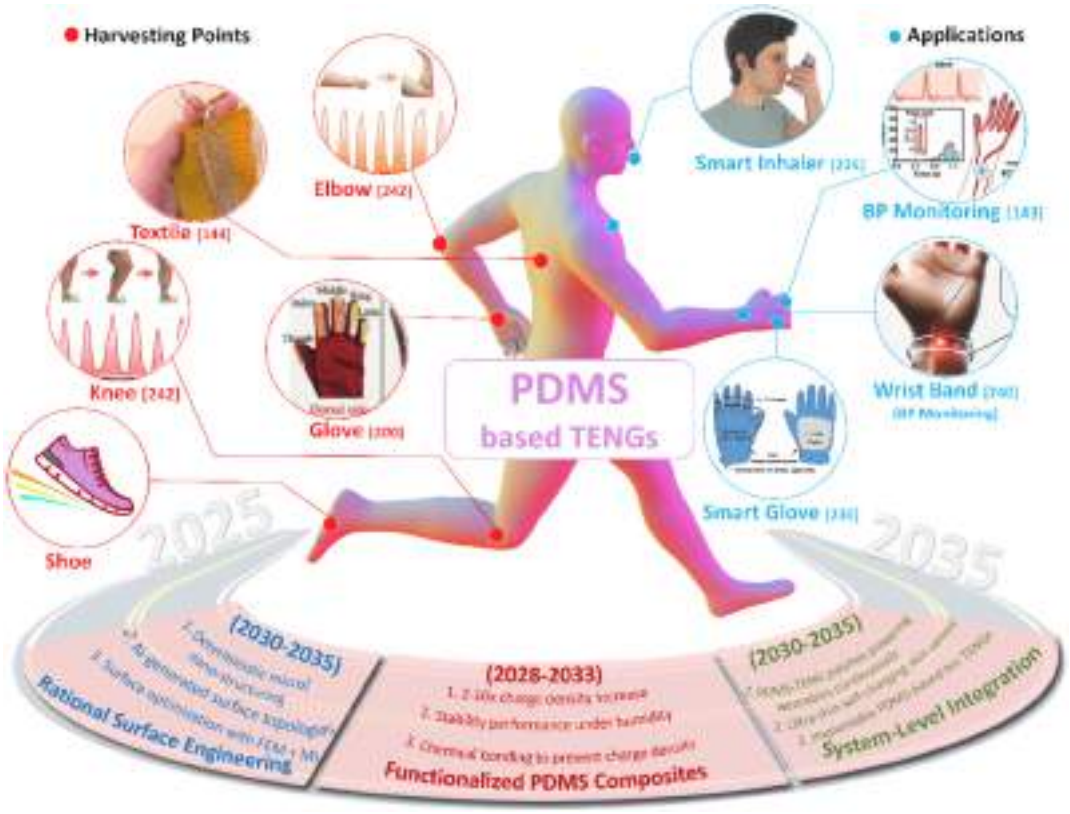


Fig. 22. Energy harvesting modes, applications, and future roadmap of PDMS-based TENGs.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] M. Zhang, A. Shahid, Y. Waleed, Lin, Sustainable power sources for portable applications, Adv. Mater. Technol. 5 (2020) e01070.
[2] D. Kim, S.A. Han, J.H. Kim, J.H. Lee, S.W. Kim, S.W. Lee, Biomolecular piezoelectric materials: from amino acids to living tissues, Adv. Mater. 32 (2020) 1906989.
[3] S. Pan, H. He, L. Xin, B. Li, X. Wu, X. Zhang, P. Wang, Super-elastic microstructured triboelectric fibers and textiles fabricated by extrusion and thermal drawing for smart-home applications, Nano Energy 122 (2024) 109259–109267.
[4] J. Zhu, M. Zhu, Q. Shi, et al., Progress in TENG technology—A journey from energy harvesting to nanoenergy and nanosystem, EcoMat 2 (2020) e12058.
[5] B. Shi, Q. Wang, H. Su, J. Li, B. Xie, P. Wang, J. Qiu, C. Wu, Y. Zhang, X. Zhou, T. W. Kim, Progress in recent research on the design and use of triboelectric nanogenerators for harvesting wind energy, Nano Energy 116 (2023) 116–135.
[6] M.M. Rastegardoost, O.A. Tafreshi, Z. Saadatnia, S. Ghaffari-Mosanenzadeh, C. B. Park, H.E. Naguib, Recent advances on porous materials and structures for high-performance triboelectric nanogenerators, Nano Energy 111 (2023) 111–134.
[7] U. Alvarado, A. Juanicorena, I. Adin, B. Sedano, I. Gutiérrez, J. de No, Energy harvesting technologies for low-power electronics, Trans. Emerg. Telecommun. Technol. 23 (2012) 728–741.
[8] Z. Lou, L. Li, L. Wang, G. Shen, Recent progress of self-powered sensing systems for wearable electronics, Small 13 (2017) 1701791.
[9] D. Newell, M. Duffy, Review of power conversion and energy management for low-power, low-voltage energy harvesting powered wireless sensors, IEEE Trans. Power Electron. 34 (2019) 9794–9805.
[10] J. Iannacci, Microsystem based Energy Harvesting (EH-MEMS): Powering pervasivity of the internet of things (IoT)–A review with focus on mechanical vibrations, J. King Saud Univ.-Sci. 31 (2019) 66–74.
[11] H. Elahi, K. Munir, M. Eugeni, S. Atek, P. Gaudenzi, Energy harvesting towards self-powered IoT devices, Energies 13 (2020) 5528.
[12] S. Roy, A.N.M. Wasekul Azad, S. Baidya, M.K. Alam, F. Khan, Powering solutions for biomedical sensors and implants inside the human body: a comprehensive review on energy harvesting units, energy storage, and wireless power transfer techniques, IEEE Trans. Power Electron. 37 (2022) 12237–12263.
[13] L. Mateu, F. Moll, Review of energy harvesting techniques and applications for microelectronics, in: VLSI Circuits and Systems II, Vol. 5837, SPIE, 2005, pp. 359–373.
[14] A. Muhtaroğlu, Micro-scale energy harvesting for batteryless information technologies, in: Energy Harvesting and Energy Efficiency: Technology, Methods, and Applications, Springer, Cham, 2017, pp. 63–85.
[15] J. Tian, X. Chen, Z.L. Wang, Environmental energy harvesting based on triboelectric nanogenerators, Nanotechnology 31 (2020) 242001.
[16] C. Chen, L. Chen, Z. Wu, H. Guo, W. Yu, Z. Du, Z.L. Wang, 3D double-faced interlock fabric triboelectric nanogenerator for bio-motion energy harvesting and as self-powered stretching and 3D tactile sensors, Mater. Today 32 (2020) 84–93.
[17] S.K. Ghosh, D. Mandal, Synergistically enhanced piezoelectric output in highly aligned 1D polymer nanofibers integrated all-fiber nanogenerator for wearable nano-tactile sensor, Nano Energy 53 (2018) 245–257.
[18] Z. Zhang, Y. Chen, J. Guo, ZnO nanorods patterned-textile using a novel hydrothermal method for sandwich structured-piezoelectric nanogenerator for human energy harvesting, Phys. E. 105 (2019) 212–218.
[19] Y. Guo, X.S. Zhang, Y. Wang, W. Gong, Q. Zhang, H. Wang, J. Brugger, All-fiber hybrid piezoelectric-enhanced triboelectric nanogenerator for wearable gesture monitoring, Nano Energy 48 (2018) 152–160.
[20] S. Siddiqui, H.B. Lee, D.I. Kim, L.T. Duy, A. Hanif, N.E. Lee, An omnidirectionally stretchable piezoelectric nanogenerator based on hybrid nanofibers and carbon electrodes for multimodal straining and human kinematics energy harvesting, Adv. Energy Mater. 8 (2018) 1–11.
[21] K. Dong, X. Peng, Z.L. Wang, Fiber/fabric-based piezoelectric and triboelectric nanogenerators for flexible/stretchable and wearable electronics and artificial intelligence, Adv. Mater. 32 (2020) 1–43.
[22] C. Jiang, C. Wu, X. Li, Y. Yao, L. Lan, F. Zhao, Z. Ye, Y. Ying, J. Ping, All-electrospun flexible triboelectric nanogenerator based on metallic MXene nanosheets, Nano Energy 59 (2019) 268–276.

[23] Z. Ren, L. Wu, Y. Pang, W. Zhang, R. Yang, Strategies for effectively harvesting wind energy based on triboelectric nanogenerators, Nano Energy 100 (2022) 107522.
[24] M.A.M. Hasan, W. Zhu, C.R. Bowen, Z.L. Wang, Y. Yang, Triboelectric nanogenerators for wind energy harvesting, Nat. Rev. Electr. Eng. (2024) 1–13.
[25] J. Li, J. Chen, H. Guo, Triboelectric nanogenerators for harvesting wind energy: recent advances and future perspectives, Energies 14 (2021) 6949.
[26] C. Rodrigues, D. Nunes, D. Clemente, N. Mathias, J.M. Correia, P. Rosa-Santos, J. Ventura, Emerging triboelectric nanogenerators for ocean wave energy harvesting: State of the art and future perspectives, Energy Environ. Sci. 13 (2020) 2657–2683.
[27] R. Li, Y. Li, Y. Zhao, Y. Li, Y. Li, Harvest of ocean energy by triboelectric generator technology, Appl. Phys. Rev. 5 (2018) 3.
[28] A.M. Nazar, K.J. Idala Egbe, A. Abdollahi, M.A. Hariri-Ardebili, Triboelectric nanogenerators for energy harvesting in ocean: a review on application and hybridization, Energies 14 (2021) 5600.
[29] B. Shan, T. Ai, K. Wang, Triboelectric nanogenerator for ocean energy harvesting: a review of technological advances and future perspectives, Int. J. Electrochem. Sci. (2024).
[30] J. Chen, J. Yang, Z. Li, X. Fan, Y. Zi, Q. Jing, Z.L. Wang, Networks of triboelectric nanogenerators for harvesting water wave energy: a potential approach toward blue energy, ACS Nano 9 (2015) 3324–3331.
[31] K. Uchino, Piezoelectric energy harvesting systems—essentials to successful developments, Energy Technol. 6 (2018) 829–848.
[32] H. Kim, Y. Tadesse, S. Priya, Piezoelectric energy harvesting, in: Energy Harvesting Technologies, Springer, Boston, MA, 2009, pp. 3–39.
[33] N. Sezer, M. Koç, A comprehensive review on the state-of-the-art of piezoelectric energy harvesting, Nano Energy 80 (2021) 105567.
[34] C. Covaci, A. Gontean, Piezoelectric energy harvesting solutions: a review, Sensors 20 (2020) 3512.
[35] Y. Hu, J. Yang, S. Niu, W. Wu, Z.L. Wang, Hybridizing triboelectrification and electromagnetic induction effects for high-efficient mechanical energy harvesting, ACS Nano 8 (2014) 7442–7450.
[36] J.V. Vidal, V. Slabov, A.L. Kholkin, M.P.S. Dos Santos, Hybrid triboelectric-electromagnetic nanogenerators for mechanical energy harvesting: a review, Nano-Micro Lett. 13 (2021) 1–58.
[37] A. Khaligh, P. Zeng, C. Zheng, Kinetic energy harvesting using piezoelectric and electromagnetic technologies—state of the art, IEEE Trans. Ind. Electron. 57 (2009) 850–860.
[38] X. Zhang, Z. Zhang, H. Pan, W. Salman, Y. Yuan, Y. Liu, A portable high-efficiency electromagnetic energy harvesting system using supercapacitors for renewable energy applications in railroads, Energy Convers. Manage. 118 (2016) 287–294.
[39] Y.H. Zhang, A. Lee, C.H. Lee, Design and application of piezoelectric and electromagnetic energy harvesters for mechanical energy harvesting in the human-body: a review, Sens. Actuators A Phys. (2024) 115207.
[40] Y. Chen, Y. Cheng, Y. Jie, X. Cao, N. Wang, Z.L. Wang, Energy harvesting and wireless power transmission by a hybridized electromagnetic–triboelectric nanogenerator, Energy Environ. Sci. 12 (2019) 2678–2684.
[41] S.M. Tripathi, A.N. Tiwari, D. Singh, Grid-integrated permanent magnet synchronous generator based wind energy conversion systems: a technology review, Renew. Sust. Energy Rev. 51 (2015) 1288–1305.
[42] R. Zhang, M. Hummelgård, J. Örtegren, M. Olsen, H. Andersson, Y. Yang, H. Zheng, H. Olin, The triboelectricity of the human body, Nano Energy 86 (2021) 106041.
[43] F.R. Fan, Z.Q. Tian, Z.L. Wang, Flexible triboelectric generator, Nano Energy 1 (2012) 328–334.
[44] W.G. Kim, D.W. Kim, I.W. Tcho, J.K. Kim, M.S. Kim, Y.K. Choi, Triboelectric nanogenerator: structure, mechanism, and applications, ACS Nano 15 (2021) 258–287.
[45] M. Burhan, R.A. Rehman, B. Khan, B.S. Kim, IoT elements, layered architectures and security issues: a comprehensive survey, Sensors 18 (2018) 1–37.
[46] Z.L. Wang, From contact electrification to triboelectric nanogenerators, Rep. Prog. Phys. 84 (2021) 096502.
[47] C. Callaty, C. Rodrigues, J. Ventura, Triboelectric nanogenerators in harsh conditions: a critical review, Nano Energy (2025).
[48] C. Zhang, W. Tang, C. Han, F. Fan, Z.L. Wang, Theoretical comparison, equivalent transformation, and conjunction operations of electromagnetic induction generator and triboelectric nanogenerator for harvesting mechanical energy, Adv. Mater. 26 (2014) 3580–3591.
[49] Z.L. Wang, Triboelectric nanogenerators as new energy technology for self-powered systems and as active mechanical and chemical sensors, ACS Nano 7 (2013) 9533–9557.
[50] J. Shao, M. Willatzen, Z.L. Wang, Theoretical modeling of triboelectric nanogenerators (TENGs), J. Appl. Phys. 128 (2020).
[51] D. Choi, Y. Lee, Z.-H. Lin, S. Cho, M. Kim, C.K. Ao, S. Soh, et al., Recent advances in triboelectric nanogenerators: from technological progress to commercial applications, ACS Nano 17 (12) (2023) 11087–11219.
[52] Z.L. Wang, Triboelectric nanogenerator (TENG)—sparking an energy and sensor revolution, Adv. Energy Mater. 10 (17) (2020) 2000137.
[53] J. Luo, W. Gao, Z.L. Wang, The triboelectric nanogenerator as an innovative technology toward intelligent sports, Adv. Mater. 33 (17) (2021) 2004178.
[54] S. Rathore, S. Sharma, B.P. Swain, R.K. Ghadai, A critical review on triboelectric nanogenerator, IOP Conf. Ser.: Mater. Sci. Eng. 377 (2018) 012186.
[55] W. Akram, Q. Chen, G. Xia, J. Fang, A review of single electrode triboelectric nanogenerators, Nano Energy 106 (2023) 108043.
[56] T. Cheng, Q. Gao, Z.L. Wang, The current development and future outlook of triboelectric nanogenerators: a survey of literature, Adv. Mater. Technol. 4 (3) (2019) 1800588.
[57] R.D.I.G. Dharmasena, S.R.P. Silva, Towards optimized triboelectric nanogenerators, Nano Energy 62 (2019) 530–549.
[58] D.G. Dassanayaka, T.M. Alves, N.D. Wanasekara, I.G. Dharmasena, J. Ventura, Recent progresses in wearable triboelectric nanogenerators, Adv. Funct. Mater. 32 (44) (2022) 2205438.
[59] R. Walden, C. Kumar, D.M. Mulvihill, S.C. Pillai, Opportunities and challenges in triboelectric nanogenerator (TENG) based sustainable energy generation technologies: a mini-review, Chem. Eng. J. Adv. 9 (2022) 100237.
[60] J. Luo, Z.L. Wang, Recent progress of triboelectric nanogenerators: from fundamental theory to practical applications, EcoMat 2 (4) (2020) e12059.
[61] J. Chen, Z.L. Wang, Reviving vibration energy harvesting and self-powered sensing by a triboelectric nanogenerator, Joule 1 (2017) 480–521.
[62] R. Huang, J. Zhu, A hybrid electromagnetic and leaf-shaped polytetrafluoroethylene triboelectric with an arc-shaped brace structure for energy harvesting, RSC Adv. 7 (2017) 49562–49567.
[63] C. Zhang, X. Lin, N. Zhang, Y. Lu, Z. Wu, G. Liu, S. Nie, Chemically functionalized cellulose nanofibrils-based gear-like triboelectric nanogenerator for energy harvesting and sensing, Nano Energy 66 (2019) 104126.
[64] Y. Gao, D. Liu, L. Zhou, S. Li, Z. Zhao, X. Yin, S. Chen, Z.L. Wang, J. Wang, A robust rolling-mode direct-current triboelectric nanogenerator arising from electrostatic breakdown effect, Nano Energy 85 (2021) 106014.
[65] T. Guo, G. Liu, Y. Pang, B. Wu, F. Xi, J. Zhao, T. Bu, X. Fu, X. Li, C. Zhang, Z. L. Wang, Compressible hexagonal-structured triboelectric nanogenerators for harvesting tire rotation energy, Extrem. Mech. Lett. 18 (2018) 1–8.
[66] L. Jin, S.L. Zhang, S. Xu, H. Guo, W. Yang, Z.L. Wang, Free-fixed rotational triboelectric nanogenerator for self-powered real-time wheel monitoring, Adv. Mater. Technol. 6 (2021) 1–7.
[67] D.S. Kong, J.Y. Han, Y.J. Ko, S.H. Park, M. Lee, J.H. Jung, A highly efficient and durable kirigami triboelectric nanogenerator for rotational energy harvesting, Energies 14 (2021) 1120.
[68] L. Lin, S. Wang, Y. Xie, Q. Jing, S. Niu, Y. Hu, Z.L. Wang, Segmentally structured disk triboelectric nanogenerator for harvesting rotational mechanical energy, Nano Lett. 13 (2013) 2916–2923.
[69] S. Shen, J. Fu, J. Yi, L. Ma, F. Sheng, C. Li, T. Wang, C. Ning, H. Wang, K. Dong, Z. L. Wang, High-efficiency wastewater purification system based on coupled photoelectricecatalytic action provided by triboelectric nanogenerator, Nano-Micro Lett. 13 (2021).
[70] S. Wang, Y. Xie, S. Niu, L. Lin, Z.L. Wang, Freestanding triboelectric-layer-based nanogenerators for harvesting energy from a moving object or human motion in contact and non-contact modes, Adv. Mater. 26 (2014) 2818–2824.
[71] S. Wang, S. Niu, J. Yang, L. Lin, Z.L. Wang, Quantitative measurements of vibration amplitude using a contact-mode freestanding triboelectric nanogenerator, ACS Nano 8 (2014) 12004–12013.
[72] Z.L. Wang, J. Chen, L. Lin, Progress in triboelectric nanogenerators as a new energy technology and self-powered sensors, Energy Environ. Sci. 8 (2015) 2250–2282.
[73] Z. Liu, H. Li, B. Shi, Y. Fan, Z.L. Wang, Z. Li, Wearable and implantable triboelectric nanogenerators, Adv. Funct. Mater. 29 (20) (2019) 1808820.
[74] H. Wang, M. Han, Y. Song, H. Zhang, Design, manufacturing and applications of wearable triboelectric nanogenerators, Nano Energy 81 (2021) 105627.
[75] V. Vivekananthan, A. Chandrasekhar, N.R. Alluri, Y. Purusothaman, G. Khandelwal, S.J. Kim, Triboelectric nanogenerators: Design, fabrication, energy harvesting, and portable-wearable applications, Nanogenerators 5 (2020) 3.
[76] S.Y. Kuang, J. Chen, X.B. Cheng, G. Zhu, Z.L. Wang, Two-dimensional rotary triboelectric nanogenerator as a portable and wearable power source for electronics, Nano Energy 17 (2015) 10–16.
[77] K.Y. Lee, M.K. Gupta, S.W. Kim, Transparent flexible stretchable piezoelectric and triboelectric nanogenerators for powering portable electronics, Nano Energy 14 (2015) 139–160.
[78] Z. Zhang, J. Zhang, H. Zhang, H. Wang, Z. Hu, W. Xuan, J. Luo, A portable triboelectric nanogenerator for real-time respiration monitoring, Nanoscale Res. Lett. 14 (2019) 1–11.
[79] Y. Zou, J. Xu, Y. Fang, X. Zhao, Y. Zhou, J. Chen, A hand-driven portable triboelectric nanogenerator using whirligig spinning dynamics, Nano Energy 83 (2021) 105845.
[80] N. Cui, J. Liu, L. Gu, S. Bai, X. Chen, Y. Qin, Wearable triboelectric generator for powering the portable electronic devices, ACS Appl. Mater. Interfaces 7 (33) (2015) 18225–18230.
[81] H. Chen, L. Bai, T. Li, C. Zhao, J. Zhang, N. Zhang, Y. Xu, Wearable and robust triboelectric nanogenerator based on crumpled gold films, Nano Energy 46 (2018) 73–80.
[82] P. Munirathinam, A. Chandrasekhar, Wearable triboelectric nanogenerator for real-time IoT-supported security applications, Sustain. Mater. Technol. 37 (2023) e00700.
[83] T. Zhou, C. Zhang, C.B. Han, F.R. Fan, W. Tang, Z.L. Wang, Woven structured triboelectric nanogenerator for wearable devices, ACS Appl. Mater. Interfaces 6 (16) (2014) 14695–14701.
[84] Y. Mao, Y. Zhu, T. Zhao, C. Jia, X. Wang, Q. Wang, Portable mobile gait monitor system based on triboelectric nanogenerator for monitoring gait and powering electronics, Energies 14 (16) (2021) 4996.

[85] C. He, W. Zhu, B. Chen, L. Xu, T. Jiang, C.B. Han, G.Q. Gu, D. Li, Z.L. Wang, Smart floor with integrated triboelectric nanogenerator as energy harvester and motion sensor, ACS Appl. Mater. Interfaces 9 (31) (2017) 26126–26133.

[86] W. Wang, Y. Zhang, G. Wu, Z. Zhao, Y. Wu, H. Zheng, Triboelectric–electromagnetic hybrid nanogenerator for harvesting blue energy and creating an ocean wave warning system, Nanoscale Adv. 6 (14) (2024) 3566–3572.

[87] X. Su, Y. Su, H. Yan, X. Zhang, G. Dai, X. Dong, J. Wu, X. Zhao, K. Zhao, Z. Li, A roller-type triboelectric nanogenerator based on rotational friction between wool and stacked interfaces for omnidirectional wind energy harvesting, Nanoscale 17 (6) (2025) 3370–3380.

[88] G. Khandelwal, T. Minocha, S.K. Yadav, A. Chandrasekhar, N.P.M.J. Raj, S. C. Gupta, S.J. Kim, All edible materials derived biocompatible and biodegradable triboelectric nanogenerator, Nano Energy 65 (2019) 104016.

[89] S. Chao, H. Ouyang, D. Jiang, Y. Fan, Z. Li, Triboelectric nanogenerator based on degradable materials, EcoMat 3 (1) (2021) e12072.

[90] Y. Mi, Y. Lu, Y. Shi, Z. Zhao, X. Wang, J. Meng, N. Wang, Biodegradable polymers in triboelectric nanogenerators, Polymers 15 (1) (2022) 222.

[91] Y. Lyu, Y. Wang, Output optimization of biodegradable triboelectric nanogenerators, Nano Energy 103 (2022) 107811.

[92] Q. Zheng, Y. Zou, Y. Zhang, Z. Liu, B. Shi, X. Wang, Z.L. Wang, Biodegradable triboelectric nanogenerator as a life-time designed implantable power source, Sci. Adv. 2 (3) (2016) e1501478.

[93] A. Baburaj, S. Banerjee, A.K. Aliyana, C. Shee, M. Banakar, S. Bairagi, S.W. Ali, Biodegradable based TENGs for self-sustaining implantable medical devices, Nano Energy (2024).

[94] G. Jian, S. Zhu, X. Yuan, S. Fu, N. Yang, C. Yan, X. Wang, C.P. Wong, Biodegradable triboelectric nanogenerator as a implantable power source for embedded medicine devices, NPG Asia Mater. 16 (1) (2024) 12.

[95] Z. Li, H. Feng, Q. Zheng, H. Li, C. Zhao, H. Ouyang, S. Noreen, Photothermally tunable biodegradation of implantable triboelectric nanogenerators for tissue repairing, Nano Energy 54 (2018) 390–399.

[96] R.S. Ganesh, H.J. Yoon, S.W. Kim, Recent trends of biocompatible triboelectric nanogenerators toward self-powered e-skin, EcoMat 2 (4) (2020) e12065.

[97] Q. Zheng, Y. Jin, Z. Liu, H. Ouyang, H. Li, B. Shi, W. Jiang, H. Zhang, Z. Li, Z. L. Wang, Robust multilayered encapsulation for high-performance triboelectric nanogenerator in harsh environment, ACS Appl. Mater. Interfaces 8 (40) (2016) 26697–26703.

[98] R. Cao, Y. Xia, J. Wang, X. Jia, C. Jia, S. Zhu, W. Zhang, X. Gao, X. Zhang, Suppressing thermal negative effect and maintaining high-temperature steady electrical performance of triboelectric nanogenerators by employing phase change material, ACS Appl. Mater. Interfaces 13 (35) (2021) 41657–41668.

[99] L. Zheng, Y. Xiang, D. Zheng, N. Luo, Degradation mechanisms and intelligent recognition of PTFE triboelectric performance under hygrothermal cycling environment, ACS Appl. Mater. Interfaces 17 (35) (2025) 49500–49509.

[100] J. Zhao, Y. Shi, Boosting the durability of triboelectric nanogenerators: a critical review and prospect, Adv. Funct. Mater. 33 (14) (2023), 2213407.

[101] L. Liu, M. Wu, W. Zhao, J. Tao, X. Zhou, J. Xiong, Progress of triboelectric nanogenerators with environmental adaptivity, Adv. Funct. Mater. 34 (2024) 2308353.

[102] W. Xu, M.C. Wong, J. Hao, Strategies and progress on improving robustness and reliability of triboelectric nanogenerators, Nano Energy 55 (2019) 203–215.

[103] S.A. Lone, K.C. Lim, K. Kaswan, S. Chatterjee, K.P. Fan, D. Choi, S. Lee, H. Zhang, J. Cheng, Z.-H. Wang, Recent advancements for improving the performance of triboelectric nanogenerator devices, Nano Energy 99 (2022) 107318.

[104] H. Li, F. Xu, T. Guan, Y. Li, J. Sun, Mechanically and environmentally stable triboelectric nanogenerator based on high-strength and anti-compression self-healing ionogel, Nano Energy 90 (2021) 106645.

[105] H. Wu, H. Li, X. Wang, A high-stability triboelectric nanogenerator with mechanical transmission module and efficient power management system, J. Micromech. Microeng. 30 (2020) 115017.

[106] R. Hinchet, W. Seung, S.W. Kim, Recent progress on flexible triboelectric nanogenerators for self-powered electronics, ChemSusChem 8 (2015) 2327–2344.

[107] H. Wu, Y. Huang, F. Xu, Y. Duan, Z. Yin, Energy harvesters for wearable and stretchable electronics: from flexibility to stretchability, Adv. Mater. 28 (2016) 9881–9919.

[108] A. Yar, A. Karabiber, A. Ozen, F. Ozel, S. Coskun, Flexible nanofiber-based triboelectric nanogenerators with high power conversion, Renew. Energy 162 (2020) 1428–1437.

[109] F.R. Fan, J. Luo, W. Tang, C. Li, C. Zhang, Z. Tian, Z.L. Wang, Highly transparent and flexible triboelectric nanogenerators: performance improvements and fundamental mechanisms, J. Mater. Chem. A 2 (2014) 13219–13225.

[110] H. Wu, C. Shan, S. Fu, K. Li, J. Wang, S. Xu, C. Hu, Efficient energy conversion mechanism and energy storage strategy for triboelectric nanogenerators, Nat. Commun. 15 (2024) 6558.

[111] J.L. Armitage, Tribology and the Triboelectric Effect: A Study into the Influence of Tribological Factors on Frictional Electrification, University of Leeds, 2022. PhD Thesis.

[112] L.S. McCarty, G.M. Whitesides, Electrostatic charging due to separation of ions at interfaces: contact electrification of ionic electrets, Angew. Chem. Int. Ed. 47 (2008) 2188–2207.

[113] A. Baburaj, S.N. Kumar, A.K. Aliyana, M. Banakar, S. Bairagi, G. Stylios, Factors affecting the performance of flexible triboelectric nanogenerators (F-TENGs) and their sensing capabilities: a comprehensive review, Nano Energy 108 (2023) 108983.

[114] G.-J. Choi, S.-H. Sohn, S.-J. Kim, I.-K. Park, Polymer composite-based triboelectric nanogenerators: recent progress, design principles, and future perspectives, Polymers 17 (2025) 1962.

[115] Z.L. Wang, Triboelectric nanogenerators as new energy technology and self-powered sensors–principles, problems and perspectives, Faraday Discuss. 176 (2014) 447–458.

[116] R. Zhang, H. Olin, Material choices for triboelectric nanogenerators: a critical review, EcoMat 2 (2020) e12062.

[117] P. Zhao, N. Soin, K. Prashanthi, J. Chen, S. Dong, E. Zhou, L. Yu, Emulsion electrospinning of polytetrafluoroethylene (PTFE) nanofibrous membranes for high-performance triboelectric nanogenerators, ACS Appl. Mater. Interfaces 10 (2018) 5880–5891.

[118] J. Liu, J. Wang, Y. Wang, Z. Wu, H. Sun, Y. Yang, W. Kang, A high-performance stretchable triboelectric nanogenerator based on polytetrafluoroethylene (PTFE) particles, Energy Environ. Mater. 8 (2025) e12814.

[119] X. Cai, Z. Liu, J. Dong, H. Li, J. Han, J. Huang, H. Chen, U-shaped tube-based liquid–solid triboelectric nanogenerator for harvesting unutilized compressed air energy, Micromachines 14 (2023) 2057.

[120] Y. Su, Y. Yang, X. Zhong, H. Zhang, Z. Wu, Y. Jiang, Z.L. Wang, Fully enclosed cylindrical single-electrode-based triboelectric nanogenerator, ACS Appl. Mater. Interfaces 6 (2014) 553–559.

[121] J.P. Lee, J.W. Lee, J.M. Baik, The progress of PVDF as a functional material for triboelectric nanogenerators and self-powered sensors, Micromachines 9 (2018) 532.

[122] B. Hedau, B.-C. Kang, T.-J. Ha, Enhanced triboelectric effects of self-poled MoS_2-embedded PVDF hybrid nanocomposite films for bar-printed wearable triboelectric nanogenerators, ACS Nano 16 (2022) 18355–18365.

[123] M.G. Evich, M.J. Davis, J.P. McCord, B. Acrey, J.A. Awkerman, D.R. Knappe, M. J. Strynar, Per- and polyfluoroalkyl substances in the environment, Science 375 (2022) eabg9065.

[124] X. Fang, J. Wang, The progress in environmental distributions and toxicological effects of perfluoroalkyl acids, Adv. Earth Sci. 25 (2010) 543.

[125] M.M. Plassmann, Environmental occurrence and fate of semifluorinated n-alkanes and perfluorinated alkyl acids present in ski waxes, Department of Applied Environmental Science (ITM), Stockholm University, 2011.

[126] J. Han, L. Kiss, H. Mei, A.M. Remete, M. Ponikvar-Svet, D.M. Sedgwick, V. A. Soloshonok, Chemical aspects of human and environmental overload with fluorine, Chem. Rev. 121 (2021) 4678–4742.

[127] D.E. Brinza, A. Stiegman, P.R. Staszak, E.G. Laue, R.H. Liang, Vacuum Ultraviolet (VUV) radiation-induced degradation of Fluorinated Ethylene Propylene (FEP) teflon aboard the Long Duration Exposure Facility (LDEF), NASA Langley Res. Center (1992).

[128] B. Améduri, H. Hori, Recycling and the end of life assessment of fluoropolymers: recent developments, challenges and future trends, Chem. Soc. Rev. 52 (2023) 4208–4247.

[129] P. Sanz-Gallén, S. Nogué, P. Munné, A. Faraldo, Hypocalcaemia and hypomagnesaemia due to hydrofluoric acid, Occup. Med. 51 (2001) 294–295.

[130] S.E. Fenton, A. Ducatman, A. Boobis, J.C. DeWitt, C. Lau, C. Ng, S.M. Roberts, Per- and polyfluoroalkyl substance toxicity and human health review: current state of knowledge and strategies for informing future research, Environ. Toxicol. Chem. 40 (2021) 606–630.

[131] B.J. Henry, J.P. Carlin, J.A. Hammerschmidt, R.C. Buck, L.W. Buxton, H. Fiedler, J. Seed, O. Hernandez, A critical review of the application of polymer of low concern and regulatory criteria to fluoropolymers, Integr. Environ. Assess. Manag. 14 (2018) 316–334.

[132] N.J. Olsavsky, V.M. Kearns, C.P. Beckman, P.L. Sheehan, F.J. Burpo, H. D. Bahaghighat, E.A. Nagelli, Research and regulatory advancements on remediation and degradation of fluorinated polymer compounds, Appl. Sci. 10 (2020) 6921.

[133] S. Li, J. Zhang, J. He, W. Liu, Y. Wang, Z. Huang, H. Pang, Y. Chen, Functional PDMS elastomers: bulk composites, surface engineering, and precision fabrication, Adv. Sci. 10 (2023) 2304506.

[134] R. Ariati, F. Sales, A. Souza, R.A. Lima, J. Ribeiro, Polydimethylsiloxane composites characterization and its applications: a review, Polymers 13 (2021) 4258.

[135] A. Šutka, L. Lapčinskis, D. He, H. Kim, J.D. Berry, J. Bai, M. Knite, A.V. Ellis, C. K. Jeong, P.C. Sherrell, Engineering polymer interfaces: a review toward controlling triboelectric surface charge, Adv. Mater. Interf. 10 (2023) 2300323.

[136] N. Luo, Y. Feng, X. Li, W. Sun, D. Wang, Q. Ye, X. Sun, F. Zhou, W. Liu, Manipulating electrical properties of silica-based materials via atomic oxygen irradiation, ACS Appl. Mater. Interf. 13 (2021) 15344–15352.

[137] J. Li, N.A. Shepelin, P.C. Sherrell, A.V. Ellis, Poly(dimethylsiloxane) for triboelectricity: from mechanisms to practical strategies, Chem. Mater. 33 (2021) 4304–4327.

[138] Y. Wang, Y. Yang, Z.L. Wang, Triboelectric nanogenerators as flexible power sources, NPJ Flex Electron 1 (2017) 10.

[139] M. Alaraby, D. Abass, A. Velázquez, A. Hernández, R. Marcos, Polytetrafluoroethylene microplastic properties, pollution, toxicity and analysis: a review, Environ. Chem. Lett. 24 (2026) 27–59.

[140] W. Yang, X. Wang, H. Li, J. Wu, Y. Hu, Z. Li, H. Liu, Fundamental research on the effective contact area of micro-/nano-textured surface in triboelectric nanogenerator, Nano Energy 57 (2019) 41–47.

[141] F.R. Fan, L. Lin, G. Zhu, W. Wu, R. Zhang, Z.L. Wang, Transparent triboelectric nanogenerators and self-powered pressure sensors based on micropatterned plastic films, Nano Lett. 12 (2012) 3109–3114.

[142] C. Cho, C. Lee, J.H. Oh, Wearable self-powered pressure-sensing device based on carbon nanotubes/porous PDMS and poly(ethylene oxide), ACS Appl. Nano Mater. 7 (2024) 5040–5050.

[143] D. Tantraviwat, M. Ngamyingyoud, W. Sripumkhai, P. Pattamang, G. Rujijanagul, B. Inceesungvorn, Tuning the dielectric constant and surface engineering of a $BaTiO_3$/Porous PDMS composite film for enhanced triboelectric nanogenerator output performance, ACS Omega 6 (2021) 29765–29773.

[144] J. Seong, B.-U. Bak, D. Lee, J. Jin, J. Kim, Tribo-piezoelectric synergistic $BaTiO_3$/PDMS micropyramidal structure for high-performance energy harvester and high-sensitivity tactile sensing, Nano Energy 122 (2024) 109264.

[145] T. Zhang, C. Yao, X. Xu, Z. Liu, Z. Liu, T. Sun, S. Huang, et al., Nanopores-templated CNT/PDMS microcolumn substrate for the fabrication of wearable triboelectric nanogenerator sensors to monitor human pulse and blood pressure, Adv. Mater. Technol. 10 (2025) 2400749.

[146] R. Barras, A. dos Santos, T. Calmeiro, E. Fortunato, R. Martins, H. Aguas, P. Barquinha, R. Igreja, L. Pereira, Porous PDMS conformable coating for high power output carbon fibers/ZnO nanorod-based triboelectric energy harvesters, Nano Energy 90 (2021) 106582.

[147] W. Shin, S. Paria, S. Mondal, G.-B. Lee, H. Kim, C. Park, C. Nah, Fabrication of a single-crystalline SnS-based piezo-assisted efficient single-electrode triboelectric nanogenerator for energy harvesting and sensing applications, J. Mater. Chem. A 11 (2023) 11471–11484.

[148] Y. Nurmakanov, G. Kalimuldina, G. Nauryzbayev, D. Adair, Z. Bakenov, Structural and chemical modifications towards high-performance of triboelectric nanogenerators, Nanoscale Res. Lett. 16 (2021) 122.

[149] X. Li, Q. Yang, D. Ren, Q. Li, H. Yang, X. Zhang, Y. Xi, A review of material design for high performance triboelectric nanogenerators: performance improvement based on charge generation and charge loss, Nanoscale Adv. (2024).

[150] M. Shanbedi, H. Ardebili, A. Karim, Polymer-based triboelectric nanogenerators: Materials, characterization, and applications, Prog. Polym. Sci. 144 (2023) 101723.

[151] J. Shao, M. Willatzen, T. Jiang, W. Tang, X. Chen, J. Wang, Z.L. Wang, Quantifying the power output and structural figure-of-merits of triboelectric nanogenerators in a charging system starting from the Maxwell's displacement current, Nano Energy 59 (2019) 380–389.

[152] Y. Zi, S. Niu, J. Wang, Z. Wen, W. Tang, Z.L. Wang, Standards and figure-of-merits for quantifying the performance of triboelectric nanogenerators, Nat. Commun. 6 (2015) 8376.

[153] S. Wang, L. Lin, Z.L. Wang, Nanoscale triboelectric-effect-enabled energy conversion for sustainably powering portable electronics, Nano Lett. 12 (2012) 6339–6346.

[154] C.K. Chung, K.H. Ke, High contact surface area enhanced Al/PDMS triboelectric nanogenerator using novel overlapped microneedle arrays and its application to lighting and self-powered devices, Appl. Surf. Sci. 508 (2020) 145310.

[155] S. Yu, Y. Sun, Y. Ni, X. Zhang, H. Zhou, Controlled formation of surface patterns in metal films deposited on elasticity-gradient PDMS substrates, ACS Appl. Mater. Interf. 8 (2016) 5706–5714.

[156] X. Cui, J. Nie, Y. Zhang, Recent advances in high charge density triboelectric nanogenerators, Int. J. Extrem. Manuf. 6 (2024) 042001.

[157] Z.-Y. Huo, Y.-J. Kim, I.-Y. Suh, D.-M. Lee, J.H. Lee, Y. Du, S. Wang, H.-J. Yoon, S.-W. Kim, Triboelectrification induced self-powered microbial disinfection using nanowire-enhanced localized electric field, Nat. Commun. 12 (2021) 3693.

[158] J. Zhou, Y. Zhang, Z. Luo, Q. Zhang, L. Wang, Tip-effect-driven field enhancement and faster mass transport and their advanced applications, Chem. Commun. (2025).

[159] Z. Lin, B. Zhang, Y. Xie, Z. Wu, J. Yang, Z.L. Wang, Elastic-connection and soft-contact triboelectric nanogenerator with superior durability and efficiency, Adv. Funct. Mater. 31 (2021) 2105237.

[160] M. Klaassen, E.G. De Vries, M.A. Masen, Friction in the contact between skin and a soft counter material: effects of hardness and surface finish, J. Mech. Behav. Biomed. Mater. 92 (2019) 137–143.

[161] N. Selway, V. Chan, J.R. Stokes, Influence of fluid viscosity and wetting on multiscale viscoelastic lubrication in soft tribological contacts, Soft Matter 13 (2017) 1702–1715.

[162] Z.H. Lin, G. Cheng, S. Lee, K.C. Pradel, Z.L. Wang, Harvesting water drop energy by a sequential contact-electrification and electrostatic-induction process, Adv. Mater. 26 (2014) 4690–4696.

[163] B.N. Chandrashekar, B. Deng, A.S. Smitha, Y. Chen, C. Tan, H. Zhang, H. Peng, Z. Liu, Roll-to-roll green transfer of CVD graphene onto plastic for a transparent and flexible triboelectric nanogenerator, Adv. Mater. 27 (2015) 5210–5216.

[164] H. Varghese, H.M.A. Hakkeem, K. Chauhan, E. Thouti, S. Pillai, A. Chandran, A high-performance flexible triboelectric nanogenerator based on cellulose acetate nanofibers and micropatterned PDMS films as mechanical energy harvester and self-powered vibrational sensor, Nano Energy 98 (2022) 107339.

[165] H.S. Hong, N.H. Ha, D.D. Thinh, N.H. Nam, N.T. Huong, N.T. Hue, T.V. Hoang, Enhanced sensitivity of self-powered NO_2 gas sensor to sub-ppb level using triboelectric effect based on surface-modified PDMS and 3D-graphene/CNT network, Nano Energy 87 (2021) 106165.

[166] X.Z. Niu, S.L. Peng, L.Y. Liu, W.J. Wen, P. Sheng, Characterizing and patterning of PDMS-based conducting composites, Adv. Mater. 19 (2007) 2682.

[167] D. Jang, Y. Kim, T.Y. Kim, K. Koh, U. Jeong, J. Cho, Force-assembled triboelectric nanogenerator with high-humidity-resistant electricity generation using hierarchical surface morphology, Nano Energy 20 (2016) 283–293.

[168] J.H. Lee, I. Yu, S. Lee, J.K. Kim, U. Jeong, Remarkable increase in triboelectrification by enhancing the conformable contact and adhesion energy with a film-covered pillar structure, Nano Energy 34 (2017) 233–241.

[169] M.L. Seol, J.H. Woo, D.I. Lee, H. Im, J. Hur, Y.K. Choi, Nature-replicated nano-in-micro structures for triboelectric energy harvesting, Small 10 (2014) 3887–3894.

[170] Y. Sun, Y. Zheng, R. Wang, T. Lei, J. Liu, J. Fan, W. Shou, Y. Liu, 3D micro-nanostructure based waterproof triboelectric nanogenerator as an outdoor adventure power source, Nano Energy 100 (2022) 107506.

[171] Y. Jung, J. Ahn, J.-S. Kim, J.-H. Ha, J. Shim, H. Cho, Y.S. Oh, Spherical micro/nano hierarchical structures for energy and water harvesting devices, Small Methods 6 (2022) 2200248.

[172] C. Lee, S. Yang, D. Choi, W. Kim, J. Kim, J. Hong, Chemically surface-engineered polydimethylsiloxane layer via plasma treatment for advancing textile-based triboelectric nanogenerators, Nano Energy 57 (2019) 353–363.

[173] O. Polat, Plasma Treatment of Polyethylene and Polydimethylsiloxane by Air, Oxygen, and Carbon Tetrafluoride and their Characterization, Master's Thesis, University of South Florida, 2022.

[174] Y.G. Kim, N. Lim, J. Kim, C. Kim, J. Lee, K.-H. Kwon, Study on the surface energy characteristics of polydimethylsiloxane (PDMS) films modified by C4F8/O2/Ar plasma treatment, Appl. Surf. Sci. 477 (2019) 198–203.

[175] G.-G. Cheng, S.-Y. Jiang, K. Li, Z.-Q. Zhang, Y. Wang, N.-Y. Yuan, J.-N. Ding, W. Zhang, Effect of argon plasma treatment on the output performance of triboelectric nanogenerator, Appl. Surf. Sci. 412 (2017) 350–356.

[176] X. Cheng, L. Miao, Z. Su, H. Chen, Y. Song, X. Chen, H. Zhang, Controlled fabrication of nanoscale wrinkle structure by fluorocarbon plasma for highly transparent triboelectric nanogenerator, Microsyst. Nanoeng. 3 (2017) 1–9.

[177] X. Cheng, B. Meng, X. Chen, M. Han, H. Chen, Z. Su, M. Shi, H. Zhang, Single-step fluorocarbon plasma treatment-induced wrinkle structure for high-performance triboelectric nanogenerator, Small 12 (2016) 229–236.

[178] W. Kim, T. Okada, H.-W. Park, J. Kim, S. Kim, S.-W. Kim, S. Samukawa, D. Choi, Surface modification of triboelectric materials by neutral beams, J. Mater. Chem. A 7 (2019) 25066–25077.

[179] H. Zhu, J. Liang, W. Long, F. Zeng, X. Zhang, Z. Chen, Enhancing PDMS-based triboelectric nanogenerator output by optimizing the microstructure and dielectric constant, J. Mater. Chem. C 12 (2024) 1782–1791.

[180] Y. Xiao, J. Lu, B. Xu, Synergistic effect study of g-C3N4 composites for high-performance triboelectric nanogenerators, *Energy Mater.* 5 (2025) N/A.

[181] M.S. Kwak, Y.-J. Park, M.P. Kim, H. Ko, Interfacial polarization for high-performance triboelectric devices: principles, strategies, and applications, ACS Appl. Mater. Interf. (2025).

[182] X. Kang, C. Pan, Y. Chen, X. Pu, Boosting performances of triboelectric nanogenerators by optimizing dielectric properties and thickness of electrification layer, RSC Adv. 10 (2020) 17752–17759.

[183] X. Pu, C. Zhang, Z.L. Wang, Triboelectric nanogenerators as wearable power sources and self-powered sensors, Nat. Sci. Rev. 10 (2023) nwac170.

[184] D.W. Kim, J.H. Lee, J.K. Kim, U. Jeong, Material aspects of triboelectric energy generation and sensors, NPG Asia Mater. 12 (2020) 6.

[185] W. Choi, J. Choi, Y. Han, H. Yoo, H.-J. Yoon, Polymer dielectric-based emerging devices: advancements in memory, field-effect transistor, and nanogenerator technologies, Micromachines 15 (2024) 1115.

[186] A. Chen, C. Zhang, G. Zhu, Z.L. Wang, Polymer materials for high-performance triboelectric nanogenerators, Adv. Sci. 7 (2020) 2000186.

[187] C.H. Park, M.P. Kim, Advanced triboelectric applications of biomass-derived materials: a comprehensive review, Mater 17 (2024) 1964.

[188] M.P. Kim, D.-S. Um, Y.-E. Shin, H. Ko, High-performance triboelectric devices via dielectric polarization: a review, Nanoscale Res. Lett. 16 (2021) 35.

[189] C. Wu, A.C. Wang, W. Ding, H. Guo, Z.L. Wang, Triboelectric nanogenerator: a foundation of the energy for the new era, Adv. Energy Mater. 9 (2019) 1802906.

[190] M. Ha, J. Park, Y. Lee, H. Ko, Triboelectric generators and sensors for self-powered wearable electronics, ACS Nano 9 (2015) 3421–3427.

[191] M.P. Kim, Multilayered functional triboelectric polymers for self-powered wearable applications: a review, Micromachines 14 (2023) 1640.

[192] S. Niu, Z.L. Wang, Theoretical systems of triboelectric nanogenerators, Nano Energy 14 (2015) 161–192.

[193] Z.L. Wang, On the first principle theory of nanogenerators from Maxwell's equations, Nano Energy 68 (2020) 104272.

[194] Z.L. Wang, On Maxwell's displacement current for energy and sensors: the origin of nanogenerators, Mater. Today 20 (2017) 74–82.

[195] L. Zhu, Q. Wang, Novel ferroelectric polymers for high energy density and low loss dielectrics, Macromolecules 45 (2012) 2937–2954.

[196] J. Yuan, S. Yao, P. Poulin, Dielectric constant of polymer composites and routes to high-k or low-k nanocomposite materials, in: X. Huang, C. Zhi (Eds.), *Polymer Nanocomposites: Electrical and Thermal Properties*, Springer, Cham, 2016, pp. 3–28.

[197] X. Chen, J.-K. Tseng, I. Treufeld, M. Mackey, D.E. Schuele, R. Li, M. Fukuto, E. Baer, L. Zhu, Enhanced dielectric properties due to space charge-induced interfacial polarization in multilayer polymer films, J. Mater. Chem. C 5 (2017) 10417–10426.

[198] M. Feng, Y. Feng, T. Zhang, J. Li, Q. Chen, Q. Chi, Q. Lei, Recent advances in multilayer-structure dielectrics for energy storage application, Adv. Sci. 8 (2021) 2102221.

[199] M. Mackey, D.E. Schuele, L. Zhu, L. Flandin, M.A. Wolak, J.S. Shirk, A. Hiltner, E. Baer, Reduction of dielectric hysteresis in multilayered films via nanoconfinement, Macromolecules 45 (2012) 1954–1962.

[200] Z. Liu, K. Wang, X. Jiang, M.S. Javed, W. Han, Overcoming current leaks in CNT/PDMS triboelectric composites by wrapping CNTs with TiO_2 insulation layer, Appl. Phys. Lett. 121 (2022) 111.

[201] Z. Liu, M. Muhammad, L. Cheng, E. Xie, W. Han, Improved output performance of triboelectric nanogenerators based on polydimethylsiloxane composites by the

capacitive effect of embedded carbon nanotubes, Appl. Phys. Lett. 117 (2020) 143902.
[202] A. Kaur, A. Gupta, C. Ying, M. Rahmani, G. Sapra, Smart wearable triboelectric nanogenerator for self-powered bioelectronics and therapeutics, Microelectron. Eng. 275 (2023) 111992.
[203] M. Vafaiee, F. Ejehi, R. Mohammadpour, CNT-PDMS foams as self-powered humidity sensors based on triboelectric nanogenerators driven by finger tapping, Sci. Rep. 13 (2023) 370.
[204] X. Xie, X. Xin, X. Hou, S. Qian, J. Hou, J. Zhang, X. Bi, W. Geng, J. Mu, J. He, X. Chou, Wide measurement range and high sensitivity spongy MWCNT/polydimethylsiloxane pressure sensor based on a single-electrode enhanced triboelectric nanogenerator, Sci. China Technol. Sci. 66 (2023) 1734–1745.
[205] W. Yang, Y. Liu, Z. Zhang, Q. Li, T. Yu, Y. Li, Scalable, flexible, and hierarchical porous conductive nanocomposites for self-powered and pressure sensing dual-mode integration, Compos. Sci. Technol. 232 (2023) 109884.
[206] N. Mohamadbeigi, N. Rafiefard, F. Ejehi, R. Mohammadpour, A. Iraji Zad, Triboelectric nanogenerator-enabled 3D microporous polydimethylsiloxane–graphene oxide nanocomposite for flexible self-powered humidity sensing applications, Energy Technol. 12 (2024) 2301136.
[207] J. Chun, J.W. Kim, W.-S. Jung, C.-Y. Kang, S.-W. Kim, Z.L. Wang, J.M. Baik, Mesoporous pores impregnated with Au nanoparticles as effective dielectrics for enhancing triboelectric nanogenerator performance in harsh environments, Energy Environ. Sci. 8 (2015) 3006–3012.
[208] L.K. Anlin, K.V. Vijoy, K. Pradeesh, S. Thomas, H. John, K.J. Saji, Effects of metal nanoparticles on the performance of PDMS based triboelectric nanogenerators, Phys. B Condens. Matter 639 (2022) 413952.
[209] C. Cong, F. Jiang, G. Wang, H. Li, H. Zhang, B. Diao, E. Zhao, S.W. Joo, S.H. Kim, X. Li, Mass-produced, dispenser-printed single-electrode triboelectric nanogenerators for wearable applications: a simple approach, J. Mater. Chem. C 12 (2024) 11603–11614.
[210] P. Chinathun, W. Wongwiriyapan, S. Rattanamai, N. Ketama, A. Treetong, T. Ikuno, G. Tumcharern, A. Klamchuen, Carbon nanotube/polydimethylsiloxane composite micropillar arrays using non-lithographic silicon nanowires as a template for performance enhancement of triboelectric nanogenerators, Nanotechnology 32 (2021) 095303.
[211] X. Fu, J. Zhang, J. Xiao, Y. Kang, L. Yu, C. Jiang, Y. Pan, H. Dong, S. Gao, Y. Wang, A high-resolution, ultrabroad-range and sensitive capacitive tactile sensor based on a CNT/PDMS composite for robotic hands, Nanoscale 13 (2021) 18780–18788.
[212] C.-R. Yang, C.-T. Ko, S.-F. Chang, M.-J. Huang, Study on fabric-based triboelectric nanogenerator using graphene oxide/porous PDMS as a compound friction layer, Nano Energy 92 (2022) 106791.
[213] E.J. Jelmy, D. Jose, K.V. Vijoy, K.J. Saji, H. John, Enhanced triboelectric performance of graphene oxide–conducting polymer hybrid modified polydimethylsiloxane composites, Mater. Adv. 3 (2022) 6897–6907.
[214] D. Meisak, A. Plyushch, M. Kinka, S. Balčiu¯nas, V. Kalendra, S. Schaefer, A. Zarkov, et al.Effect of particle size on the origin of electromechanical response in BaTiO3/PDMS nanogenerators, ACS Appl. Electron. Mater. 6 (2024) 7464–7474.
[215] J. Jiao, Y. Su, C. Wang, M. Zhou, X. Yang, H. Zhang, G. Sui, Novel flexible friction layer constructed from ZnO in situ grown on ZnSnO3 nanocubes toward significantly enhancing output performances of a triboelectric nanogenerator, ACS Appl. Energy Mater. 6 (2023) 1283–1293.
[216] J.M. Yun, G. Kwon, J.-H. Choi, K.-H. Kim, Advancing high-performance piezoelectric nanogenerators: simple electric field switching for orientated and aligned BaTiO3/PDMS composite, ACS Appl. Mater. Interfaces 16 (2024) 55598–55608.
[217] S. Hajra, A.M. Padhan, M. Sahu, P. Alagarsamy, K. Lee, H.J. Kim, Lead-free flexible Bismuth Titanate-PDMS composites: a multifunctional colossal dielectric material for hybrid piezo-triboelectric nanogenerator to sustainably power portable electronics, Nano Energy 89 (2021) 106316.
[218] G. Suo, Y. Yu, Z. Zhang, S. Wang, P. Zhao, J. Li, X. Wang, Piezoelectric and triboelectric dual effects in mechanical-energy harvesting using BaTiO3/polydimethylsiloxane composite film, ACS Appl. Mater. Interf. 8 (2016) 34335–34341.
[219] P. Paydari, N. Manavizadeh, A. Hadi, J. Karamdel, The morphology effect of embedded ZnO particles-based composite on flexible hybrid piezoelectric triboelectric nanogenerators for harvesting biomechanical energy, J. Sol-Gel Sci. Technol. 105 (2023) 337–347.
[220] X. He, X. Mu, Q. Wen, Z. Wen, J. Yang, C. Hu, H. Shi, Flexible and transparent triboelectric nanogenerator based on high performance well-ordered porous PDMS dielectric film, Nano Res. 9 (2016) 3714–3724.
[221] W. Oh, S. Hajra, S. Divya, S. Panda, Y. Oh, Z. Jaglic, P. Pakawanit, T.H. Oh, H. J. Kim, Contact electrification of porous PDMS-nickel ferrite composites for effective energy harvesting, Mater. Sci. Eng. B 292 (2023) 116397.
[222] P. Zhang, W. Zhang, L. Deng, H. Zhang, A triboelectric nanogenerator based on temperature-stable high dielectric BaTiO3-based ceramic powder for energy harvesting, Nano Energy 87 (2021) 106176.
[223] I.S.S. Vamsi, N.K. Das, S. Badhulika, FeNbO4 nanoparticles-PDMS based triboelectric nanogenerator as self-powered sensor for human motion and artefact position tracking applications, Mater. Sci. Semicond. Process. 169 (2024) 107945.
[224] D. Ali, B. Yu, X. Duan, H. Yu, M. Zhu, Enhancement of output performance through post-poling technique on BaTiO3/PDMS-based triboelectric nanogenerator, Nanotechnology 28 (2017) 075203.
[225] K. Lee, S. Park, K. Jung, I.Y. Moon, J.H. Ryu, K.M. Kim, Enhanced performance of polydimethylsiloxane-based triboelectric nanogenerator using BaTiO3/MWCNTs, J. Cryst. Growth 649 (2025) 127952.
[226] S. Sriphan, T. Charoonsuk, T. Maluangnont, N. Vittayakorn, High-performance hybridized composited-based piezoelectric and triboelectric nanogenerators based on BaTiO3/PDMS composite film modified with Ti0.8O2 nanosheets and silver nanopowders cofillers, ACS Appl. Energy Mater. 2 (2019) 3840–3850.
[227] S. Feng, H. Zhang, D. He, Y. Xu, A. Zhang, Y. Liu, J. Bai, Synergistic effects of BaTiO3/multiwall carbon nanotube as fillers on the electrical performance of triboelectric nanogenerator based on polydimethylsiloxane composite films, Energy Technol. 7 (2019) 1900101.
[228] F.F. Hatta, M.A.S. Haniff, M.A. Mohamed, Enhanced-performance triboelectric nanogenerator based on polydimethylsiloxane/barium titanate/graphene quantum dot nanocomposites for energy harvesting, ACS Omega 9 (2024) 5608–5615.
[229] N. Buatip, D. Munthala, P. Amonpattaratkit, P. Pakawanit, X. Hu, W. Jongpinit, P. Janphuang, C. Wan, C. Bowen, S. Pojprapai, Piezo-tribo-electric nanogenerator based on BCZT/MCNTs/PDMS piezoelectric composite for compressive energy harvesting, Mater. Res. Bull. 173 (2024) 112686.
[230] D.H. Kim, B. Dudem, J.S. Yu, High-performance flexible piezoelectric-assisted triboelectric hybrid nanogenerator via polydimethylsiloxane-encapsulated nanoflower-like ZnO composite films for scavenging energy from daily human activities, ACS Sust. Chem. Eng. 6 (2018) 8525–8535.
[231] V.S. Kavarthapu, M.V. Paranjape, P. Manchi, A. Kurakula, J.K. Lee, S.A. Graham, J.S. Yu, Wireless alerts and data monitoring from BNNO-MWCNTs/PDMS composite film-based TENG integrated inhaler for smart healthcare application, Small 20 (2024) 2403218.
[232] K. Lee, H. Han, J.H. Ryu, S. Kang, K. Jung, Y.-K. Kim, T. Song, S. Mhin, K.M. Kim, Laser-driven formation of ZnSnO3/CNT heterostructure and its critical role in boosting performance of the triboelectric nanogenerator, Carbon 212 (2023) 118120.
[233] C. Lee, C. Cho, J.H. Oh, Highly flexible triboelectric nanogenerators with electrospun PVDF-TrFE nanofibers on MWCNTs/PDMS/AgNWs composite electrodes, Compos. Part B Eng. 255 (2023) 110622.
[234] N. Rafiefard, S. Fardindoost, M.K. Kisomi, L. Shooshtari, A. Irajizad, S. Seddighi, R. Mohammadpour, D. Vashaee, High-performance flexible and stretchable self-powered surface engineered PDMS-TiO2 nanocomposite based humidity sensors driven by triboelectric nanogenerator with full sensing range, Sens. Actuat. B Chem. 378 (2023) 133105.
[235] X. Meng, D. Hui, S. Ge, S. Zhou, X. Hu, D. Lin, W. Liu, Nanopillar-and nanocone-structured SrTiO3/PDMS films for triboelectric nanogenerators, ACS Appl. Nano Mater. 7 (2024) 14193–14202.
[236] G. Jian, Q. Meng, Y. Jiao, L. Feng, H. Shao, F. Wang, F. Meng, Hybrid PDMS-TiO2-stainless steel textiles for triboelectric nanogenerators, Chem. Eng. J. 417 (2021) 127974.
[237] W. He, Y. Qian, B.S. Lee, F. Zhang, A. Rasheed, J.-E. Jung, D.J. Kang, Ultrahigh output piezoelectric and triboelectric hybrid nanogenerators based on ZnO nanoflakes/polydimethylsiloxane composite films, ACS Appl. Mater. Interf. 10 (2018) 44415–44420.
[238] S. Mei, B. Xu, J. Wan, J. Chen, Preparation of CNT/CNF/PDMS/TPU nanofiber-based conductive films based on centrifugal spinning method for strain sensors, Sensors 24 (2024) 4026.
[239] Y. Qian, M. Sohn, W. He, H. Park, K.R.V. Subramanian, D.J. Kang, A high-output flexible triboelectric nanogenerator based on polydimethylsiloxane/three-dimensional bilayer graphene/carbon cloth composites, J. Mater. Chem. A 8 (2020) 17150–17155.
[240] C. Jirayupat, W. Wongwiriyapan, P. Kasamechonchung, T. Wutikhun, K. Tantisantisom, Y. Rayanasukha, T. Jiemsakul, et al., Piezoelectric-induced triboelectric hybrid nanogenerators based on the ZnO nanowire layer decorated on the Au/polydimethylsiloxane–Al structure for enhanced triboelectric performance, ACS Appl. Mater. Interfaces 10 (2018) 6433–6440.
[241] M. Matsunaga, J. Hirotani, S. Kishimoto, Y. Ohno, High-output, transparent, stretchable triboelectric nanogenerator based on carbon nanotube thin film toward wearable energy harvesters, Nano Energy 67 (2020) 104297.
[242] S.S. Panicker, S.P.R. Rajeev, V. Thomas, Harnessing multi-effect energy harvesting: triboelectric, piezoelectric, and flexoelectric synergies of hBN in PDMS nanogenerators for high-performance wearable devices, J. Mater. Sci. 1–23 (2025).
[243] S. Ozden, N.S. Dutta, K. Randazzo, T. Tsafack, C.B. Arnold, R.D. Priestley, Interfacial engineering to tailor the properties of multifunctional ultralight weight hBN-polymer composite aerogels, ACS Appl. Mater. Interf. 13 (2021) 13620–13628.
[244] M. Noor-A-Alam, M. Nolan, Engineering ferroelectricity and large piezoelectricity in h-BN, ACS Appl. Mater. Interf. 15 (2023) 42737–42745.
[245] M. Noor-A-Alam, H.J. Kim, Y.-H. Shin, Dipolar polarization and piezoelectricity of a hexagonal boron nitride sheet decorated with hydrogen and fluorine, Phys. Chem. Chem. Phys. 16 (2014) 6575–6582.
[246] Y. Nan, D. Tan, J. Zhao, M. Willatzen, Z.L. Wang, Shape-and size dependent piezoelectric properties of monolayer hexagonal boron nitride nanosheets, Nanoscale Adv. 2 (2020) 470–477.
[247] M. Droth, G. Burkard, V.M. Pereira, Piezoelectricity in planar boron nitride via a geometric phase, Phys. Rev. B 94 (2016) 075404.
[248] H. Rostami, F. Guinea, M. Polini, R. Roldán, Piezoelectricity and valley chern number in inhomogeneous hexagonal 2D crystals, NPJ 2D Mater. Appl 2 (2018) 15.
[249] A. Tripathy, B. Saravanakumar, S. Mohanty, S.K. Nayak, A. Ramadoss, Comprehensive review on flexoelectric energy harvesting technology: mechanisms, device configurations, and potential applications, ACS Appl. Electron. Mater. 3 (2021) 2898–2924.

[250] S.I. Kundalwal, V.K. Choyal, V. Choyal, Flexoelectric effect in boron nitride–graphene heterostructures, Acta Mech. 232 (2021) 3781–3800.
[251] P.V. Minh, L.M. Thai, N.T. Dung, A. Tounsi, N.T.C. Nhung, D.V. Thom, An overview of the flexoelectric phenomenon, potential applications, and proposals for further research directions, Int. J. Mech. Mater. Des. 19 (2023) 903–925.
[252] B.G. Hyun, Y.-S. Jun, J.-H. Lee, M. Hamidinejad, Z. Saadatnia, S. Ghaffari-Mosanenzadeh, H.E. Naguib, C.B. Park, Incorporation of hexagonal boron nitride and multi-walled carbon nanotube for motion sensing triboelectric nanogenerators, Compos. Part B Eng. 272 (2024) 111193.
[253] R. Umapathi, M. Rethinasabapathy, V. Kakani, H. Kim, Y. Park, H.K. Kim, G. M. Rani, H. Kim, Y.S. Huh, Hexagonal boron nitride composite film based triboelectric nanogenerator for energy harvesting and machine learning assisted handwriting recognition, Nano Energy 136 (2025) 110689.
[254] A. Mahapatra, R.S. Ajimsha, D. Deepak, R. Aggarwal, S. Kumar, R. Venkatesh, S. S. Roy, P. Misra, Textile-integrated MoS_2-PDMS single electrode triboelectric nanogenerator for vibrational energy harvesting and biomechanical motion sensing, Nano Energy 116 (2023) 108829.
[255] W. He, M. Sohn, R. Ma, D.J. Kang, Flexible single-electrode triboelectric nanogenerators with MXene/PDMS composite film for biomechanical motion sensors, Nano Energy 78 (2020) 105383.
[256] C. Jiang, X. Li, Y. Ying, J. Ping, A multifunctional TENG yarn integrated into agrotextile for building intelligent agriculture, Nano Energy 74 (2020) 104863.
[257] A. Mirsepah, L. Shooshtari, R. Mohammadpour, A. Esfandiar, A. Irajizad, Wearable broadband MoS2 photodetector for dual heart rate and UV detection powered by PDMS-MXene TENG, Chem. Eng. J. 499 (2024) 155953.
[258] Q. Zhou, Y. Zheng, K. Kinoshita, T. Ikuno, Effect of adding ionic liquid into polydimethylsiloxane on the output performance of triboelectric nanogenerator, Jpn. J. Appl. Phys. 63 (2024) 121004.
[259] L. Zhu, P. Xu, B. Chang, J. Ning, T. Yan, Z. Yang, H. Lu, Hierarchical structure by self-sedimentation of liquid metal for flexible sensor integrating pressure detection and triboelectric nanogenerator, Adv. Funct. Mater. 34 (2024) 2400363.
[260] K. Zhao, J. Zhou, Z. Gao, J. Zhang, Y. Ye, J. Wu, Y. Ding, B. Zhang, Triboelectric nanogenerator fabricated from high-charge-density, wear-resistant MoS2 nanosheet-PDMS composite for energy harvesting and motion detection, ACS Appl. Electron. Mater. 7 (2025) 1985–2000.
[261] L. Lan, T. Yin, C. Jiang, X. Li, Y. Yao, Z. Wang, S. Qu, Z. Ye, J. Ping, Y. Ying, Highly conductive 1D-2D composite film for skin-mountable strain sensor and stretchable triboelectric nanogenerator, Nano Energy 62 (2019) 319–328.
[262] J. Lin, Y. Tsao, M. Wu, T. Chou, Z. Lin, J. Wu, Single-and few-layers MoS2 nanocomposite as piezo-catalyst in dark and self-powered active sensor, Nano Energy 31 (2017) 575–581.
[263] X. He, H. Guo, X. Yue, J. Gao, Y. Xi, C. Hu, Improving energy conversion efficiency for triboelectric nanogenerator with capacitor structure by maximizing surface charge density, Nanoscale 7 (2015) 1896–1903.
[264] L. Shan, L. Yaqian, Z. Xianghong, L. Enlong, Y. Rengjian, L. Qiming, C. Xiang, C. Huipeng, G. Tailiang, Bioinspired kinesthetic system for human-machine interaction, Nano Energy 88 (2021) 106283.
[265] X. Zhao, Z. Ounaies, A facile method to enhance the flexibility and triboelectric output of PDMS using ionic liquid-coated single-wall carbon nanotubes, Nano Energy 94 (2022) 106908.
[266] W. Qing, W. Yan, Y. Zhou, Q. Zhou, Q. Wang, Y. Zhu, K. Wei, L. Zhao, H. Zou, S. Liu, Flexible single-electrode triboelectric sensor array for material recognition and trajectory tracking, Chem. Eng. J. (2025).
[267] S. Chen, W. Xing, H. Wang, W. Cheng, Z. Lei, F. Zheng, P. Tao, et al., A bottom-up approach to generate isotropic liquid metal network in polymer-enabled 3D thermal management, Chem. Eng. J. 439 (2022) 135674.
[268] S.-Q. Yi, H. Sun, Y.-F. Jin, K.-K. Zou, J. Li, L.-C. Jia, D.-X. Yan, Z.-M. Li, CNT-assisted design of stable liquid metal droplets for flexible multifunctional composites, Compos. Part B Eng. 239 (2022) 109961.
[269] G. Li, Z. Li, H. Hu, B. Chen, Y. Wang, Y. Mao, H. Li, B. Zhang, Recent progress in self-healing triboelectric nanogenerators for artificial skins, Biosensors 15 (2025) 37.
[270] H. Xu, X. Wang, J. Niu, Y. Nan, J. Pu, H. Zhou, J. Duan, Y. Huang, B. Hou, Construction of MXene/PDMS-based triboelectric nanogenerators for high-performance cathodic protection, Adv. Mater. Interf. 9 (2022) 2102085.
[271] Y. Liu, E. Li, Y. Yan, Z. Lin, Q. Chen, X. Wang, L. Shan, H. Chen, T. Guo, A one-structure-layer PDMS/MXenes based stretchable triboelectric nanogenerator for simultaneously harvesting mechanical and light energy, Nano Energy 86 (2021) 106118.
[272] Y. Cai, X.-N. Zhang, G.-G. Wang, G.-Z. Li, D.-Q. Zhao, N. Sun, F. Li, H.-Y. Zhang, J.-C. Han, Y. Yang, A flexible ultra-sensitive triboelectric tactile sensor of wrinkled PDMS/MXene composite films for E-skin, Nano Energy 81 (2021) 105663.
[273] S.P. Selvam, S. Cho, Energizing bacterial sensing: unveiling new horizons with Ti2CN-ionic liquid triboelectric nanogenerator, Chem. Eng. J. 475 (2023) 146434.
[274] P. Lv, L. Shi, C. Fan, Y. Gao, A. Yang, X. Wang, S. Ding, M. Rong, Hydrophobic ionic liquid gel-based triboelectric nanogenerator: next generation of ultrastable, flexible, and transparent power sources for sustainable electronics, ACS Appl. Mater. Interf. 12 (2020) 15012–15022.
[275] D. Xu, Z. Jing, H. Wang, W. Yang, P. Xu, D. Niu, P. Ma, Advanced flexible self-healing triboelectric nanogenerators for applications in complex environments, Nanoscale 17 (2025) 7713–7737.
[276] S.-J. Wang, L. Wang, H.-Z. Su, Z.-C. Wu, Q.-G. Zhang, W. Fan, X.-L. Jing, Room temperature self-healing and recyclable PDMS elastomers with superior mechanical properties for triboelectric nanogenerators, Chin. J. Polym. Sci. 42 (2024) 1566–1577.
[277] Z. Peng, Y. Zhang, C.L.R. Choi, P. Zhang, T. Wu, Y.K. Chan, Continuous roller nanoimprinting: next generation lithography, Nanoscale 15 (2023) 11403–11421.
[278] M. Cheng, X. Chen, J. Zhang, X. Ye, B. Ai, Closing the loop in next-generation sensing through shadow sphere lithography, plasmonics, and artificial intelligence, Chem. Soc. Rev. (2025).
[279] Y. Su, D. Yin, X. Zhao, T. Hu, L. Liu, Exploration of advanced applications of triboelectric nanogenerator-based self-powered sensors in the era of artificial intelligence, Sensors 25 (2025) 2520.
[280] A. Babu, D. Mandal, Roadmap to human c convergence, ACS Appl. Energy Mater. 7 (2024) 822–833.
[281] R. Zhang, Machine learning-assisted triboelectric nanogenerator-based self-powered sensors, Cell Rep. Phys. Sci. 5 (2024).
[282] M. Ji, Z. Wang, J. Wu, L. Huang, M. Zheng, G. Cheng, H. Cai, J. Luo, H. Zhou, Z. L. Wang, Machine learning assisted triboelectric nanogenerator technology for intelligent sports, Sci. Adv. 11 (2025) eadz3515.
[283] H. Sheng, Y. Ma, H. Zhang, J. Yuan, F. Li, W. Li, E. Xie, W. Lan, Integration of supercapacitors with sensors and energy harvesting devices: a review, Adv. Mater. Technol. 9 (2024) 2301796.
[284] M.C.A. de Oliveira, A.L. Freire, I.C.M. Candido, Y. Messaddeq, J. Lapointe, J. Fréchette, R. Vallée, H.P. de Oliveira, Flexible micro supercapacitor and triboelectric nanogenerator based on laser induced graphene for self-powered applications, Next Energy 4 (2024) 100141.
[285] H. Afshar, F. Kamran, F. Shahi, Recent progress in energy harvesting technologies for self-powered wearable devices: the significance of polymers, Polym. Adv. Technol. 36 (2025) e70187.
[286] Y. Luo, M.R. Abidian, J.-H. Ahn, D. Akinwande, A.M. Andrews, M. Antonietti, Z. Bao, et al., Technology roadmap for flexible sensors, ACS Nano 17 (2023) 5211–5295.